

Degree Project in Industrial Design Engineering

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Improving the User Experience of Smart Locks

A Study on Implementing Inclusive Design on Keypads

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Abstract

As smart locks become increasingly more common, the need for inclusive and accessible keypad design grows. This thesis investigates how keypad interfaces can be designed to accommodate a diverse range of users, including individuals with varying physical, sensory, and cognitive abilities. The project was conducted on behalf of a company specializing in locks and access systems, the project follows the Double Diamond design process—Discover, Define, Develop, and Deliver.

Through a combination of literature review, user testing, interviews, and evaluations, the study identifies common usability challenges such as unclear symbols, poor visibility, and insufficient tactile feedback. These insights informed a set of design recommendations, compiled into a handbook and applied to two conceptual product prototypes. The results emphasize the importance of intuitive layouts, clear visual and tactile cues, and responsive feedback. By addressing these aspects, the proposed solutions aim to enhance usability for a broader user base and support the development of more inclusive products.

Keywords

Inclusive design, Keypad usability, Smart locks, Accessibility, User-centered design, Physical interaction, Sensory capabilities, Cognitive capabilities, Motor capabilities, Visual feedback, Tactile feedback, Design recommendations, Double Diamond Process

Sammanfattning

I takt med att smarta lås blir allt vanligare ökar behovet av inkluderande och tillgänglig design för knappsatser. Denna rapport undersöker hur knappsatser kan utformas för att vara användbara för en bred användargrupp, där personer med olika fysiska, sensoriska och kognitiva förutsättningar inkluderas. Projektet har genomförts på uppdrag av ett företag som utvecklar produkter och lösningar för lås och passersystem, och följer Double Diamond-processens fyra faser: Discover, Define, Develop och Deliver.

Genom en kombination av litteraturstudier, användartester, intervjuer och utvärderingar identifierades vanliga utmaningar vid användning av knappsatser, såsom otydliga symboler, bristande synlighet och avsaknad av taktil återkoppling. Dessa insikter låg till grund för en uppsättning designrekommendationer som sammanställdes i en handbok och tillämpades i två konceptuella produktprototyper. Resultaten betonar vikten av intuitiva layouter samt tydlig visuell och taktil återkoppling. Genom att ta hänsyn till dessa aspekter syftar de föreslagna lösningarna till att förbättra användbarheten för en bredare användargrupp och främja utvecklingen av mer inkluderande produkter.

Nyckelord

Inkluderande design, Användbarhet, Knappsats, Smarta lås, Tillgänglighet, Användarcentrerad design, Fysisk interaktion, Sensoriska förmågor, Kognitiva förmågor, Motoriska förmågor, Visuell återkoppling, Taktil återkoppling, Designrekommendationer, Double Diamond-processen

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List of Acronyms

AI	Artificial Intelligence
CAD	Computer-Aided Design
CJM	Customer Journey Map
EAA	European Accessibility Act
FDM	Fused Deposition Modeling
IEC	International Electrotechnical Commission
ISO	International Organization for Standardization
LED	Light Emitting Diode
RFID	Radio-Frequency Identification
RQ	Research Question
SLA	Stereolithography
SDG	Sustainable Development Goals
UV	Ultraviolet
WHO	World Health Organization

1 Introduction

This chapter will provide an introduction to the project, and it will describe the context and problem of the project, as well as its purpose, goals, delimitations and structure.

1.1 Background

Smart locks for domestic use are becoming increasingly popular. According to a survey on smart home security conducted by Statista (2022), the penetration rate of smart locks increased from 0.5 percent in 2017 to 1.9 percent in 2021 worldwide and is expected to climb to 5.4 percent by 2026. This indicates that smart locks will reach even larger user groups, naturally representing different user capabilities.

This master thesis project is conducted at a company that offers various products and solutions related to locks, entrances, and access systems. Many of these products utilize entrance codes and keypads. The company now seeks to explore if and how their keypads could be improved based on inclusive design. The project will focus on a reference product, a smart lock for domestic use, as this product reaches a wide and diverse user group.

1.2 Problem Description

The company has expressed a desire and need to incorporate a more inclusive design mindset during the development process, aiming to create products that are more accessible to a diverse user base. Currently, products are often tested by smaller groups that do not represent a wide range of functional capabilities, which risks missing valuable feedback.

As the popularity of these products increases and the population continues to age, it is crucial to adapt the design to be more inclusive (Statista, 2022). The European Accessibility Act (EAA) (European Parliament and Council of the European Union, 2019) addresses this issue by imposing new accessibility standards for products and services that are essential for people to maintain their independence. Although the reference product in this project is not directly affected by this legislation, the company provides products and solutions to a wide range of markets, making the EAA highly relevant and important to follow.

1.3 Purpose

The purpose of this thesis is to increase knowledge on how keypads can be designed to be more inclusive and accessible for a diverse user group. By exploring various factors that contribute to keypad design, this work aims to provide insights that can enhance usability for as many users as possible.

To achieve this purpose, the following research questions (RQ) will be addressed:

1. What common obstacles or limitations do users face when interacting with keypads?
2. What key factors need to be considered when designing keypads for inclusive use?
3. How can inclusive design be applied to existing products to improve keypad usability?

1.3.1 Contribution Related to the United Nations Sustainable Development Goals

This project contributes to three of the United Nations Sustainable Development Goals (SDGs), particularly SDG 9 (Industry, Innovation and Infrastructure), SDG 10 (Reduced Inequalities), and SDG 11 (Sustainable Cities and Communities) (see Figure **1**). By applying inclusive design to smart lock keypads, the project promotes the development of more accessible and user-friendly technology for the future. The project encourages innovation and promotes inclusive and social sustainability for industrial products (SDG 9), where keypads are commonly used. The thesis addresses barriers and suggests solutions for individuals with diverse physical, sensory, and cognitive abilities (SDG 10), and contributes to safer and more inclusive urban environments (SDG 11). The project aligns with the United Nations ambition to ensure that technological progress benefits all members of society, regardless of capabilities (United Nations, 2015).



Figure 1: The United Nations Sustainable Development Goals 9, 10 & 11 (United Nations, 2015)

1.4 Goal

The goal of this project is to identify the user needs and capabilities that most significantly affect the user experience when interacting with a keypad. The project aims to develop recommendations for designing keypads for smart locks in an inclusive way, improving usability and user experience for a wide and diverse user group. This goal is divided into the following three sub-goals:

- Investigate and identify user needs.
- Investigate current keypad designs to identify strengths and weaknesses.

- Develop design recommendations and physical prototypes.

The deliverables of this project are:

- A handbook containing the design recommendations, presented with visualizations.
- Two concepts showing how the recommendations can been applied to the reference product.
- A physical model of a smart lock, implementing the design recommendations for the keypad.

This project will benefit two main groups: the company the project is conducted at, and the customers who buy and use the products. The company will receive valuable feedback on the usability of their current products and recommendations for future improvements. If these recommendations are implemented, customers will benefit from a product designed for a wider range of users, ideally enhancing its usability for all.

1.5 Delimitations

To define the project's scope and maintain focus, delimitations were introduced by the company at the beginning of the project and later added by us during its continuation. The delimitations are as follows:

- Limited to products with keypads.
- Only the area around the keypad is considered on the reference product.
- Not included is the exploration and development of new fonts.
- Not included is the exploration and development of sound signals.
- Not included is the exploration and development of light signals.
- Not included are electronics and software.
- Not included are IP classifications, materials properties or manufacturing methods.

The delimitations above were established to align the project with the company's expectations and to ensure a focused and manageable scope. The project includes the overall interaction with the keypad, including visual elements such as symbols, layout, and feedback. However, light and sound signals were not studied in depth, as the company already has an ongoing study on the subject, and uses company-specific sound signals. Similarly, font development was excluded as the company already has established fonts in their design guidelines. Since the project focuses on the physical interface, areas such as electronics, software, material properties, and manufacturing methods were also excluded as they fall outside the scope of user interaction.

1.6 Structure of the Thesis

This thesis is structured as follows:

- **Chapter 2:** Presents the methodology and methods used to conduct this project. It outlines the research, data collection techniques, and analysis methods employed throughout the study.
- **Chapter 3:** Covers the Discover stage, where the subject is examined through various research methods. This chapter includes subchapters detailing the desktop research, testing of reference product, and observational studies conducted to understand the current state and user needs.
- **Chapter 4:** Details the Define stage, summarizing insights from the Discover stage to create a problem statement. This chapter combines the findings, identifies key user needs, and challenges that the project aims to address.
- **Chapter 5:** Focuses on the Develop stage, where different potential solutions are generated and evaluated. This chapter describes the ideation process, concept development, and prototyping activities undertaken to explore various design options.
- **Chapter 6:** Describes the Deliver stage, where the final solution is chosen, validated, and presented as the result of the project. This chapter includes the testing and refinement of the chosen solution, as well as the final design recommendations and prototypes.
- **Chapter 7:** Discusses the execution of the project and its results. It provides an analysis of the project's outcomes, the effectiveness of the design solutions, and reflections on the research process.
- **Chapter 8:** Presents the conclusions drawn from the project. It summarizes the key findings, discusses the implications for the company and the broader field of inclusive design, and suggests directions for future research.

2 Method

The methodology of this project follows the Double Diamond process, a design and innovation framework consisting of four key phases: *Discover, Define, Develop,* and *Deliver*. This process balances divergent (exploratory) and convergent (focused) thinking, as illustrated in Figure **2** (Design Council, 2005).

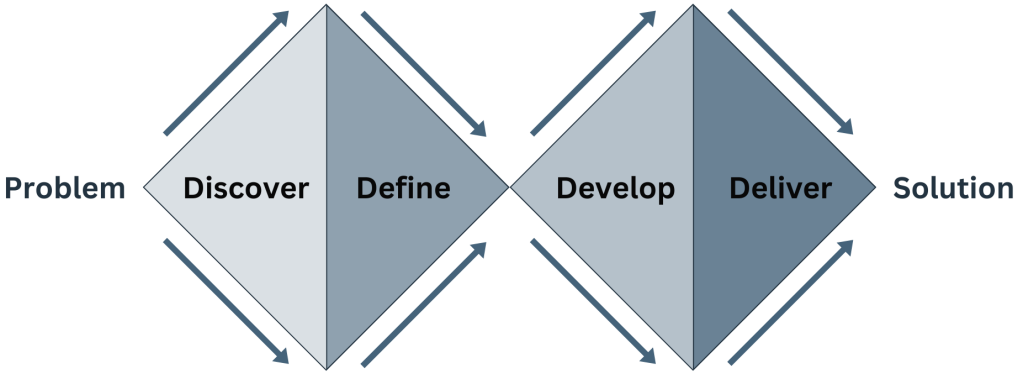


Figure 2: Illustration of the double diamond process

The Double Diamond process provides guidance on the activities to be performed and helps in dividing the project timeline. It consists of the following phases:

- **Discover:** This phase involves exploring the problem space and gathering insights through various research methods. It includes desktop research, testing of the reference product, and observational studies to understand the current state and user needs.
- **Define:** In this phase, the insights from the Discover stage are compiled to create a clear problem statement. It involves identifying key user needs and challenges that the project aims to address.
- **Develop:** This phase focuses on generating and evaluating different potential solutions trough ideation, concept development, and prototyping.
- **Deliver:** The final phase involves choosing, refining, and presenting the final solution.

In addition to using the Double Diamond process as a guide, a semi-flexible Gantt chart was used to keep track of time and activities throughout the project.

2.1 Discover

This section covers the methods used during the Discover phase of the project. The aim of this phase was to explore inclusive design, evaluate keypad solutions and understand user needs.

2.1.1 Desktop Research

To create a comprehensive understanding of the project, a desktop research was conducted by exploring the internet and searching for relevant keywords. The search engine used for this was *Google Scholar*,

leading to databases such as *ResearchGate* and *ScienceDirect*. Literature was gathered on several main topics: *inclusive design, user capabilities, standards and guidelines, ergonomics and anthropometry, keypad design, keypad technology, smart locks,* and *existing solutions*.

To understand how inclusive design is currently applied, relevant standards and guidelines were reviewed, along with research on ergonomics and anthropometry. Studies on ergonomics and usability provided insight into preferred keypad sizes, shapes, and layouts. The research also explored user priorities when interacting with keypads, including the importance of feedback.

Given the widespread use of keypads, it was also important to examine existing solutions on the market and the technologies available. This included analyzing different types of keypads and their technical properties to understand their potential and limitations.

2.1.2 Testing of Reference Product

To gain a better understanding of the reference product and the user experience it provides, a series of user tests were conducted to identify potential limitations. Due to security and access restrictions, external participants could not be invited. As a result, the test group consisted of seven individuals already present at the office, who are not involved in the development of the tested product. This setup allowed the study to be carried out in a realistic environment while maintaining a neutral perspective among participants. The test group had varied experience with the reference product, ranging from those who had the product at home and those who had never encountered it. All participants were able-bodied, without any reduced functional capability.

The tests were conducted in the company's office, where the smart lock was mounted on a door. Participants were asked to unlock the door five times under different conditions. Each time, they entered a six-digit code followed by pressing an unlock key with a symbol of an unlocked padlock. The entrance code was changed between rounds to prevent memorization.

The different conditions were as follows:

1. **Baseline:** During the first round, the participant was asked to unlock the door without receiving further instructions.
2. **Standing straight:** During the second round, the participant was not allowed to bend down or angle their body to better view the keypad.
3. **Wearing gloves:** During the third round, the participant was asked to wear gloves when unlocking the door.

4. **Reduced dexterity:** During the fourth round, the participant was asked to attach a form of exoskeleton made of paper that reduced their fine motor skills.

5. **Closed eyes:** During the last round, the participant was asked to close their eyes when unlocking the door.

After each unlocking task, participants rated their experience on a scale of 1-5 based on the following questions:

1. How well could you see the numbers and symbols on the keypad?

2. How easy did you find it to locate the correct keys?

3. How easy did you find it to press the correct keys?

4. How sure did you feel that you pressed the correct keys?

During the tests, one person was responsible for asking the questions, while another person wrote down the participants responses and took notes on observations. The responses were then analyzed to identify common limitations and experiences with the smart lock.

2.1.3 Interview with Customer Support Manager

To collect insights from customer support, an interview was conducted with a Customer Support Manager regarding the reference product. The interview aimed to gather information on the product's usage, user preferences, and common feedback. The interview was semi-structured, allowing for both predefined questions and follow-up questions based on the manager's responses. The interview was recorded, transcribed and analyzed to extract insights regarding the product's performance and user experience. The list of interview questions is presented in Appendix A.

2.1.4 Interviews with Trigger Material

To gather information on people's experiences, perceptions, and preferences regarding keys and keypads, a series of interviews were conducted using trigger material. The study involved 11 participants aged between 27 and 64 years. Participant 1 (P1) had severely reduced motor and dexterity capabilities and used a wheelchair, assisted by a personal assistant. Participant 2 (P2) was visually impaired, while participants 3 to 11 considered themselves able-bodied.

The trigger material, referred to as the "trigger board," consisted of an MDF board with six different keypads mounted on it, see Figure **3**. The trigger board was designed to prompt participants to think about keys and keypads through interaction.

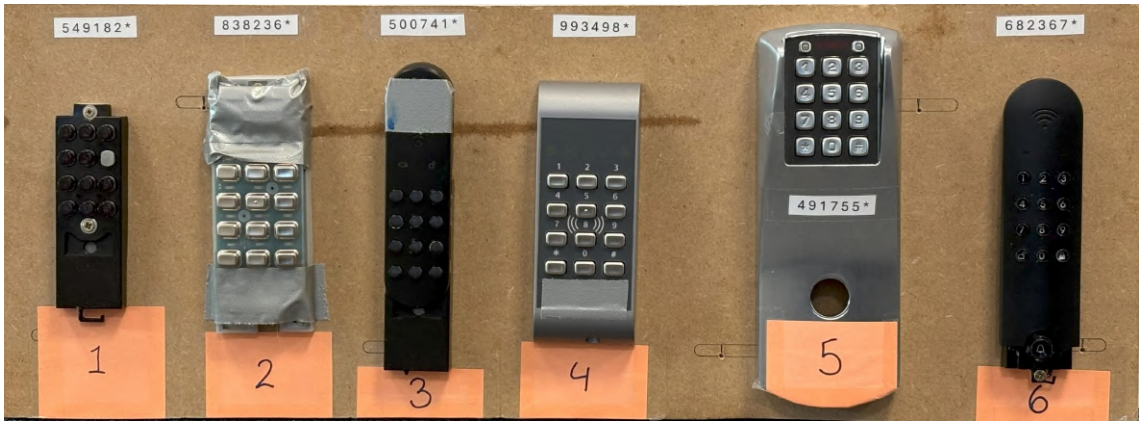


Figure 3: Trigger board

The trigger board was placed in a standing position on a table or shelf, positioning the keypads at a height similar to that of a door handle to simulate a realistic context. Participants were asked to enter randomly generated codes assigned to the keypads, with brief questions posed between each entry.

Following the keypad interaction, participants underwent a shorter interview focusing on the tested keypads. They were asked questions about their preferences and to compare the keypads with each other. None of the keypads on the trigger board were connected to electricity, eliminating any auditory or LED light feedback that the original product might have provided. The interview guide for this segment is attached in Appendix B.

Lastly, another interview was conducted in which participants were asked about the use of keypads in their daily lives, both in relation to doors and access, as well as other situations such as payment terminals and phones. The interview guide for this segment is attached in Appendix C. The trigger board remained close to the participants during both interviews to help them recall and feel how keypads can look and make them feel.

The responses from both interview segments were recorded, transcribed and analyzed to identify common experiences, perceptions, and preferences regarding keys and keypads.

2.1.5 Layout Comparison Tests

To explore users' attitudes towards different keypad layouts, a test was conducted, collecting quantitative data. The test involved 17 participants from various occupations and backgrounds, all identified as able-bodied. The procedure of the test is presented below:

- **Keypad Layouts:** Ten different keypad layouts were created, varying in the arrangement of the keys. These layouts were presented with both round and square keys to determine if participants had different preferences based on the key shape. The layouts are presented in Appendix D.

- **Preferred Layouts:** Participants were asked to look at and pretend to interact with the keypads, which were printed on paper. They then chose their favorite layout(s) based on which one they would prefer to use on a daily basis.
- **Special Function Keys:** After selecting their preferred layout, participants were tasked with designing one or more special function keys for unlocking and locking a connected door. They were provided with black, blue, green, and red pens to draw the key(s) on the keypad, deciding its position and appearance.

All tests were conducted in private to avoid influencing future participants. During the tests, one person asked the questions while another took notes on the participants' comments and preferences. The papers with the chosen keypads and drawn keys were collected and compiled together with the notes to identify common preferences for keypad layouts and special function key designs.

2.2 Define

This section covers the methods used during the Define phase of the project. The aim of this phase was to narrow down the insights gathered during the Discover phase to identify key problems and opportunities.

2.2.1 Clustering

Clustering was used to organize the collected information in a structured way. This was done by writing down findings on post-it notes and placing them on a wall to get a clear view of everything. By clustering the data, themes could be identified. This visual approach helped in recognizing patterns and connections between different areas, which eased the compilation of insights. The clustering can be seen in Appendix H.

2.2.2 Customer Journey Maps

Customer Journey Maps (CJM) were used to visualize a user's interactions with a product or service over time, as described by Wikberg Nilsson, Ericson, and Törlind (2015). These maps were created based on insights from the discovery phase and helped identify pain points and opportunities for improvement. Two CJMs were created (see Appendix I), one where the users went to work, and one where the user was at home. The CJMs followed two personas, one able bodied and one wheelchair user.

2.2.3 Needs Analysis

A needs analysis, which can be seen in Appendix J, was conducted using insights from the Discovery phase and the CJM. The needs were categorized into physical, cognitive, and emotional needs to

gain a comprehensive understanding of the user's requirements. Physical needs cover aspects such as ergonomics and accessibility. Cognitive needs cover aspects such as simple navigation and clarity. Emotional needs cover the user's feelings and satisfaction when interacting with the product.

2.3 Develop

This section covers the methods used during the Develop phase of the project. The aim of this phase was to generate and refine solutions based on the defined objectives. This involved ideation, prototyping, and testing various ideas to address the identified problems.

2.3.1 Ideation

To quickly visualize and compare ideas, both sketching and modeling clay were used. Initially, potential key and product shapes, as well as various layouts, were freely sketched on paper. Templates matching the dimensions of the reference product were printed to facilitate accurate sketching of potential solutions. Following the sketching phase, modeling clay was used to construct the sketches, providing a tangible sense of their physical form. Clay keys were also added to 3D-printed models of the reference product to evaluate their appearance and tactile feel in a real-world context. The sketches and clay models were analyzed to identify promising design concepts based on their visual expressions and tactile feel.

2.3.2 Concept Exploration

To explore a concept of numbered keys developed during the ideation phase, multiple versions were created. CAD models were designed to represent the concept in various sizes, shapes, and key distances. The shapes included circles, squares, vertical ovals, and vertical rectangles. The sizes ranged from widths of 9 mm to 15 mm, while the spacing between the keys varied from 1 mm to 2.5 mm to 4 mm. These models were then 3D-printed to allow for physical comparison. Additionally, a model featuring the different shapes as clickable keys with a width of 11 mm was constructed. Figures **4** and **5** illustrate the size range and distance range for round keys.

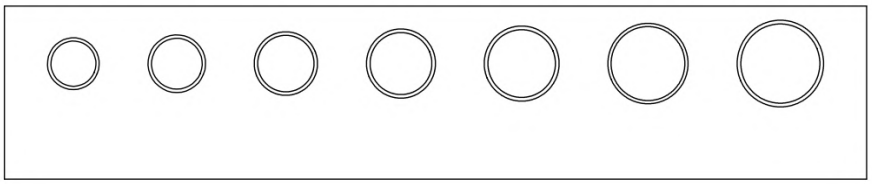


Figure 4: Illustration of size range, round keys

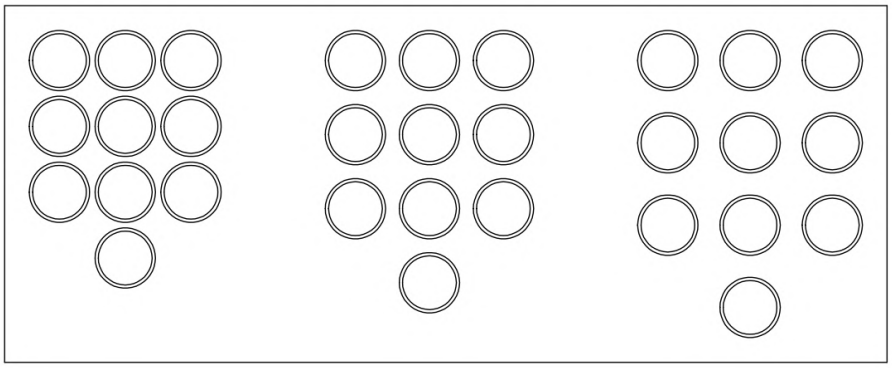


Figure 5: Illustration of key distance range, round keys

2.3.3 Concept Testing

To gain insights into potential user preferences, a series of quick tests were conducted, allowing participants to interact with the 3D-printed models. The procedure of the test is presented below:

- **Shape and Size Selection:** Initially, participants were presented with four different key shapes (circles, squares, vertical ovals, and vertical rectangles) in various sizes. They were asked to feel the models and identify their preferred shapes and sizes for a key on a keypad connected to a door.
- **Key Spacing Selection:** Once participants selected their preferred shape, they were presented with 3D-printed models displaying different distances between the keys. Participants were then asked to feel these models and indicate their preferred key spacing.
- **Clickable Key Verification:** Participants were introduced to models featuring clickable keys in the different shapes to verify if their preferred shape remained consistent when the keys were clickable.
- **Comparison with Reference Product:** Lastly, participants were presented with the reference product to try out and compare with the models, sharing any opinions they had.

The tests were carried out with two test groups. The first group consisted of seven seniors aged between 84 and 92 years, with varying capabilities and health statuses, including conditions such as shaking hands, poor vision, limited use of one hand, previous strokes, and wheelchair dependency.

The second group consisted of 7 participants aged between 26 and 60 years, all of whom considered themselves able-bodied.

The results of the concept tests were analyzed by compiling the participants preferences and comments into summary tables, allowing a structured comparison between different shapes, sizes, and spacings.

Patterns and recurring themes were identified across both test groups, highlighting common preferences and usability concerns.

2.3.4 Exploring Angles

To determine the optimal angles at which numbers and symbols are most visible, a low fidelity test was conducted. The setup was based on the keypad height of the reference product positioned at 115 cm. As such, the resulting angle preferences may not be directly translated to keypads mounted at other heights.

A cardboard plate was attached to a wall by its top edge, allowing it to pivot from 0° (flat against the wall) to 90° (horizontal), as shown in Figure **6**. Printed keypads in three different sizes were then attached to the cardboard, with number heights ranging from 3 mm to 5 mm.



Figure 6: Angled cardboard

To determine suitable angles for most users, two extremes were examined: one very tall user at 210 cm and one wheelchair user reaching a height of 130 cm. These scenarios were simulated by a single participant using a stepping stool and an office chair, respectively. In both cases, the viewing distance from the eyes to the cardboard surface was maintained at approximately 40 cm, see Figure **7**.

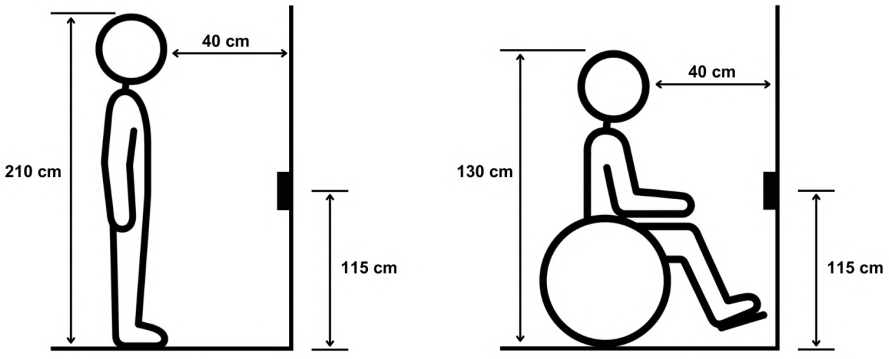


Figure 7: Illustration of test simulation

The cardboard was gradually tilted from 0° to 90°. At each angle, the participant rated the readability of the numbers using the following color-coded scale:

- **Green:** Numbers are clearly visible and readable
- **Yellow:** Numbers are visible but require some effort to read
- **Red:** Numbers are unreadable

This rating process was repeated for all three number sizes and both user height scenarios. The angles rated as green in both scenarios were identified and used to define a recommended viewing angle range suitable for a wide range of users.

2.3.5 Symbol Preference Study

To investigate the symbols most commonly associated with unlocking and locking, a brief in-person preference study was conducted to collect quantitative data. The study involved 15 participants between the ages 25 and 64, all able bodies with varying occupations. The procedure is presented below:

- **Unlocking Symbols:** Participants were first presented with a paper containing six symbols commonly associated with unlocking or indicating that you can enter. They were asked to select the symbol they most strongly associated with a door being unlocked.
- **Locking Symbols:** Participants were then presented with a paper with six symbols commonly associated with locking or indicating that you cannot enter. They were asked to select the symbol they most strongly associated with a door being locked.
- **Combination:** Lastly, participants were shown all twelve symbols together and asked to combine one unlocking symbol and one locking symbol to create a pair that they felt most clearly represented the concepts of unlocking and locking.
- **Actual Symbol Scale:** After selecting their preferred symbol combination, participants were shown the same symbols again, this time at the actual size they were intended to be used. This was done to see whether their initial preference held when the symbol details were reduced in scale.

The participants' responses were noted down manually during the session and later compiled into a summary table to identify the most frequently selected symbols and preferred combinations. The symbols shown to the participants are presented in Appendix E.

2.3.6 Concept Development

To further develop the concept by implementing the recommended keys and angles on the reference product, more sketching was done. The aim was to determine two designs that could be refined in the Deliver phase.

2.4 Deliver

This section outlines the methods employed during the Deliver phase of the project. The objective of this phase was to finalize and refine the key deliverables defined in Chapter 1.4: a handbook containing design recommendations, the application of these recommendations to two conceptual designs and physical models.

2.4.1 Handbook

The purpose of the handbook is to serve as a visual complement to the report, providing guidance for product developers and other relevant stakeholders involved in keypad design. It also functions as a reference for future design decisions.

The handbook is structured around three main themes, identified as the most relevant to inclusive design within the scope of this project. Each theme is introduced with a brief explanation of its significance, followed by detailed design recommendations. To enhance clarity and practical applicability, best practice examples are included throughout.

2.4.2 Application of Recommendations

To effectively demonstrate the proposed design recommendations, they were applied to two conceptual designs based on the reference product. The purpose of using the reference product as a foundation was to evaluate whether the recommendations genuinely enhanced user interaction for individuals with varying abilities. Developing two distinct concepts allowed for the demonstration of the recommendations' flexibility, one concept representing a more extensive design intervention, and the other a less invasive modification.

Both concepts were modeled in 3D, using SolidWorks, to present realistic representations in terms of context, scale, and materials. These digital models were also used to produce physical prototypes at scale by 3D printing.

3 Discover

This chapter includes the findings from the projects Discover phase, including desktop research, assessment of the reference product, trigger material and interviews and a layout comparison.

3.1 Desktop Research

This section presents the findings from the desktop research as detailed in Chapter 2.1.1. The findings are organized into sub-chapters, each focusing on a subject relevant to the thesis.

3.1.1 Inclusive Design

Inclusive design is a design philosophy that addresses the diverse needs and capabilities of users, including the elderly and disabled. Its goal is to create products or services accessible to and usable by as many people as possible without special adaptations (Persad et al., 2007). The principles of inclusive design are central to addressing the company's need and desire to create more inclusive and accessible products. By implementing inclusive design and engaging with people of varied capabilities, the company can ensure that their products are consciously designed to be usable by a diverse user base.

Keates and Clarkson (2004) categorize the arguments for inclusive design into ethical and business perspectives. Ethically, many users are excluded from everyday products and services due to insufficient support. This includes older adults, people with disabilities, and people who lack prior knowledge. Inclusive design minimizes exclusion and the approach can help potentially excluded groups maintain their independence and quality of life.

From a business standpoint, Keates and Clarkson (2004) argue that inclusive design helps companies expand and retain market share by catering to a diverse customer base. It ensures compliance with accessibility and anti-discrimination laws, avoiding legal issues. Additionally, it offers competitive advantages, such as improved products and a positive, inclusive brand image.

3.1.2 User Capabilities

User capability refers to the range of abilities an individual possesses that enables them to perform tasks effectively (Persad et al., 2007). In relation to consumer products, Persad et al. (2007) broadly categorize relevant capabilities into three areas:

- **Sensory Capabilities:** Sensory capabilities include vision, hearing, and touch. Design considerations for vision include visual acuity, contrast sensitivity, color perception, usable visual field, and depth perception. For hearing and touch, considerations include the ability to detect different sound frequencies, sound localization, and tactile feedback.

- **Cognitive Capabilities:** Cognitive capabilities involve mental processing, understanding, and memory. Relevant aspects include working memory, long-term memory, language comprehension, and problem-solving..
- **Motor Capabilities:** Most products require upper limb motor capabilities, such as fine dexterity, finger function, force exertion, two-handed coordination, and reach ranges. For fixed products like control panels and displays, total body movement may be required, including bending ranges and overall mobility

Understanding the sensory, cognitive, and motor capabilities of different users is essential for designing keypads that are accessible to as many users as possible. This supports the purpose of improving usability by identifying and addressing the specific needs and limitations of various user groups.

Functional capabilities may be reduced due to various causes and can be permanent or temporary. The elderly often experience reduced functional capabilities, with conditions such as hearing loss, cataracts, and osteoarthritis being common according to the World Health Organization (WHO) (2024). According to WHO (2015), approximately 5% of the Swedish population aged 65-74 years, and 25% aged 75 years or older have limitations in daily activities, such as using a telephone, shopping for groceries or using a map. As the proportion of the world's population over 60 years is estimated to reach 22% by 2050 (WHO, 2015), it is crucial to consider this group and their varying capabilities when designing new products.

According to statistics compiled by Funka (2023), around 36% of the Swedish population has some degree of reduced functional capability, and nearly 11% have a combination of two or more disabilities. Selected statistics from the compilation are presented below:

Sensory Capabilities

- 4,9 % of adults have difficulty seeing even with glasses or contact lenses.
- At least 30,000 people are visually impaired or completely blind.
- 18,4 % of adults have difficulty hearing.

Cognitive Capabilities

- 13,3 % of adults have difficulty reading.
- 14 % of the population is estimated to have low cognitive ability, defined as having an IQ between 70 and 85.

Motor Capabilities

- 5,9 % of adults have a mobility impairment.
- 3,4 % of adults use an aid or need assistance moving due to mobility impairments.
- 1/4 of people over 45 are estimated to have been diagnosed with osteoarthritis.

3.1.3 The User Pyramid

The 'user pyramid' (Benktzon, 1993) is an approach to inclusive design that illustrates the range of potential users and their abilities. Users are divided into three categories based on their functional capabilities, from full capability at the bottom to severe disabilities at the top, see Figure **8**.

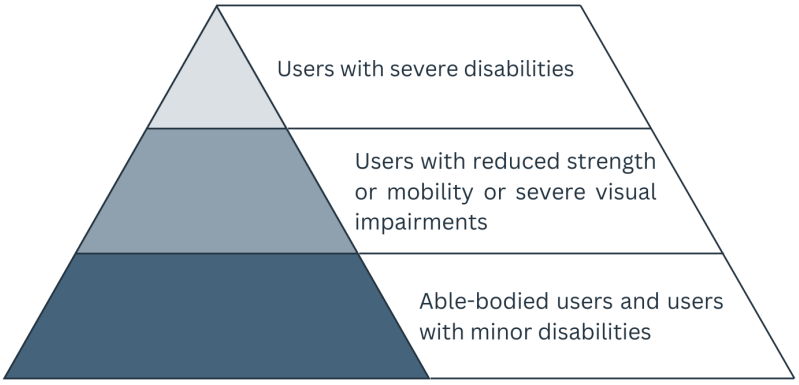


Figure 8: The User Pyramid illustrating capability categories (adapted from Benktzon, 1993)

When using the pyramid to design inclusively, designers must choose between a 'top-down' or 'bottom-up' approach (Keates and Clarkson, 2004), illustrated in Figure **9**. The top-down approach designs products for users with the least functional capabilities first, aiming for designs that are also appreciated by others. The bottom-up approach starts with an existing mainstream product for able-bodied users and modifies it to be more inclusive.

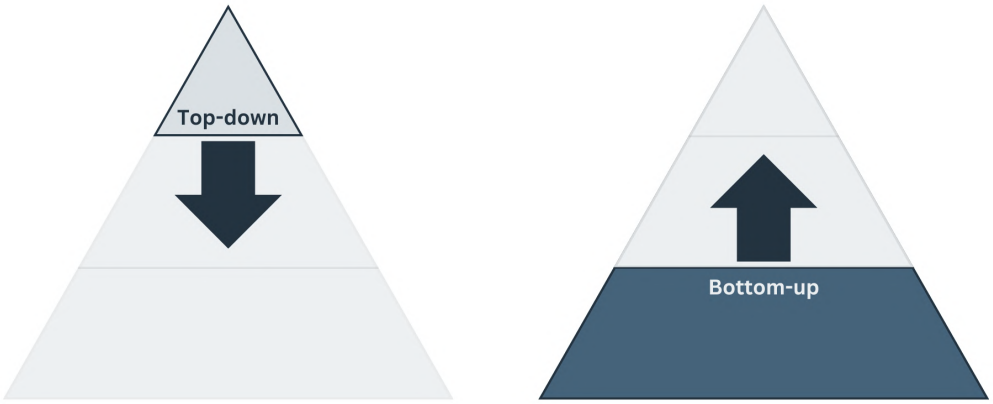


Figure 9: Illustration of the top-down and bottom-up approaches

While the user pyramid helps consider a wide range of user needs, it also presents challenges (Keates and Clarkson, 2004). The top-down approach risks creating over-specialized products that appeal only to a small group, even if intended for the mainstream market. The bottom-up approach may not fully meet the needs of users beyond the base of the pyramid. Keates and Clarkson (2004) also point out that neither approach automatically results in a fully inclusive product. Therefore, it is best to select the method most suitable for the intended users, whether the product is mainstream or assistive, rather than compromising between approaches and ending up with a product that does not fit either market fully.

With its two approaches, the company can effectively implement the user pyramid. They can either redesign and adapt their current products to be more accessible using the bottom-up approach, or work top-down when creating new products. This flexibility supports the company's goal of creating more inclusive and accessible products.

3.1.4 Standards and Guidelines

Inclusive design ensures products are accessible to as many users as possible, regardless of capability. To facilitate this and ensure compliance with legal and industry requirements, various standards and guidelines have been established. By adhering to the established standards and guidelines the company can ensure that their keypads meet necessary accessibility requirements.

The ISO/IEC Guide 71:2014(E), titled "Guide for addressing accessibility in standards," provides a comprehensive framework for incorporating accessibility into standards. Developed by the International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC), this guide outlines accessibility goals and describes human abilities and characteristics to help standard developers address the needs of diverse users (ISO and IEC, 2014).

For a keypad on a smart lock, the following aspects from the guide could be considered:

- **Provide multiple means of information:** Ensure various ways for users to access the information needed to operate the product. For example, incorporating text-to-speech functionality can assist users with visual impairments.
- **Minimize unnecessary complexity:** Design the keypad to be straightforward and intuitive, with clear icons and simple navigation. This helps users of all abilities operate the device without confusion.
- **Perceivability and Understandability:** Use high-contrast colors and large, readable fonts to improve visibility. Provide auditory feedback for actions to assist users with visual impairments.

Additionally, offer clear and concise instructions to ensure the product is easy to understand and use.

- **Error Tolerance:** Design the keypad to allow users to recover from mistakes easily. Ensure accidental touches or inputs do not result in significant disruptions, enhancing the overall user experience.

Another applicable standard is EN 301 549 (European Telecommunications Standards Institute [ETSI], 2021), a European standard defining accessibility requirements for information and communication technology (ICT) products and services. The standard covers both software, such as web pages and mobile applications, and hardware, including personal computers and information kiosks. Relevant aspects from EN 301 549 include:

- **Tactile Characteristics:** There should be distinguishable tactile characteristics to identify by touch, including raised symbols and different textures.
- **Operability:** The design should be operable with limited strength and dexterity. Buttons should not require excessive force and should be appropriately placed for ease of use.
- **Visual Characteristics:** There should be high contrast between buttons and their background, large and clear fonts, and backlighting that does not cause glare.

Both ISO/IEC Guide 71:2014(E) and EN 301 549 emphasize the importance of making keys distinguishable both physically and visually. A study by Muraki et al. (2010) assessed the ease of pressing cell phone keys with the thumb and provided design recommendations to improve usability. The study compared the experiences of young vs. elderly and male vs. female participants. The cell phones used in the tests had widths ranging from 38-44 mm, similar to the reference product, which is about 45 mm wide.

The findings revealed that elderly participants were more likely to press the wrong keys, especially those in the rightmost column. They miskeyed 2.5 times more often than younger participants, with the numbers "2", "3", and "9" being the most frequently mispressed. Additionally, elderly participants took nearly twice as long to operate the keys compared to younger participants. This underscores the need to consider delayed information processing, reduced dexterity, and larger thumb sizes when designing keypads.

The study recommended that keys should be spaced sufficiently apart to be pressed easily and highlighted the importance of tactile feedback (Muraki et al., 2010), aligning with the standards set by ISO/IEC Guide 71:2014 and EN 301 549.

3.1.5 Ergonomics and Anthropometry

When designing products and systems for a diverse user base, one could consider various aspects of ergonomics as a tool for increased usability. Ergonomics can be defined as the study of people in their working environment, the goal of ergonomic design is to eliminate discomfort and minimize the risk of injury due to work (UNC, 2024). This section will explore the key concepts of cognitive ergonomics, followed by an examination of physical ergonomics and anthropometry.

Cognitive Ergonomics

Rose and Eklund (2019) describe cognitive ergonomics as the study of how we perceive, interpret, and process information. It focuses on designing components to be easily understood and safe. Key aspects to consider include:

- **Importance:** Place the most important buttons, controls, or functions in the most convenient locations.
- **Use Frequency:** Position the most frequently used controls or functions in the most accessible spots.
- **Order:** Arrange buttons or controls that are intended to be used in a specific sequence in that order and group them together.

For instance, the direction of movement on a control should match the effect on the function being controlled. If something increases, it should move to the right (+) or up, and if it decreases, it should move to the left (-) or down. This principle supports the idea that products should provide clear and immediate feedback for each action, helping users understand that their input has been registered.

To enhance cognitive ergonomics, it is essential to minimize cognitive load and provide clear, intuitive feedback. Sensory confirmation plays a crucial role in ensuring users can confidently enter their codes without excessive focus. Keys should be easy to locate and use without conscious thought. If a key layout is unfamiliar or confusing, users must actively think about where to press, increasing cognitive load. This is especially important for situations where users interact with the device quickly or under stress (e.g., entering a PIN on a keypad while in a hurry). Adding a distinguishing feature, like a tactile marker on the central key, can improve navigation by touch (Gaspar et al., 2017), often seen on the ”5” key.

Haptic feedback, such as a slight vibration or click sensation upon pressing a key, reinforces tactile confirmation. This was shown to be the number one preference in key feedback based on an article referring to car radio buttons (Gaspar et al., 2017). Auditory signals, like a soft beep, allow users to verify input without needing to look at the keypad. Visual feedback, such as backlit keys or indicator

lights, helps users track their progress during code entry (Oulasvirta, Kim, and Lee, 2018). Without feedback, users may press harder than necessary or press multiple times, leading to errors (Gaspar et al., 2017). To prevent errors and frustration, keypads could incorporate features like a "Clear" or "Cancel" button, enabling users to correct mistakes without restarting the entire process. Implementing slow-release activation, where a key registers only after a slight hold, can reduce accidental presses. Including a hysteresis effect (difference between press and release force) can also prevent unintended double presses (Oulasvirta, Kim, and Lee, 2018).

Cognitive impairments, such as memory loss and attention difficulties, is common among the elderly. Additionally, a lack of technical expertise or experience can influence their initial attitude toward using a smart product, especially when it replaces a familiar item like a door lock (Ismatullaev, Saduakas, and Kim, 2022). Therefore, it is essential to provide user-friendly interfaces and allow extended interaction time to ensure ease of use.

Physical Ergonomics

Physical interaction with keypads primarily involves the hand, particularly the fingers. Norris and Wilson (2000) conducted a study with 148 participants aged 2 to 86, measuring finger push strength. They found that the thumb generated almost double the force compared to the index finger, regardless of the direction of force. Additionally, there was a notable difference in strength between sexes in adults over 16 years.

Since smart door locks are often used in quick, repetitive, and sometimes emergency scenarios, their physical design should focus on efficiency, comfort, and reliability. Therefore, it is beneficial for keypads to incorporate tactile feedback (e.g., a small click or bump when a key is pressed) (Oulasvirta, Kim, and Lee, 2018). This helps users confirm their input without needing to visually check each press. If the force is too high, the button feels stiff and hard to use, if too low, it feels flimsy (Gaspar et al., 2017). A slight force-displacement curve (similar to tactile mechanical keyboards) can improve accuracy without requiring excessive force. If keys require too much force, it can cause fatigue, especially for users with limited dexterity or strength. A low-to-moderate force requirement (0.5-1N) is ideal for comfort while preventing accidental presses (Oulasvirta, Kim, and Lee, 2018).

Keys could have concave surfaces to improve finger placement and prevent slipping. Raised edges or textured surfaces help users differentiate keys by touch alone (Denno et al., 1992). A study on keyboards found that larger key sizes (12-15mm width) improve accessibility, especially for users with limited dexterity (Oulasvirta, Kim, and Lee, 2018). Small keys increase the chance of pressing the wrong key, frustrating the user (Gaspar et al., 2017).

Anthropometry

In inclusive design, it is important to consider not only physical abilities, such as finger strength, but also anthropometric data. Anthropometric data refers to the systematic measurement of the human body's physical dimensions, such as height, width, limb length, and joint range (ILO Encyclopaedia, 2011).

When designing keys and keypads, hand and finger measurements are relevant to ensure the product is comfortable and accessible, enhancing usability and reducing the risk of errors. Studies have shown that most people have similar finger sizes regardless of dexterity capability. The width of the index finger normally ranges from 20 to 22 mm, and the thumb normally ranges from 24 to 27 mm (Feeney, 2002). This data is presented in Table **1**, where M represents male participants and W represents female participants.

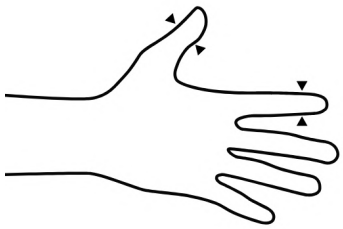


Figure 10: Index and Thumb Distal Joint Placement

Table 1: Hand Dimensions: Distal Joint Width of Participants (Feeney, 2002)

Finger [mm]	**Disabled M**	**Non-disabled M**	**Disabled W**	**Non-disabled W**
Index Joint Width	21.9	21.7	20.1	20.2
Thumb Joint Width	27.0	26	24.2	24.7

3.1.6 Keypad Design

Keypads are a common part of everyday life, found in environments such as workplaces, retail settings, and personal devices. Their usability is influenced by design elements like key size and spacing. Even small variations can affect ease of use, which becomes particularly relevant when considering a wide range of user needs in inclusive design.

Keypads come in various designs and mechanics. Some provide tactile feedback, where the user feels a "click" when pressing down, which enhances the user experience. In contrast, linear keys, like those on a TV remote control, offer a smooth press without tactile feedback, requiring more conscious control

and trust in the product. Touch-sensitive keys, such as those on touchscreens, lack travel distance and tactile feedback, leading to higher error rates as users cannot rely on physical sensations to confirm their input (Oulasvirta, Kim, and Lee, 2018).

Key Design, Size and Distance

A study aimed at proposing an optimal computer keyboard design for elderly users evaluated different keyboard designs (Hsieh, Ho, and Lee, 2022). Key findings indicated that circle and square keys outperformed vertical rectangle keys in terms of completion time, accuracy, and satisfaction. Larger keys (12.5-17 mm) decreased error rates the most, with 15 mm being optimal for elderly users. The smallest key size (7.5 mm) had the highest error rate. The study also found that 3 mm key spacing resulted in significantly shorter task completion times compared to keys without spacing. Although 3 mm was optimal for 15 mm buttons, 1 mm spacing was also satisfactory (Hsieh, Ho, and Lee, 2022).

When designing keypads, there are more factors to consider than key size and spacing. To accommodate users with dexterity impairments, keypads should have lateral spacing, the horizontal distance between the centers of adjacent keys, of at least 12 mm (Mooney, Nussbaum, and Smith-Jackson, 2002). Key travel distance, the distance a key travels downward when pressed, is also crucial. A short travel distance (1.5-2 mm) is ideal for fast, repeated presses, while deeper travel (3-4 mm) provides better feedback. Low-profile keys should compensate for reduced travel with stronger haptic or auditory feedback (Oulasvirta, Kim, and Lee, 2018).

Keypad Layout

In 1983, Goodman et al. conducted a study on human factors in keypad design. Although the keypad was originally intended for use with a Telidon system, many findings are still applicable today. The research found that test subjects preferred a 3×3+1 key arrangement over a linear layout, see Figure **11**. This was primarily due to its familiarity, which made it more comfortable to use (Goodman et al., 1983). Users rely on spatial memory and haptic exploration to navigate button layouts. Standardized layouts (like a numeric keypad) improve learnability and reduce errors (Gaspar et al., 2017).

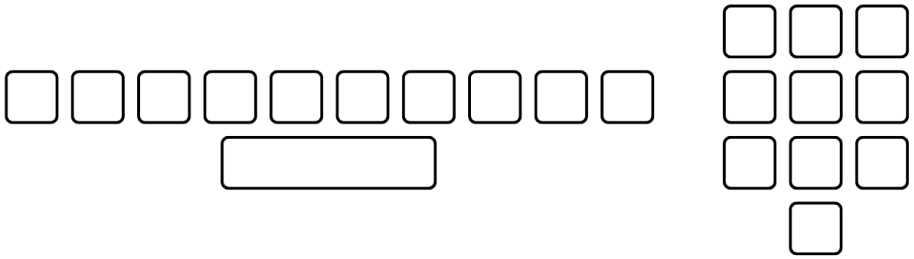


Figure 11: Linear and 3×3+1 key arrangements

An other study done by Mooney, Nussbaum, and Smith-Jackson (2002) also emphasized that special function keys should be distinguishable by size, shape, and orientation to improve accessibility for visually impaired users. Oulasvirta, Kim, and Lee (2018) highlight that grouping frequently used keys together improves efficiency, and separating function keys prevents errors. These studies underscore the importance of simplicity and intuitive design in creating user-friendly interfaces. This can be seen in modern payment terminals, where the accept, cancel, and undo keys are placed on the side of the numeric keys and have distinguishing colors, see Figure **12**.



Figure 12: Keypad on payment terminal

3.1.7 Keypad Technology

Keypads are integral components in various applications, from everyday consumer electronics to specialized industrial and medical equipment. Their design and technology significantly influence functionality, reliability, and user experience. This section explores different types of keypads relevant for smart locks.

Mechanical Keyboards

Mechanical keyboards use individual springs and switches for each key, offering distinct tactile and auditory feedback (PC Mag, n.d.-a). Each key provides a "clicky" feel and sound that enhances typing accuracy and speed. Mechanical switches are durable, with a lifespan of up to 100 million executions. These keys are commonly used in gaming and professional keyboards where responsiveness and longevity are crucial (Cherry, 2025).

Membrane Keyboards

Membrane keyboards come in two designs: dome switch and flexible plastic sheet. Dome switch keyboards use a single rubber-like membrane underneath the keys, making contact with a circuit

board when pressed. This design is common in QWERTY keyboards and various keypads but provides less tactile sensation (PC Mag, n.d.-b). Flexible plastic sheet keyboards are more resilient to dirt and spillage. They use a flexible plastic sheet with a printed key pattern and conductive ink. When pressed, the top layer contacts the circuit at the bottom. This type is widely used in self-service applications and often includes feedback mechanisms like beeps or LEDs (PC Mag, n.d.-b).

Silicone Keypads

Silicone keypads consist of a single piece of silicone webbing over switches, providing a tactile surface. Produced via compression molding, they offer uniform tactile response and resistance to fluids and environmental damage. When pressed, they complete a circuit. These keypads are durable, cost-effective, and have a long lifespan (Silicone Dynamics, 2014). They allow for precise key design and haptic feedback (N&H Technology GmbH, 2023).

Stainless Steel Keyboards

Stainless steel keyboards are robust and vandal-proof, ideal for public spaces, transport, and industrial environments. They resemble conventional PC keyboards and offer customization options such as key layout, printing, and backlighting. The keys are labeled by laser engraving or etching, ensuring durability. These keyboards are designed for use with thick work gloves and are resistant to dirt and damage, providing reliable performance in harsh conditions (N&H Technology GmbH, 2021).

Touch Keypads

Touch keypads or keyboards are essentially touchscreens with virtual keys, such as those on smartphones or tablets. Users input text or numbers by tapping on the screen.

3.1.8 Smart Locks

A smart lock is a digital lock that allows homeowners to remotely control and monitor their connected doors or windows using a smartphone app. It provides keyless entry and the ability to hand out virtual keys to family, friends, and service professionals (Yale, 2022). In addition to the smartphone app, smart locks can often be operated with keypads for PIN entry and RFID tags.

There are two main types of smart door locks, full-replacement locks, and semi-replacement locks. Full-replacement models replace the door's existing interior and exterior hardware on both sides. The reference product in this thesis is of this type, a common design is illustrated in Figure **13**. In contrast, semi-replacement models only replace the lock knob and attach to the door's existing deadbolt upon installation, making them more suitable for apartments and rented homes.

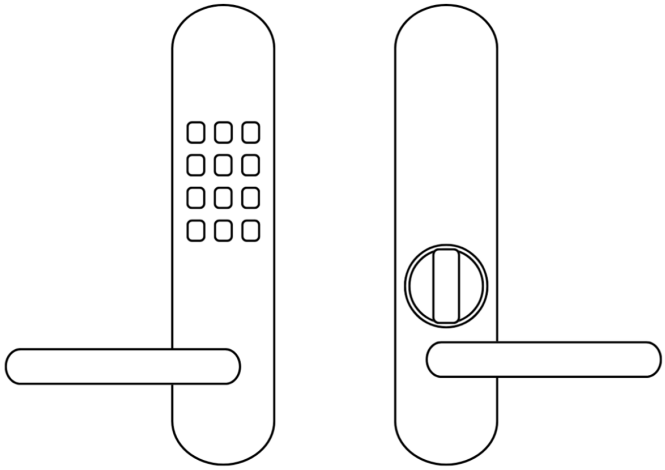


Figure 13: Illustration of parts of a full-replacement lock

3.1.9 Existing Solutions

The usability of a keypad is a critical design aspect in smart door locks as it directly affects the user experience, security and accessibility. Intuitive design allows the user to enter codes quickly and accurately while poor design can lead to mispresses, frustration or security vulnerabilities.

The most common keypad layout is the traditional 3x3+1 grid, used in most smart locks and standard phone keypads. It is intuitive and familiar to users, reducing the learning curve and potential confusion. However, classic keys and layouts can show wear patterns, posing a security risk.

Touch keypads, which lack physical keys, do not show wear and can enhance security by allowing randomized number placements to prevent fingerprint tracking. The layout can be similar to traditional keypads or have a circular design, see Figure **14**. However, touch keypads may struggle in cold or wet conditions and do not work with normal gloves.

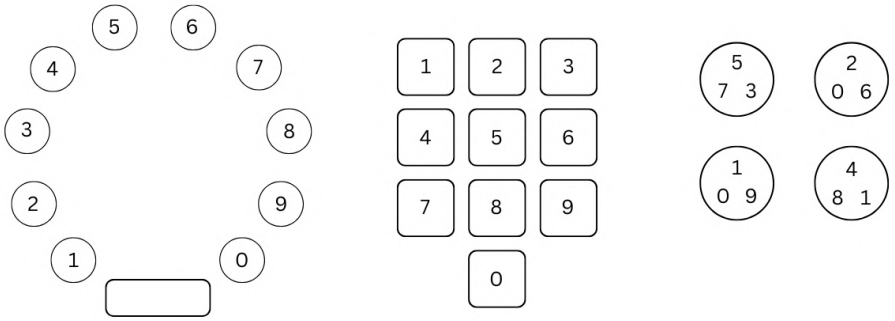


Figure 14: Keypad layouts

Many smart locks use backlighting to enhance the visibility of keys and symbols, essential for low-light conditions. Classic keys often use edge-lit buttons, where LEDs provide a subtle glow around each key and/or on the numbers and symbols. Touch pads use full surface illumination, where the entire keypad lights up when touched.

Chapter 3.1 provides a theoretical foundation for all three research questions by outlining key principles, user capabilities, design standards, ergonomic considerations, and keypad-specific design factors that must be addressed to create inclusive, accessible, and user-friendly keypads.

3.2 Assessing the Reference Product

To gain a comprehensive understanding of the reference product and the user experience it provides, both user testing and company insights were considered. This chapter presents the findings from the testing of the reference product as detailed in Chapter 2.1.2, and the interview held with customer support as detailed in Chapter 2.1.3.

3.2.1 Testing of Reference Product

The following section presents the findings from the usability tests conducted on the smart lock, as described in Chapter 2.1.2. Both empirical findings and preliminary analysis are presented based on the evaluation of the reference product. Observations are combined with early interpretations to identify usability challenges. The tests involved seven participants who unlocked a door under five different conditions. The time spent on each test was approximately 20 minutes per participant.

The average rating of the reference product for each question during the rounds is presented in Table **2** below. After each question, participants explained their ratings and answered follow-up questions.

Table 2: Average ratings of reference product.

Question	**Round 1**	**Round 2**	**Round 3**	**Round 4**	**Round 5**
How well could you see the numbers and symbols on the keypad?	2,4	-	-	-	-
How easy did you find it to locate the correct keys?	3,6	3,4	3,3	3,6	2,7
How easy did you find it to press the correct keys?	4,7	5	2,6	3,5	2,9
How sure did you feel that you pressed the correct keys?	4,7	4,2	2,7	4,4	2,9

Round 1 - Baseline

During the first round, all participants bent down at the waist or crouched until the keypad was at eye level when entering the code. The participants unanimously reported that the numbers and symbols on the keys could not be seen in their entirety when standing straight and looking down at the keypad. The numbers and symbols were cut off by the housing and the keys' edges. Figure **15** shows the keypad at eye level, and Figure **16** shows the perspective of a person approximately 170 cm tall. The reason for the cut-off numbers and symbols is that they are printed on the back of the keys, placing them inside the product housing, obscuring them.



Figure 15: Keypad at eye level

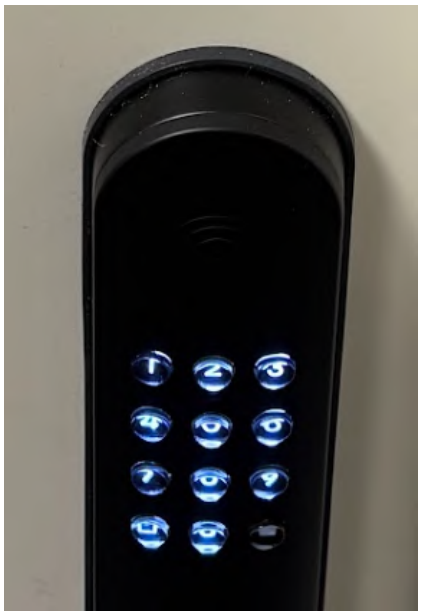


Figure 16: Keypad from a perspective

Six participants expressed confusion due to the lack of fully visible symbols, declaring that they did not know or see that there was an unlock button, which had to be pressed after the code was entered to unlock the door.

The results of the second round, where participants were not allowed to bend down, did not differ significantly from the first round.

Round 3 - Wearing Gloves

During the third round, all participants reported that they found it significantly harder to press the correct buttons while wearing gloves. Expressing that the larger contact area of the finger made it difficult to see the keys, see Figures **17** and **18**. Four participants stated that the gloves also removed their ability to feel the keys and their positions, creating a concern about accidentally pressing multiple keys.



Figure 17: Entering code wearing glove using thumb

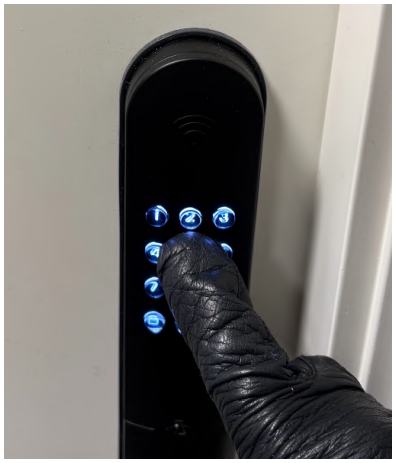


Figure 18: Entering code wearing glove using pointer finger

Round 4 - Reduced Dexterity

In this round, participants' dexterity was significantly reduced by the attached exoskeleton. Most participants found it challenging because the setup prevented them from using their thumb, the preferred digit. The reduced dexterity and inability to close the hand resulted in awkward hand positioning, requiring the use of the longest finger, see Figures **19** and **20**. Six participants mentioned that they had trouble pressing the correct keys as their fingers could not be separated and easily slid off the keys.



Figure 19: Hand position wearing exoskeleton

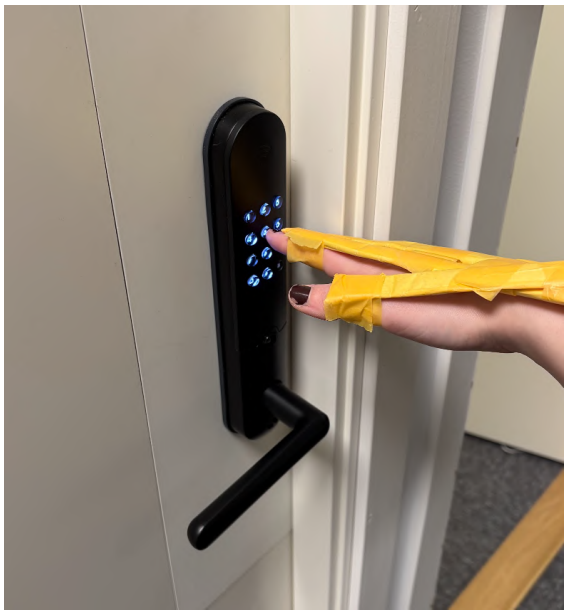


Figure 20: Entering code wearing exoskeleton

Round 5 - Closed eyes

When participants were asked to close their eyes before entering the code, all expressed how they relied on muscle memory and previous experience with the keypad layout. Most counted the keys, starting at "1" in the top left corner, to find the correct ones. None of the participants used or mentioned the tactile marking on key "5". When asked about the marking, multiple participants said that it was small and placed too close to the sharp edge of the key, making it hard to notice. Figure **21** shows a close-up of the keypad where the tactile marking is visible. Although all participants expressed counting as their way to identify the numbered keys, two participants stated that the unlock and lock keys were harder to find due to their integration with the numbered keys and identical shape and proportions.



Figure 21: Close-up of keypad

Based on the observations and comments, the following key takeaways emerged. Participants desired clearer numbers and symbols on the keys, as well as clearer lock and unlock keys. One participant suggested separating the lock and unlock keys from the numeric keys to improve usability, especially when wearing gloves or with impaired vision. The similarity in appearance between the lock and unlock keys made them difficult to distinguish, and it was not obvious to press the unlock key after entering the code. Most participants used their thumb to enter the code, often out of habit from using their phone. Using the thumb also allowed for more precision, as they could brace their other fingers against the side of the lock for support.

3.2.2 Interview with Customer Support Manager

Customer support not only helps users but also provides the company with valuable insights into how the product is used and perceived. To leverage this knowledge, an interview was conducted with a

Customer Support Manager about the reference product.

The Customer Support Manager confirmed that the product is primarily purchased for domestic use, particularly for standalone villas, but it is also used for shared spaces in apartment complexes and construction sites. Usability requirements vary depending on the context. He noted that the adjustable sound level is highly appreciated, users on construction sites prefer an audible door lock, while parents of young children prefer silent operation.

Additionally, the interviewee mentioned feedback regarding the size and spacing of the keys. Users with larger hands or those wearing gloves have experienced difficulties entering the code. Another recurring feedback point was that the lock's built-in doorbell is rarely used. Owners reported that their guests often do not notice this feature and instead knock on the door or use the house's existing doorbell if it is still present. The doorbell key is placed below the numeric keys and does not automatically light up during interaction with the lock, making it hard to recognize, see Figure **22**.



Figure 22: Image of the lock from two perspectives, circled doorbell

3.2.3 Key Insights

Chapter 3.2 primarily addresses RQ1 by identifying usability challenges encountered with the reference product, such as poor visibility, unclear feedback, and difficulty using the keypad under various physical constraints. These findings also contribute to RQ3 by highlighting how design improvements, like clearer symbols, better tactile feedback, and more distinguishable function keys, can enhance accessibility for users with different physical and cognitive abilities.

Listed below are both empirical findings and a preliminary analysis based on the assessment of the reference product:

- Symbols placed on the rear of keys were not visible from a standing position, causing confusion,

especially when locating the unlock and lock keys.

- The “5” key’s tactile marking was too small and poorly positioned, making it difficult to detect by touch.
- Participants struggled to locate and press keys accurately with gloves or reduced dexterity, emphasizing the need for larger keys and better tactile feedback.
- Lock and unlock keys were visually and physically identical to numeric keys, leading to uncertainty and input errors.
- Improvements are needed in symbol placement, tactile features, and function key differentiation to support a broader range of users.

3.3 Trigger Material with Interviews

This section presents the findings from interviews conducted with the trigger board to gather information on participants’ experiences, perceptions, and preferences regarding keys and keypads, as detailed in Chapter 2.1.4. The durations of the interviews varied from 30 minutes to 1 hour per participant.

The results are organized into sections based on themes that emerged during the interviews. Each section includes a summary of participants’ responses, supported by quotes and observations. The original and translated quotes are attached in Appendix F.

3.3.1 Use and placement of numbers and symbols

When faced with keypads lacking numbers or symbols (like keypad no. 2) and keypads with very faint numbers and symbols (like keypad no. 1 and no. 3), all participants hesitated before entering the code. The absence of clear indicators made them unsure of where the correct keys were placed. They all assumed the keypads had a 3x3+1 layout, even though it was not explicitly stated.

> *”I can’t see anything on the first one [keypad no. 1]. Absolutely nothing. I can feel the keys, so I can count where they are. ... Sometimes it’s stubbornness that helps you get through, and now, here key 5 is marked, even though it’s poorly marked, so you can figure out where the other keys are” – P2 [Man, 60’s, visually impaired]*

> *”There was no text on it [keypad no. 2], so you didn’t really know what you were clicking on. Sometimes the numbers go from top to bottom, and sometimes the other way around, so I thought it was a bit unclear. ... Now I just assumed it starts at the top, but you don’t know for sure.” – P5, [Man, 20’s, able-bodied]*

The lack of visible numbers and symbols created both a delay and a feeling of uncertainty among all participants as they had to stop and actively think about how a keypad typically looks. This extra effort created frustration, even if the participants were used to seeing and using similar keypads.

> *"If I saw keypad 2 when I was stressed, I would get so angry, because I think clarity is very important when you're stressed. I would just look at it and think, 'What is this?!'" – P5 [Man, 30's, able-bodied]*

> *"I don't like keypad 1 and I don't like keypad 2, and I don't understand how I can have so much trouble with the fact that there are no numbers on them?!" – P3 [Woman, 30's, able-bodied]*

Both P1 [Woman, 40's, severely reduced motor capabilities] and P2 [Man, 60's, visually impaired] emphasized the importance of having a tactile marking on key 5 for all keypads on the trigger board. They agreed that the marking was most distinct on keypads 2 and 4, which they appreciated. While the marking was present on keypads 1, 3, and 6, they noted that it was not optimal. The markings on these keypads were small and not centered on the key, making it harder for users to recognize and feel secure that they have found the correct key.

In addition to the existence and visibility of the numbers and symbols, their placement also played a significant role. Four participants noted that the arrangement on keypad 4, where the numbers and symbols are positioned above the corresponding keys, added an extra step of thinking. This setup was perceived as less intuitive, making it harder to determine which key each number belonged to. A close-up of keypad number four is shown in Figure **23**.



Figure 23: Close-up of keypad number four.

> *"Sometimes when the numbers are above the keys like this, I find it a bit unclear. If the numbers are placed under the keys or above the keys, I don't know if the five is connected to [key] five or if it is connected to [key] two" - P11 [Woman, 30's, able-bodied]*

"I would probably say that I would have wanted the numbers on the keys and not above. Now there's a millisecond where I wonder which number belongs to which button." - P4 [Woman, 20's, able-bodied]

Apart from the placement of the numbers on keypad no. 4, six participants were confused by the printed bracket-like symbols around the number eight (see Figure **23**). These brackets were intended to symbolize the keypad's tag function, but the participants did not interpret them this way. Theories about their purpose ranged from representing the middle for people with dyscalculia, to indicating different types of alarms. However, there was a consensus that if any number should have a special marking, it should be number five, since this key is usually distinguished by a tactile marking.

"Maybe an emergency button. I can imagine this [keypad] being in an elevator, if you need to go to a specific floor, so if you get stuck or can't get out, you might press button 8 to send a signal somewhere." - P2 [Man, 60's, visually impaired]

"Around key 8, I see a lot of parentheses, which I don't know why. ... the large parentheses should be in the same place as key 5, since it is always marked." - P1 [Woman, 40's, severely reduced motor capabilities]

Another interesting observation was that four of the participants pressed the wrong keys when finishing the code with the asterisk, even when the numbers and symbols were clearly visible. They pressed the hashtag in the bottom right corner instead of the asterisk in the bottom left. When asked about this, it appeared they had acted on a feeling rather than actually looking.

"Oh, I just assumed the star was there [on the hashtag], it feels intuitive, then I've been entering the wrong code every time." - P4 [Woman, 20's, able-bodied]

"I would click on it last, and then it felt most natural to click on the key that is placed last, to finish, you know. ... I didn't check what the symbol was ... But it feels more instinctive to click on the last one, I think." - P5 [Man, 30's, able-bodied]

On keypad 6, which is identical to the reference product, the asterisk and hashtag have been replaced with symbols of a locked and unlocked padlock, as shown in Figure **21** in Chapter 3.2.1. This change caused some confusion and frustration, as the symbols are placed where the asterisk and hashtag are typically found on standard keypads. The issue was intensified by the fact that the symbols were not visible unless viewed from eye level.

"But you can't see the star or the square! Have they switched places? [sees the lock symbols] One is filled in and the other isn't. ... The meaning of the symbols is very unclear." - P7 [Woman, 30's, able-bodied]

"Where the hell is the star? The star isn't there? [sees the lock symbol] Yeah, I could have figured that out, but it would have made me irritated." - P11 [Woman, 30's, able-bodied]

These observations highlight the significant impact that the visibility and placement of numbers and symbols on keypads have on user experience. The absence of these indicators led to hesitation, uncertainty, and frustration among participants. Additionally, placements such as numbers above the keys or unfamiliar special markings like brackets added cognitive load and confusion for the users. These findings show the importance of clear, intuitive design in keypads to ensure a good user experience, particularly in situations where quick and accurate input is necessary.

3.3.2 Use of Contrasts

Contrast, or the lack thereof, was a recurring theme. Five participants strongly preferred a clear differentiation between keys and the background, as it enhanced usability in dim lighting and helped guide users to the correct keys. Contrast is particularly crucial for individuals with visual impairments, enabling them to locate keys with confidence. When contrast was insufficient, participants expressed confusion and distrust in the product. Strong visual distinction was beneficial not only for everyday use but also in stressful or distracted situations.

"...one should consider that in a keypad like this, there should be a contrasting color around the keypad, otherwise it [the keys] blends in." - P2 [Man, 60's, visually impaired]

"...on the third keypad, I have to search more because the keys are close together and the same color throughout. ... this one [keypad no. 4], is quick in a crisis, because the keys are a different color than the background." - P8 [Man, 30's, able-bodied]

Not only was the contrast between the background and the keys important, but the contrast of the symbols was frequently remarked. Participants generally preferred a clear color distinction between the symbol and the background. All participants expressed relief when using keypad no. 4 due to the ease of seeing the symbols, whereas keypad no. 1 and no. 3 were often perceived as having no numbers or symbols because of the low contrast. Keypad no. 5 had clear symbols, but the lack of contrast provoked both participant P2 and P3.

"Contrast is nice. I can not understand how someone can enter things [pin codes] without contrast [distinctive symbols], like the 5 [keypad no. 5], why is there not a contrasting color in the number [embossed on the key]? But on the 4 [keypad no. 4], there is a dark gray background and white text, that is great." - P3 [Woman, 30's, able-bodied]

"...in a doorway where the outdoor lighting has broken, I would never be able to get in. [Speaking of symbol contrast and visibility on keypad no. 6]" - P10 [Man, 60's, able-bodied]

Follow-up interviews conducted after the trigger material test further discussed the importance of contrast in everyday settings. Depending on outside lighting and color usage, displays and touchscreens can vary significantly in usability. When physical feedback is no longer available, it is crucial to consider the use of contrasting colors. Participant P2 also highlighted problems with using red on a pink background, light blue on a dark blue background, and dark gray on a light gray background.

> *"...on one of those modern keypads, they are very hard to read due to the contrast. For example, searching for names is very difficult if you can't see. ... It's very difficult to read because there is no contrast against the background. It's always [dark] gray letters on a [light] gray background. And it's impossible, for practically anyone."- P1 [Woman, 40's, severely reduced motor capabilities]*

> *"Yes, it also depends on the general lighting. If it's too bright, it causes glare, but if it's like now [overcast], it might work. However, I still can't see the screen properly." - P2 [Man, 60's, visually impaired]*

In summary, the use of contrasts is essential for the usability of keypads, both with physical keys and touchscreens. Clear differentiation between the keys, symbols, and background significantly improves the user experience, making it easier to locate and identify keys confidently.

3.3.3 Key and Keypad Characteristics

When participants entered codes into the keypads, one of the first comments was about whether the keys felt wobbly or not. Eight of the eleven participants noted a wobble in the keys of keypads 1 and 2, where the keys were significantly more protruding. This wobble was perceived negatively, as it was associated with poor quality. Additionally, the wobble made the keys harder to press, as the keypad did not guide the keys straight down when pressed. When the keys allowed any movement other than straight down, one participant mistakenly believed that the keypad registered clicks by pulling the key in any direction. Another participant also noticed a wobble in keypad 4, noting that it felt like all the keys were connected behind the casing, causing them to slightly shift when pressing any key.

> *"Yes, overall disgusting, it [the key] sticks out like a joystick, you don't want that. It shouldn't be able to move sideways at all." - P4 [Woman, 20's, able-bodied]*

> *"The keypad [keypad no. 4] doesn't feel the same, maybe not the same quality as number 3 [keypad no. 3]. Number 3 was very soft and the keys felt sturdier. And then it also feels like the keys are connected on number 4, it feels a bit off, I think. ... It feels like the keys are attached to some plate behind the cover, it feels a bit cheap." - P5 [Man, 30's, able-bodied]*

When it came to pressing the keys, participants' opinions varied. Two participants preferred the soft and smooth click on keypads 3 and 6. In contrast, three participants disliked this feeling, preferring a

more bouncy and resistant feel. The remaining six participants did not express any explicit opinions about this aspect. However, almost all participants agreed that the feedback on keypads 3 and 6 could be improved for a better overall experience.

> *"The keys here [keypad no. 3] are nice. I like the keys, they have a nice feedback. It's a bit less aggressive than the other two [keypads no. 1 and 2], I mean." - P6 [Woman, 30's, able-bodied]*

> *"I think I like the feedback on keypad 4, but I want a bit more sturdiness. When you press the keys on keypads 3 and 6, it feels like the button goes straight in, but on keypad 4, it feels a bit wobbly. Keypad 3 was probably more comfortable than keypad 6. So, the softness of keypad 3 but the feedback from keypad 4." - P5 [Man, 30's, able-bodied]*

> *Keypad 6 was sluggish in the keys. I don't know if I pressed the button at all." - P11 [Woman, 30's, able-bodied]*

When it came to the keys' geometrical shape, only one participant had a clear preference, favoring square keys over round ones. They found the separate keys more distinguishable since the corners could be felt.

> *"I must say that if we're talking about shape as well, I think rectangles are better than round ones. I think there's a clearer contrast between the keys then." - P11 [Woman, 30's, able-bodied]*

For the other participants, the distance between the keys played a more significant role than their shape. They all wanted enough space to avoid accidentally pressing adjacent keys, however, this preference differed based on the participants' finger and hand sizes.

> *"...and the spacing should also be good, I think number 4 [keypad no. 4] has good spacing, so you don't feel the keys next to it when you press." - P3 [Woman, 30's, able-bodied]*

> *"I think the keys [keypad no. 6] are a bit tight, they are a bit close to each other, I might accidentally click on more than one." - P4 [Woman, 20's, able-bodied]*

When it came to the height of the keys, four participants reacted negatively to the excessive height of the keys on keypads 1 and 2, see Figure **24**. They felt it made the keypad more unpleasant and added instability to the keys.

> *"Yes, keypad 1 was a terrible experience. The keys are too high, they tip over when you press them." - P7 [Woman, 30's, able-bodied]*

> *"It goes faster if it doesn't protrude that much. Keypad 3 is faster to press than keypad 1 and keypad 2." - P6 [Woman, 30's, able-bodied]*



Figure 24: Keypad 1-4

However, one participant preferred the protruding keys on keypad no. 1 and no. 2. P1, who has very limited motor skills, found it much easier to locate and press the more protruding keys. Apart from the absence of numbers and symbols, they found keypad 2 to be the easiest to use. This highlights a key challenge in inclusive design: what benefits some users may create barriers for others. While most participants either didn't care or preferred lower-profile keys for stability and comfort, the more protruding keys improved accessibility for users with motor impairments.

Keypad no. 1, 3, and 6 had glossy acrylic keys, while keypads 2 and 4 featured glossy metal keys with a slight indent. Keypad 5 had metal keys with embossed numbers, providing a slight texture. Five participants commented on the surface finish of the keys. Preferences varied among the participants: three found the glossy keys a bit slippery and would have preferred a slightly textured surface, while two appreciated the smooth surface, as it allowed their fingers to move quickly.

> *"However, the keys [keypad no. 2] were a bit slippery, your finger slips a bit." - P1 [Woman, 40's, severely reduced motor capabilities]*

> *"I would have liked some texture on the keys too, because I liked feeling of something, I don't know if it was the numbers or the texture." - P3 [Woman, 30's, able-bodied]*

> *"I like keypad 3 and 6, that they have this slightly glossy surface. You can slide over them quickly." - P6 [Woman, 30's, able-bodied]*

> *"The feeling of the material [keypad no. 3] is nicer than keypad 2" - P9 [Woman, 60's, able-bodied]*

Apart from the keypads on the trigger board, touchscreens where keys are displayed were also discussed. P1 experienced great trouble using any touchscreens smaller than a tablet, as they require precision, which is challenging for them. Another participant also expressed dislike for touch functions, citing the lack of immediate feedback as their main concern.

"Something I personally find very difficult is touch keys. It's one of the reasons why I don't use a smartphone. I have a lot of trouble with these touch keys, you accidentally press two buttons at the same time. ... Just because I'm a bit jittery, it goes wrong, so it's good to keep that in mind." - P1 [Woman, 40's, severely reduced motor capabilities]

"Yes, I remember having difficulty when touch was new, so when there's a slight delay, you get confused about whether you've pressed or not. Those were bad experiences. And with these induction cooktops, where you place your finger, I find it annoying, I want the button feedback." - P3 [Woman, 30's, able-bodied]

These findings highlight the diverse preferences and challenges faced by users when interacting with different keypads. Finding a perfect universal solution is challenging, as preferences and needs vary greatly among individuals. While some participants appreciated the protruding keys for their accessibility benefits, others found them inconvenient and unstable. Similarly, the shape and spacing of keys can be a matter of personal preference, influenced by factors such as finger and hand size.

3.3.4 Feedback

None of the keypads had any electronics during the test, removing the possibility of evaluating them based on their use of light and sound. The only feedback available was from the mechanical clicks and the tactile feel of the keys. Using this as a baseline, participants discussed the experienced feedback from the keys.

All participants agreed that feedback was crucial to trust the keypad and ensure their key press was registered. When the feedback was instant and clear, participants felt confident in using the product. Tactile feedback, especially the travel distance (how far the button can be pressed down) and the distinct click sensation, was particularly important for building confidence.

"...then the feedback is very clear, both in sound and feel, and it really is a step you press. It goes to a certain depth, not deeper or shallower. [Keypad no. 4]" - P6 [Woman, 30's, able-bodied]

"But keypad 6, ugh, the keys, pressing there was like a swamp. ... It gives no response, you can press and press and press and nothing happens. ... This feels like an old remote control that has given up." - P2 [Man, 60's, visually impaired]

The mention of the click was recurring, either as a tactile sensation or as auditory confirmation, often appreciated in combination. Six participants highlighted the importance of having multiple types of feedback to increase usability and the feeling of certainty.

"The click confirms to me, yes, you have pressed [key] three" – P8 [Man, 30's, able-bodied]

> *"Yes, especially if it's dark, you want to feel that you've pressed [a key], with tactile feedback. And if it also made a beep, it would be incredibly clear." – P4 [woman, 20's, able bodied]*

Although no light or sound signals were used during the test, there was a desire for these features to be included in the products to enhance the user experience, particularly under varying conditions such as dim light.

> *"And then I want feedback, it should light up around the button. ... they light up green when I click so that I can see. That's good feedback. I like feedback, maybe even both sound and light." – P4 [woman, 20's, able bodied]*

> *"And if it's a bit dark, there's no chance of seeing [keypad no.6]. ... Unless they [the keys] light up, of course, but you don't know that." – P10 [Man, 60's, able-bodied]*

Participants' discussions highlighted that effective feedback is a key factor in ensuring user confidence and trust in the keypad. While tactile feedback was essential, the absence of additional sensory cues, such as light and sound, was noticeable. By integrating multiple forms of feedback, the overall user experience and reliability of the keypad can be significantly improved.

3.3.5 Usability and Security

User feedback highlighted several important factors not fully covered in the interview guides, such as emergency situations and security.

In high-stress situations, clarity and ease of use become critical. Both P8 and P9 emphasized that they would prefer keypad 4 in an emergency due to its clear layout, allowing them to quickly identify and press the keys without hesitation. This underscores the importance of minimizing cognitive load and maximizing immediate recognition in urgent scenarios.

> *"Keypad 4 is the clearest because you can see the keys and the spacing is clear. I would prefer to use this one if there was a fire." – P8 [Man, 30's, able-bodied]*

Another concern raised by P10 was the possible buildup of dirt in the embossed numbers of keypad 5. Beyond being unhygienic, this buildup could reveal frequently used keys, making it easier to identify passcodes.

> *"Yes, I've had this type [keypad no. 5] at work, and since more people use it, there are dirty fingers and other things that get into the numbers. You can almost see what the code is." – P10 [Man, 60's, able-bodied]*

Similar patterns were observed on other keypads, where painted keys wore out over time or where materials absorbed and showed grease residue from users' fingers, shown in Figure **25** and **26**. This

suggests that material choice and surface finish are essential not only for user experience but also for security, as they can influence how easily passcodes can be revealed.



Figure 25: Keypad with grease on keys



Figure 26: Keypad with the color rubbed off

One participant, P3, expressed a preference for a hybrid approach to access control in their apartment complex. Currently, a key is required both to enter the building and access individual apartments. They liked the idea of using a keypad with an entry code for the main entrance but preferred a traditional key for their own apartment.

> *"I would probably like to have a code at the entrance, because it's nice to be able to give one to visitors. But maybe it should only work until 10 p.m., and after that, you have to use a key. ... And for my apartment door, I want a key because then I know exactly how many exist and who has one, then I know who can get in." – P3 [Woman, 30's, able-bodied]*

This response highlights the need for flexible access solutions that balance convenience and security. By allowing users to customize access based on time or context, keypad systems can better accommodate individual preferences while maintaining a clear sense of control over entry permissions.

3.3.6 Design Preferences

Five participants commented on the overall look and design of the keypads. Most of the comments focused on keypads 3 and 6, which were preferred for their appearance, and keypad 4, which was found to be the least attractive. The participants appreciated when they felt that time and design considerations had been invested in the product.

> *"Keypad 6 is nice to look at. I probably wouldn't want keypad 4 on my own door, I want a nice and stylish design on it" - P5 [Man, 30's, able-bodied]*

> *"Keypad 6 is the best looking, it has the nicest keys, they are round and feel a bit designed." - P3 [Woman, 30's, able-bodied]*

> *"Keypad 4 was the easiest, but it's boring. It's not nice looking in any way." - P7 [Woman, 30's, able-bodied]*

"Keypad 4 is ugly and feels outdated." - P9 [Woman, 60's, able-bodied]

Although keypad no. 6 stood out for its visual appeal, its functionality did not meet user expectations. Similarly, keypad no. 3 was criticized for its poor usability. Overall, while keypad 4 was favored for its usability, its design was not well-received.

When asked about how they would design a keypad based on their own preferences, participants provided the following insights:

"Then I want, number one [keypad no.1], but with contrast between the key and the background, and the number and the background. Then I want the number on the key, not somewhere next to it, and then it should be in a different color, there should be contrast. Then there should be an angle, so that it is angled upwards. And then I want feedback, it should light up around the button, it should be tactile. And absolutely no wobble." - P4 [Woman, 20's, able-bodied]

"I would want it to be clear, each key should be clear about what it is and what it does. I think each key should be easy to see. ... And then it's important with feedback, that you know you're clicking the key. And then it should be nice to look at. ... And that get a good feeling, that it doesn't feel wobbly." - P5 [Man, 30's, able-bodied]

P2, who is visually impaired, was content with keypad 4 and would only improve it by making the numbers and symbols tactile.

"No, I think that if only the numbers on keypad 4 had been tactile, it would have been fine. And that it is at the right height and that there is lighting around it. And that you could use your tag, especially if you have something to carry." - P2 [Man, 60's, visually impaired]

This feedback highlights the importance of balancing functionality with design in user interfaces. Striking the right balance between form and function is crucial to enhance user satisfaction and ensure that the product will actually be purchased and used by customers.

3.3.7 Alternative Solutions

As technology continues to advance, more solutions labeled as "modern" are being implemented, with touchscreens being one of the most commonly used in new interactive products. Despite their popularity, it is important to consider their inclusivity. In the interviews following the trigger material tests, some participants highlighted various types of keypads they encounter in their daily life and the issues associated with them. Participant P1, who has reduced motor skills and dexterity limitations, explained that standard smartphone-sized touchscreens are too small for them to use, and that they require tablet-sized touchscreens that allow larger keys. Similarly, P2, who has limited vision, agreed

from the perspective that the screen is too small to see. He shared an example of a colleague, also with limited vision, who was unable to pay for his lunch because the card terminal had a small touchscreen. This underscores the importance of incorporating alternative access solutions to increase inclusivity.

> *"Well, I don't really like these new ones with smooth buttons, they don't have enough contrast. The new payment terminals... I don't like them, you become so damn dependent on others." - P2 [Man, 60's, visually impaired]*

Incorporating multiple access solutions in a product is beneficial for everyone, as user capabilities often vary and a single solution rarely fits all. Two participants mentioned that they call themselves on their intercom instead of using a key or entering a code to gain entry. Participant P1 uses a special keypad, shown in Figure **27**, which opens the door to their building wirelessly.

> *"Yes, now I use an automatic door opener, and I have an automatic keypad instead of a key. Otherwise, with those modern keypads, they are very hard to read because of the contrast. For example, searching for a name is very difficult if you can't see well." - P1 [Woman, 40's, severely reduced motor capabilities]*



Figure 27: Wireless Keypad

Three participants expressed a desire for tags or access cards as an additional method to enter their homes. They wanted to reduce the number of steps required for entry and achieve a more seamless interaction. They argued that both keys and codes demand more effort and cause added stress. Participant P5 even wished for no interaction at all.

> *"I just remembered that we have a keypad like this at the office, but it also has a tag, so it's a nice interaction with the keypad since I don't need to enter the code and remember anything. – P3 [Woman, 30's, able-bodied]"*

> *"The best thing would be if it could recognize who you are in some other way." - P5 [Man, 30's, able-bodied]*

In contrast, participant P2 expressed a desire to enhance interaction by reintroducing the human element. P2, who has impaired vision, wished to be able to call someone for guidance and help instead of having to search on a screen. Alternatively, they wanted to be able to speak to a voice and use voice control.

By integrating multiple access methods, such as physical tags, voice control, and tactile keypads, products can better serve users with varying abilities. It is beneficial to focus on balancing convenience, accessibility, and user autonomy when creating interactive products.

3.3.8 Key Insights

The findings from Chapter 3.3 contribute to all three research questions by revealing the diverse challenges users face when interacting with keypads (RQ1), identifying critical design elements that support inclusive use (RQ2), and highlighting specific features, such as contrast, tactile feedback, and layout, that can accommodate users with varying physical and cognitive abilities (RQ3).

Key insights and preliminary analysis based on the trigger material and interviews are listed below:

- Participants hesitated when symbols were missing, faint, or placed above the keys. This increased cognitive load and reduced confidence during interaction.
- While most keypads have a tactile marking on the “5” key, only those with centered markings were considered helpful, especially by users with visual or motor impairments.
- Low contrast between keys, symbols, and backgrounds made keypads difficult to use, particularly in low-light conditions or for users with visual impairments.
- Wobbly or overly protruding keys were perceived as low quality and harder to press. Participants preferred keys with clear tactile feedback and moderate travel.
- Concave surfaces were appreciated to guide the finger.
- Familiar icons and consistent positioning improved recognition for special function keys.
- Participants emphasized the need for tactile, visual, and auditory feedback to confirm input, especially in stressful or low-visibility situations.

3.4 Layout Comparison

This section presents the findings from the layout comparison test, detailed in Chapter 2.1.5. The papers the layouts were printed on and drawn on by the participants have been scanned and are presented in Appendix G.

Out of the 17 participants, 16 chose a standard 3x3+1 as one of, or their only favorite. The reasoning behind this was that it felt familiar and reminded them of their phones. All but one reacted negatively to the more explorative layouts such as 1, 2, 5, 8 and 9, seen in Appendix D. Preferences for the key shape varied among participants. Some preferred square keys as they reminded them of the numpad on a keyboard, while others preferred round keys as they reminded them of their phone. However, the majority did not express a strong preference for the shape in regards to the layout.

The participants were also asked to design special function keys for unlocking and locking a connected door. The preferences for these keys are presented below:

- **Single Key vs. Separate Keys:**
  - Twelve participants preferred having two separate keys for unlocking and locking, to ensure clarity and avoiding mistakes. When drawing two keys, they all aimed to keep the look symmetrical.
  - Five participants designed only one key to perform both functions, expressign that was sufficient.
- **Color Preferences:**
  - Twelve participants designed their keys to be green for unlocking and red for locking, or for the keys to light up with green or red lights. Showing a common association of these colors with the respective functions.
- **Symbol Preferences:**
  - Fourteen participants added symbols on their keys. The most common symbols were an unlocked padlock for unlocking and a locked padlock for locking. Other symbols includes open and closed doors.
- **Key Placement:**
  - Six of the participants who designed two keys chose to place the unlock key on the right side of the keypad, describing the placement and function similar to an enter key on a keyboard, which comes last.
  - The other six of the participants who designed two keys chose to place the unlock key on the left side of the keypad, viewing it as the first command they are making, similar to writing from left to right.
- **Shape Preferences:**
  - Five of the participants designed to have their unlock and lock keys in another geometrical shape compared to the numbered keys, making them more distinguishable.

- Three of the participants that designed their unlock and lock keys to have the same geometrical shape as the numbered keys chose to make them larger than the numbered keys, making them more distinguishable.

The results from Chapter 3.4 contribute to RQ1 and RQ2 by demonstrating that familiarity and clarity in keypad layout significantly influence usability. The strong preference for the standard 3x3+1 layout and clearly distinguishable function keys underscores the importance of intuitive design in reducing user errors and cognitive load.

4 Define

This chapter presents insights and conclusions from the projects Define phase. This phase involves summarizing and analyzing identified problem areas to provide insights and direction for the project's continuation.

4.1 Insights

Based on user tests, interviews, and background research, Clustering was carried out (see Appendix H), Customer Journey Maps were created (see Appendix I), and a Needs Analysis was performed (see Appendix J). These activities yielded numerous insights, presented below and summarized in the conclusions.

4.1.1 Physical Needs

According to the theory presented in Chapters 3.1.2, 3.1.4, and 3.1.5, the actuation force is an important aspect of a key's design. The required force must not be too large, as excessive force leads to accessibility issues for those with limited strength and motor capabilities, while also not being too low, which can cause the keys to appear flimsy and of poor quality. This was supported during the interviews with the trigger board, where 8 of 11 participants expressed a dislike for keys they perceived as wobbly and unstable.

4.1.2 Cognitive Needs

The theory presented in Chapters 3.1.4 and 3.1.5 highlights the importance of designing products to minimize cognitive load. To achieve this, designs should be intuitive, provide clear and immediate feedback, and use legible fonts, numbers, and symbols. Empirical evidence from Chapter 3.2.1 and 3.3 supports these insights. Participants reported feeling uncertain and hesitated when numbers and symbols were not clearly visible. They expressed a strong preference for clear numbers and symbols that clearly convey their function, thereby reducing uncertainty. Additionally, participants emphasized the importance of feedback from the keys, which was crucial for their confidence that the key press had been registered.

4.1.3 Emotional Needs

Receiving clear and immediate feedback was crucial for users to feel trust and confidence. Participants expressed frustration when input feedback was unclear. From the tests and interviews presented in Chapter 3.3 and Chapter 3.2.1, it became evident that reliable and responsive keys with immediate confirmation significantly increased user satisfaction.

4.1.4 Capabilities

In Chapter 3.1.2, sensory, cognitive, and motor capabilities were identified as the most relevant to keypad design. The standards regarding accessibility, presented in Chapter 3.1.4, also addressed these areas, focusing on cognitive needs and motor and sensory aspects, such as tactility and visibility.

Empirical results from Chapter 3.3 showed that the participant with reduced vision struggled with low contrast between keys and backgrounds, unclear symbols, and small or poorly placed tactile indicators. Inadequate visual cues led to slower navigation and increased errors. The participant with limited dexterity preferred keys that protruded more, had tactile borders, and were easier to locate and press without fine motor control. Users with full vision and typical motor skills favored keys with speed and responsiveness but also struggled with low contrast and unclear symbols.

In Chapter 3.2.1, it was shown that wearing gloves and reducing the users' ability to feel and use fine motor skills led to more uncertainty, as participants relied on tactile feedback when pressing a key.

Many users have overlapping impairments, such as visual and motor limitations. According to Funka's 2023 statistics (Chapter 3.1.2), a large segment of the population lives with multiple impairments, making it essential to design for broad groups rather than targeting individual user groups. The keypad must support effective use regardless of sensory, cognitive, or motor variability.

4.1.5 Keypad Layout

As stated in Chapter 4.1.2, a design should be intuitive, and the layout of a keypad is an important aspect of this. The theory presented in Chapters 3.1.5 and 3.1.6 emphasizes that a keypad should be familiar and that standardized keypads reduce errors. Studies have found that participants preferred the standard 3x3+1 layout compared to others. This was supported by the layout comparison test presented in Chapter 3.4, where 16 of 17 participants strongly preferred the 3x3+1 layout and reacted negatively to non-conventional ones.

Apart from the placement of numbered keys, the placement of numbers and symbols can also affect the intuitiveness of the keypad. In Chapter 3.3.1, P11 noted a brief delay when labels were placed above instead of on the keys, showing that even minor deviations can add unnecessary friction.

4.1.6 Key Shape

From Chapter 3.1.6, one study referred to the shape of the keys, favoring square and round keys over vertical rectangle keys. However, since the study was conducted on keyboards, it is not fully applicable to keypads connected to door locks. In Chapter 3.3.3, only one participant had a preference regarding

shape, favoring square over round keys. Additionally, in Chapter 3.4, the preferences varied, not leading to a clear result. Therefore, the most suitable shape for keypad keys remains undetermined.

4.1.7 Key Size

Smaller keys increase the risk of pressing the wrong key for users with larger hands, gloves, or reduced dexterity (Chapters 3.2.1, 3.2.2, and 3.3). However, large, closely packed keys reduce key distinction. The optimal size must balance surface area with sufficient spacing (Chapter 3.1.6). As there are no clear size recommendations from the collected research and the keys on the trigger board were approximately the same size, the most suitable size remains undetermined.

4.1.8 Key Protrusion

Empirical results from Chapter 3.3 showed that most participants preferred keys with low to moderate protrusion, as these were perceived as more stable and of higher quality. In contrast, keys with high protrusion were often described as wobbly and of worse quality, which undermined user trust. One participant with severely reduced motor skills found the more protruding keys easier to use, as the increased height made them easier to locate and distinguish from one another. However, this participant also found keypad no. 4, featuring keys with a 2 mm protrusion, sufficiently accessible, suggesting that moderate protrusion may offer a good compromise between stability and accessibility.

4.1.9 Keys with Special Functions

The theory presented in Chapter 3.1.6 suggests that special function keys should be clearly distinguishable from numbered keys to improve accessibility, particularly for users with visual impairments. This can be achieved through variations in size, shape, and/or orientation. The literature also indicates that separating function keys from numeric keys helps prevent user errors.

These theoretical insights were confirmed by the empirical results from the testing of the reference product (Chapter 3.2.1). When participants were asked to enter a code with their eyes closed, the identical appearance and spacing of the special function keys caused uncertainty and hesitation. In Chapter 3.4, when participants were asked to sketch their ideal special function keys, approximately half chose to differentiate them through shape, size, or position. The majority also added color to indicate function, further separating them from the numeric keys and improving recognizability. Additionally, most participants preferred having one dedicated key per function rather than a single key with multiple functions. This design choice contributed to a greater sense of certainty that the correct key had been used.

4.1.10 Grip

As stated in Chapter 3.1.5, theory suggests that keys with a concave surface can improve finger placement and prevent slipping by guiding the finger toward the center of the key. Additionally, raised edges or textured surfaces can help users differentiate keys by touch, which is particularly beneficial for users with reduced dexterity. Empirical findings from Chapters 3.2.1 and 3.3.3 revealed mixed preferences regarding surface texture. Glossy surfaces were sometimes perceived as slippery, increasing the risk of accidental slips—especially for users with limited motor control. However, some participants preferred glossy surfaces over textured ones, as they allowed for quicker finger movement when entering codes and were perceived as easier to clean and less prone to showing wear.

Interestingly, the negative feedback on glossy keys was only directed at those with flat surfaces. Keypad no. 4, which featured glossy keys with a slight concave shape, received more positive responses. This suggests that the shape of the surface, specifically concavity, can counteract some of the negative experiences associated with glossy finishes, offering tactile guidance as well as allowing quick movements.

4.1.11 Visibility

The standards and guidelines presented in Chapter 3.1.4 emphasize the importance of large, legible fonts and high contrast to improve visibility and accessibility. A keypad should ensure strong contrast both between symbols and keys, and between the keys and their background, to support users with visual impairments and enhance overall usability. These theoretical recommendations were reinforced by empirical findings from Chapters 3.2.1, 3.3.1, and 3.3.2. Participants reported delays and uncertainty when numbers and symbols were not clearly visible. Many expressed that high contrast between symbols and keys was essential for confident and efficient code entry. The contrast could be achieved through various means, including color differentiation, surface finish, backlighting, or a combination of all.

While the reference product featured high contrast and good visibility when viewed at eye level, the placement of numbers and symbols partially obscured them when viewed from a standing position. This highlights the importance not only of contrast, but also of thoughtful symbol placement to ensure consistent visibility across different user perspectives.

4.1.12 Use of Tactile Markings

Chapters 3.1.4 and 3.1.5 highlight the importance of tactile markings, such as a raised dot on the central key (typically the “5” key), as a key accessibility feature in keypad design. These markings assist visually impaired users in identifying the keypad as a numeric layout and improves navigation by touch. Empirical findings from Chapters 3.2.1 and 3.3.1 support this theoretical guidance. While all

tested keypads included a tactile marking, the usability of these markings varied significantly depending on their size and placement. On some keypads, the marking was positioned along the bottom edge of the key, making it harder to detect and therefore less effective. One visually impaired participant emphasized the importance of both the presence and the placement of the tactile marking, stating that it should be centered on the key. This placement aligns with user expectations and muscle memory, making it easier to locate, identify and use effectively.

4.1.13 Other Insights

Since the design recommendations created in this project will be applied on a reference product already on the mainstream market, the bottom-up approach of the user pyramid presented in Chapter 3.1.3 is the most suitable.

Membrane keypads, such as those used in keypads no. 2 and no. 4 on the trigger board, were favored by participants for their clear and rapid tactile feedback. In contrast, the reference product uses a silicone keypad with acrylic keys mounted on top, resulting in more subdued tactile feedback. This comparison highlights the importance of tactile responsiveness in perceived usability.

Although the primary focus of this study is keypad usability, several participants expressed interest in alternative access solutions, including RFID tags, mobile applications, and biometric systems. Offering multiple access options aligns with inclusive design principles and can accommodate a broader range of user preferences and needs (Chapter 3.1.4).

The built-in doorbell of the reference product was often overlooked. Many users were unaware of its purpose, pointing to a need for more visible label, placement, and general discoverability (Chapter 3.2.2).

4.2 Summary of Key Findings and Design Recommendations

Based on the insights gathered in Chapter4.1, several challenges and opportunities for improvement have been identified. The table below summarizes the most critical findings and translates them to design recommendations or recommendations for continued development.

Table 3: Problems Identified and Recommendations/Design Opportunities

Category	**Problems Identified**	**Recommendations/ Design Opportunities**
Cognitive Load	Unclear symbols and lack of feedback causes hesitation and uncertainty.	Use intuitive layouts, clear symbols and provide immediate feedback.
Emotional Confidence	Users feel uncertain and insecure when feedback is delayed or unclear.	Provide responsive and reliable feedback to build user trust and satisfaction.
User Capabilities	Visual and motor impairments significantly affects the usability of current tested keypads.	Design for inclusivity by providing tactile markings, high contrast and distinct keys.
Keypad Layout	Non-standard layouts leads to confusion increases the cognitive load.	Use a familiar 3x3+1 layout for intuitive use and recognizability.
Key Shape & Size	No clear size differences were investigated, no clear shape preferences were identified.	Explore key shapes and sizes further.
Key Protrusion	Excessive or insufficient protrusion affects perceived quality and accessibility.	Use a moderate protrusion, around 2 mm, to ensure both stability and and distinction.
Special Function Keys	Users struggle to distinguish function keys from numbered keys when placed together.	Differentiate special function keys by shape, size, color and/or placement.
Grip & Surface Texture	Flat glossy surfaces are slippery, especially with reduced dexterity.	Use concave surfaces or rougher surface textures for flat keys.
Visibility	Low contrast and poor symbol placements reduces readability.	Use high contrast, large fonts, and consider symbol placement to ensure visibility from multiple angles.
Tactile Markings	Poor placement of tactile markings reduces their effectiveness.	Center tactile markings (e.g. on the ”5” key) to align with user expectation and improve navigation.

5 Develop

This chapter presents the ideas and decisions made during the projects Develop phase. It includes the process from initial ideation to concept prototyping, detailing the steps taken to refine and validate the design.

5.1 Ideation

During the ideation phase ideas are generated and refined. This section presents the ideas produced as described in Chapter 2.3.1.

5.1.1 Sketching

The sketching phase explored various design elements such as key shapes, special function keys, and product shapes. This stage is important for visualizing initial ideas and iterating on them to find the best solutions.

When sketching the key shapes, the exploration revolved around how to increase the visibility of the keys and their symbols, as well as physical comfort for the user when pressing. Various shapes were considered to enhance both ergonomics and aesthetics. Figure **28** shows different key shapes that were sketched and considered.

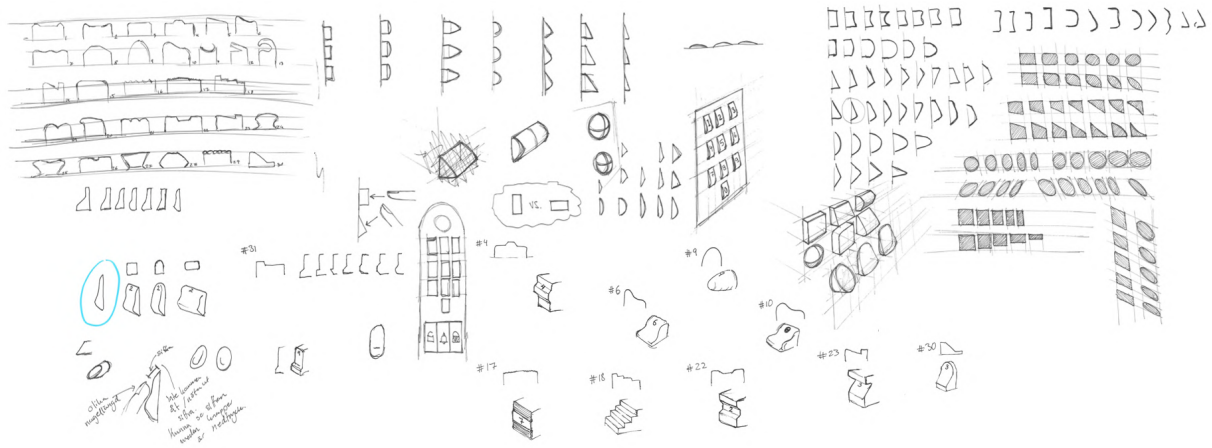


Figure 28: Explorative sketches of key shapes

One of the conclusions that came from the project's Discover and Define phase was that special function keys should be distinguished and separated from the numbered keys. This differentiation could be achieved through size, shape, and color, allowing users to recognize them easily. Various executions of this concept were sketched to explore different options. Figure **29** shows various sketches of special function keys.

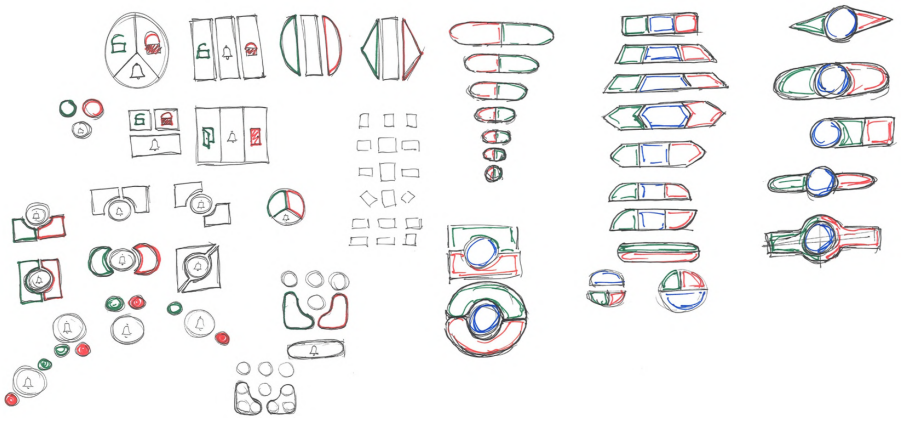


Figure 29: Explorative sketches of special function keys

Another idea to improve the visibility of the keys and symbols was to angle the surface of the whole keypad. This approach aimed to enhance the user's viewing angle and accessibility. Figure **30** shows various sketches exploring the reference product's profile shape.

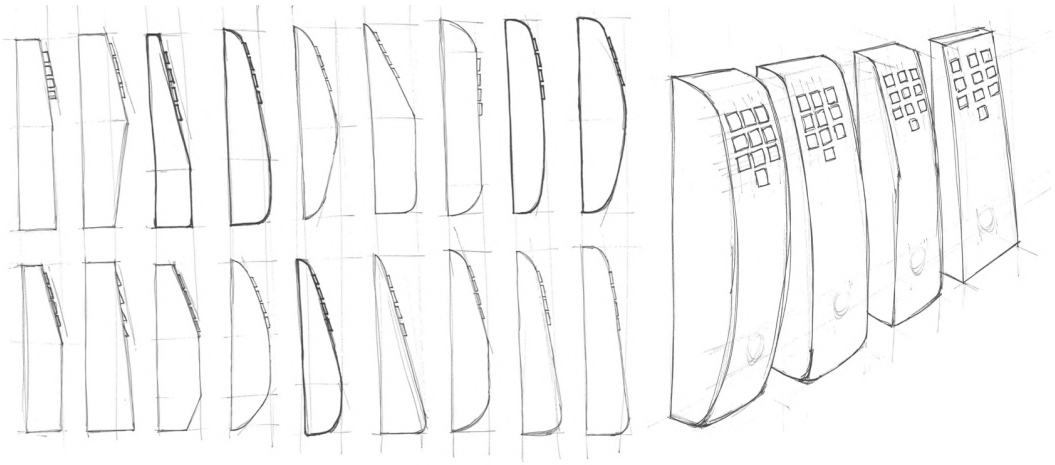


Figure 30: Sketches of product profile

5.1.2 Clay Modeling

To gain a better understanding of how the sketched keys would look and feel, clay models were created. These models focused on the shapes that appeared most interesting and relevant to achieving the visibility and comfort goals. The tactile feel of these clay models was important for determining the most ergonomic and visually appealing designs.

The clay keys that felt the most comfortable were made in complete sets and attached to a 3D-printed model of the reference product. This allowed for a more realistic evaluation of the keys in their context. The keys were arranged in a 3x3+1 keypad layout, providing a set-up for feeling and evaluating them. Figure **31** shows the clay models of the keys.



Figure 31: Clay models of keys

5.2 Concept Prototypes

Based on the clay models it was determined that keys with a concave surface provided a better grip and helped guide the users finger to the middle of the key. This also helped with distinguishing the keys by feel. From the clay models, the key shape could not be evaluated fairly as the sizes and depth of the concave surface differed from key to key.

To compare different shapes, sizes, the distance between keys 3D-printed models were created, see Figure **32-34**. Two test groups were asked to evaluate the different keys based on the models, as detailed in Chapter 2.3.3.



Figure 32: Key prototypes with varying shapes and sizes



Figure 33: Prototype with clickable keys in varying shapes

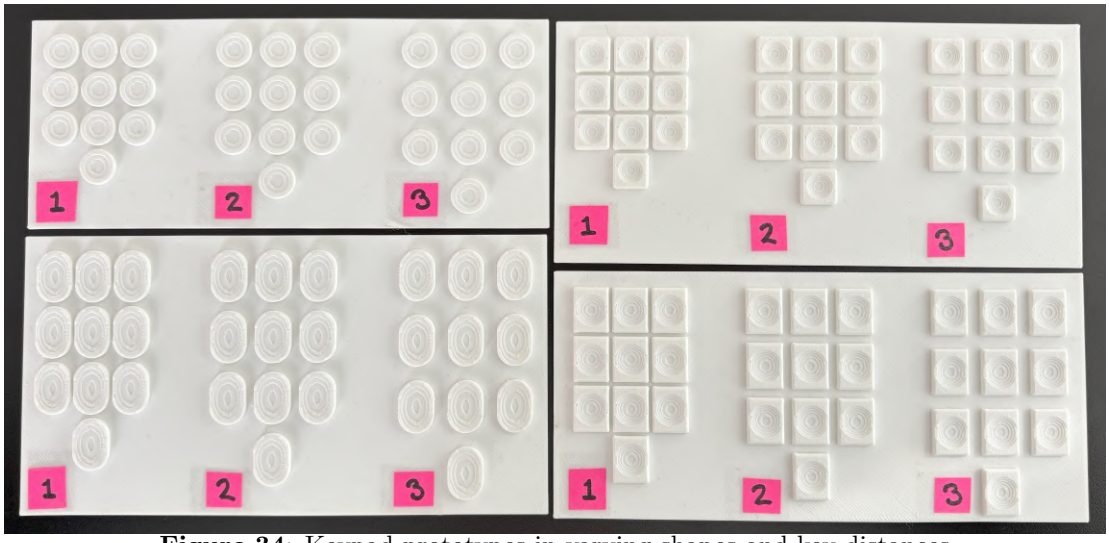


Figure 34: Keypad prototypes in varying shapes and key distances

5.2.1 Test Group 1

The first test group included seven seniors aged between 84 and 92 years, with varied physical and sensory capabilities. The participants interacted with 3D-printed key models to determine their preferences in shape, size, and spacing. Their insights also included reactions to clickable keys and the reference product. A summary of the participants preferences and comments are presented in Table **5** below.

Table 4: Summary of Preferences – Test Group 1

P	**Shape Preference and Comments**	**Size Preference and Comments**	**Spacing Preference and Comments**	**Reference Product Feedback**
P1	Oval – Wants to reach the whole surface, prefers a shape like the fingertip.	Size 7 – Wide fingers, prefers large contact area.	Spacing 3 – Clear distinction between keys.	Very small keys like on phones, hard to see symbols. Likes the spacing.
P2	Oval – Follows finger shape nicely, feels natural.	Size 4	Spacing 2 – Prefers close spacing to minimize finger movement.	Too small, accidentally presses multiple keys.
P3	Rectangle & Oval – Indicates clearly where to press.	Size 5	Spacing 2–3	Keys are too small.
P4	Oval – Follows finger shape.	Size 7 – The bigger the better.	No clear preference, but generally liked larger spacing.	Very small keys.
P5	Oval & Rectangle – Most natural to press with finger.	Size 4–5 – Wants to feel full key and edges.	Spacing 2 – Feels the best.	Cute but very small keys.
P6	Oval – Feels the best.	Size 5 (3–5 acceptable) – Prefers 5 but others are okay.	Spacing 3 – Wants to clearly distinguish each key.	Keys are very small, would often press wrong ones.
P7	Oval – Likes that it follows finger shape.	Size 3–5	Spacing 3	Keys are too small, but spacing is good.

Overall, participants in Test Group 1 preferred oval-shaped keys, noting that this form aligns well with the natural shape of their fingertips. Most preferred larger key sizes (primarily sizes 4 to 7), with widths ranging from 12-15 mm, as these were easier to locate and press. Regarding key spacing, sizes 2 and 3 (2.5 mm and 4 mm) were generally preferred, offering tactile clarity. Feedback on the reference product was largely critical, particularly regarding key size, which was commonly described as too small to use comfortably or accurately.

5.2.2 Test Group 2

The second test group included seven participants aged between 26 and 60 years, all able bodied. The participants interacted with 3D-printed key models to determine their preferences in shape, size, and spacing. Their insights also included reactions to clickable keys and the reference product. A summary of the participants preferences and comments are presented in Table **5** below.

Table 5: Summary of Preferences – Test Group 2

P	**Shape Preference and Comments**	**Size Preference and Comments**	**Spacing Preference and Comments**	**Reference Product Feedback**
P8	Oval – Fits the finger the best	Size 1-2	Spacing 2 – Not to small and not to big.	They work fine but they are a bit small.
P9	Oval – Feels natural.	Size 5 – The others are too small or too big.	Spacing 3 – Minimizes risk of mispressing when stressed.	Feels like a joke, they are way too small to be functional.
P10	Oval	Size 2-3 – Fits their fingers.	Spacing 2	Keys feel okay but the symbols are a bit small.
P11	Square & Oval	Size 2-3	Spacing 2 - Enough to distinguish the keys without separating them too much.	Looks good, but it's hard to see the symbols.
P12	Rectangle & Oval	Size 2-3 – Rectangle, Size 1-2 – Oval	Spacing 3 – Wants to feel the finger go down in between the keys.	It works, but they feel a bit small.
P13	Round - Do not like to feel corners.	Size 4 - Do not want to feel the keys edges.	Spacing 2 - Feels natural.	Really likes the round keys, but not optimal size.
P14	Square - Likes to feel corners and edges.	Size 1-3 – All feel good.	Spacing 2	The visibility of the numbers is terrible, does not like round keys.

Overall, participants in Test Group 2 showed a preference for oval-shaped keys, similar to Test Group 1, as they fit the finger well and feel natural. However, there were some variations in shape preferences, with a few participants favoring square or round keys for their distinct tactile feel. Most participants preferred the smaller keys (primarily sizes 1-3), with widths ranging from 9-11 mm as they matched the participants finger sizes. Spacing size 2 (2.5 mm) was generally preferred, as it provided enough distinction between the keys without feeling to large. The feedback on the reference product was mixed, but leaned towards the negative. Many participants noted the lacking visibility of the symbols, as well as the keys feeling a bit small.

5.2.3 Key Insights

The results from section 5.2 contributes to RQ2 and RQ3 by evaluating how different key shapes, sizes, and spacings affect usability and comfort for users with varying physical capabilities. It also addresses RQ1 by identifying specific limitations in the reference product, such as small key size and poor symbol visibility, and testing design alternatives that improve tactile clarity and accessibility.

The evaluation of the concept prototypes provided the following insights:

- **Shape Preferences:** Both test groups predominantly favored oval-shaped keys, as these align well with the natural shape of fingertips and provide a comfortable feel. However, there were notable variations, with some participants in Test Group 2 preferring square or round keys for their distinct tactile feel.
- **Size Preferences:** Participants in Test Group 1 generally preferred larger key sizes (sizes 4 to 7), with widths ranging from 12-15 mm, as these were easier to locate and press. Conversely, Test Group 2 participants favored smaller key sizes (sizes 1 to 3), with widths ranging from 9-11 mm, which matched their finger sizes better.
- **Spacing Preferences:** Key spacing preferences varied slightly between the groups. Test Group 1 preferred spacing sizes 2 and 3 (2.5 mm and 4 mm), offering tactile clarity and clear distinction between keys. Test Group 2 predominantly favored spacing size 2 (2.5 mm), providing enough distinction without feeling too large.
- **Reference Product Feedback:** Feedback on the reference product was largely critical across both groups. Participants commonly described the keys as too small to use comfortably or accurately. Additionally, visibility of symbols on the keys was a recurring issue, with many participants finding them hard to see.

The clear preference for oval-shaped keys across both test groups highlights the ergonomic benefits of this shape, whilst the variations in size and spacing preferences between the two test groups indicate

that there is no one-size-fits-all. The critical feedback on the reference product, particularly regarding key size and symbol visibility, emphasizes the need for improvements in these areas.

5.3 Angle Exploration

To investigate the best viewing angles for a keypad at a 115 cm height a low fidelity test was conducted, as described in Chapter 2.3.4. The results are approximate and presented in Table **6**, the green ranges for the two cases have been combined and are shown in Figure **35**.

Table 6: Angle range ratings by number height and user height

Number Height [mm]	**User Height [cm]**	**Green Range [°]**	**Yellow Range [°]**	**Red Range [°]**
3	210	28–90	12–27	0–11
3	130	0–60	61–90	–
4	210	20–90	10–19	0–9
4	130	0–75	76-90	–
5	210	13–90	8–12	0–7
5	130	0–75	76-90	–

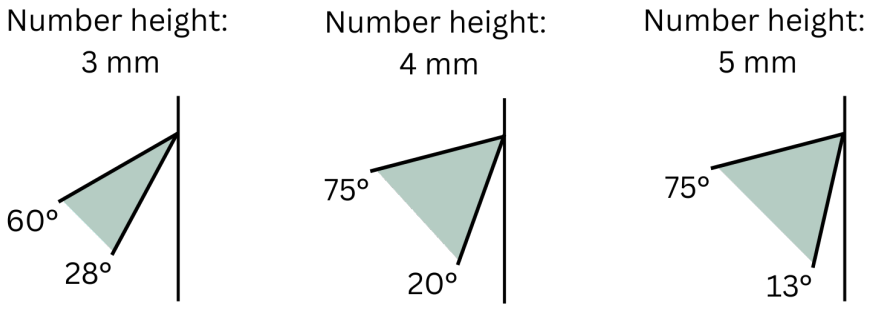


Figure 35: Green angle ranges.

The results contributes to RQ1 and RQ3 by investigating optimal viewing angles for keypad interaction, particularly for users of different heights.

5.4 Symbol Preference Study

The result of the in person symbol preference study described in Chapter 2.3.5 provided insight into which symbols users find most intuitive for representing *lock* and *unlock* functions. The study explored user preferences when selecting symbols individually, when selecting them as a pair, and wether those preferences changed when the symbols were presented in their actual scale.

For clarity, the symbols have been numbered as seen in Figure **36** and are referenced accordingly in Table **7**.

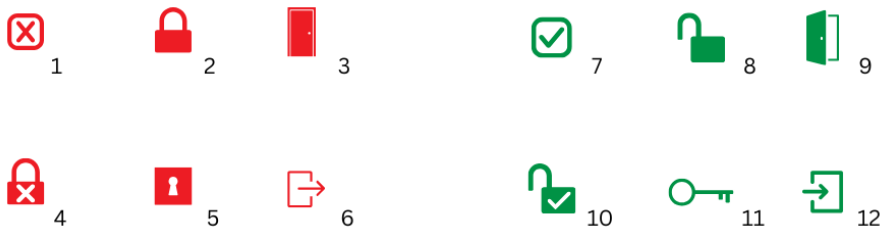


Figure 36: Numbered Lock and Unlock Symbols

Table 7: Symbol Preference Study Results Results

P	Lock Symbol	Unlock Symbol	Combination of Symbols	In Actual Scale
P1	2	8	2 & 8	2 & 8
P2	2	9	2 & 8	2 & 8
P3	2	8	2 & 8	2 & 8
P4	2	8	2 & 8	2 & 8
P5	2	11	2 & 11	2 & 8
P6	2	10	2 & 10	2 & 8
P7	2	10	2 & 8	2 & 8
P8	2	8	2 & 8	2 & 8
P9	5	11	2 & 8	2 & 8
P10	2	10	2 & 10	2 & 8
P11	4	10	4 & 10	4 & 10
P12	2	8	2 & 8	2 & 8
P13	2	11	2 & 8	2 & 8
P14	2	8	2 & 8	2 & 8
P15	2	10	2 & 8	2 & 8

The result shows an overwhelming favor for the padlock symbols, where 14 out of 15 chose the solid colored padlock as the preferred combination. When asked about the motivation for their choices, the participants emphasized the importance of clear messaging to decrease confusion.

"The door symbols [no. 6, 9 and 12] looks like emergency exits" - P3 [Man, 30's, able-bodied]

"The X on no. 4 looks like something is wrong" - P6 [Man, 40's, able bodied]

"It should be super clear, and the X symbol on no. 4 is unclear without context." - P8
[Woman, 60's, able bodied]

The results varied when evaluating the symbols individually versus in combination. Symbol 10 was generally well received, with the check mark interpreted as a successful operation, while Symbol 4 caused confusion and distrust, with the X interpreted as a failed operation. Three participants who initially preferred detailed symbols like Symbol 10 and Symbol 11 changed their preference after seeing the symbols in a smaller size, as they then appeared crowded.

5.4.1 Key Insights

Preferences for the unlock symbol were relatively evenly spread over symbols 8, 10, and 11 when chosen separately. However, this changed when asked to combine it with a lock symbol, the vast majority then chose Symbol 8 as the unlock symbol, combined with Symbol 2 for the lock function. The findings addresses RQ2 and RQ3 by identifying which lock and unlock symbols users find most intuitive and trustworthy.

Key insights on symbol preferences from the study are presented below:

- 14 out of 15 participants selected a solid padlock (Symbol 2) for locking and an open padlock (Symbol 8) for unlocking when asked to choose a pair.
- Participants emphasized the importance of instantly recognizable symbols.
- Scale affects perception, detailed icons became harder to interpret when scaled down.
- Participants often associated green with unlocking and red with locking, suggesting that color can enhance symbol clarity.

5.5 Concept Development

An oval design was selected for the keys based on the concept prototype test results described in Chapter 5.2, where 12 out of 14 participants preferred the oval shape. The oval design also complements the overall shape of the reference product, ensuring visual consistency and harmony.

The same test provided insight into the preferred key widths. Senior participants favored wider keys, ranging from 12 to 15 mm, while younger participants preferred widths between 9 and 11 mm. Both groups emphasized the importance of adequate spacing between keys. Based on this feedback, a balance between key size and spacing was determined to maintain usability and aesthetic appeal. The final key dimensions were set to 10 mm in width with a 3 mm spacing between keys.

After determining the shape and width, a brief test was conducted to evaluate the optimal length-to-width ratio. Three variants were 3D printed, each differing in the length of the straight section between the rounded ends, ranging between 1 mm, 2 mm, and 3 mm, see Figure **37**. These prototypes were compared and the 2 mm variant was selected as the most visually and ergonomically balanced.

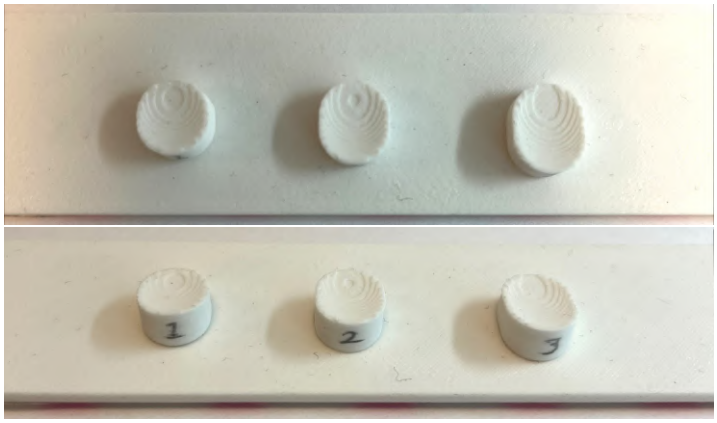


Figure 37: Width to length ratio comparison of keys

Following the initial evaluations and studies, further concept development focused on the design and placement of the special function keys. In line with the previously established 3x3+1 oval key layout, the design of the special function keys was developed to visually integrate with the numerical keys while remaining distinct. Initial concepts were rapidly sketched, followed by digital renderings for comparison, see Figure **38**.

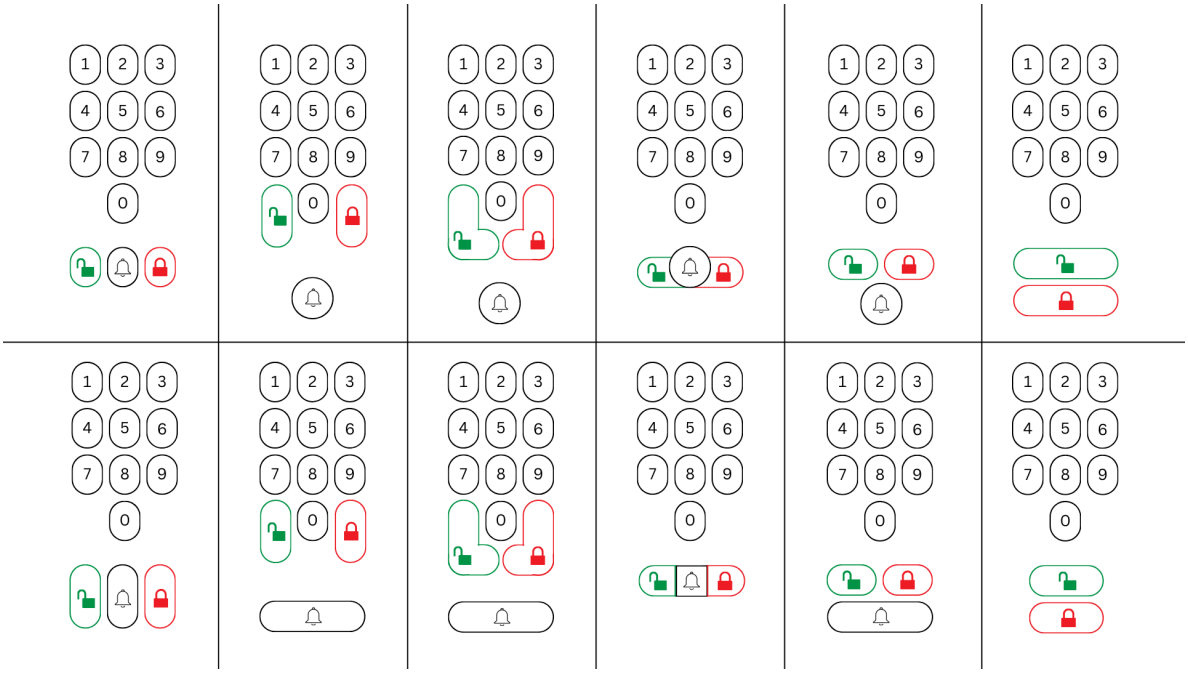


Figure 38: Special function keys ideation

The oval key shape was maintained for these special keys to preserve a coherent visual language throughout the product. However, modifications were made to ensure that they were distinguishable. The results of the layout comparison test (Chapter 3.4) and the symbol preference study (Chapter 5.4) indicated a strong user preference for a combination of color and symbols to enhance intuitiveness, therefore, the designs incorporated red and green coloring, along with padlock icons.

6 Deliver

This chapter presents the project's outcomes by translating research into tangible results that directly address the three research questions. These outcomes are designed to identify common obstacles users face when interacting with keypads (RQ1), highlight key factors when designing for inclusive use (RQ2), and demonstrate how inclusive design recommendations can be applied to existing products (RQ3).

6.1 Handbook

To support inclusive and user-friendly keypad design, a handbook of recommendations, found in Appendix K, has been compiled. These are grounded in the theory and data from Chapters 4 and 5, and organized into three focus areas: *Visibility*, *Clarity*, and *Tactility*. The handbook is intended for designers and developers working with physical keypads at the company, with the aim of creating inclusive and user-centered solutions. It serves as a visual and practical complement to the report, offering guidance supported by best-practice examples to assist future design decisions. A page from the handbook is shown in Figure **39**.



Figure 39: Excerpt from handbook

While not all suggestions suit every context, they aim to guide designers toward more intuitive interactions. When full implementation isn't possible, thoughtful adaptation with empathy for diverse users is encouraged.

6.1.1 Visibility

To improve the overall usability of a keypad, visual elements must be easy to see at a glance. Effective visibility reduces confusions, supports faster interaction and accommodates a wider range of users and environments. Key considerations are summarized in Table **8**.

Table 8: Design recommendations for enhancing keypad visibility.

Recommendation	**Reasoning**
Ensure high contrast between symbols and key surfaces, particularly in brightly lit environments.	Low contrast made it difficult for users to locate and identify keys, leading to confusion and frustration.
Visually separate keys from the background using contrasting colors, surface finishes, or perimeter lighting.	Enhances key recognition and reduces cognitive load by improving visual distinction.
Integrate backlit keys.	Improves usability in low-light conditions and enhances overall visibility.
Position symbols on, or near, the front-facing surface of keys.	Symbols placed on the rear of the key were only visible from specific angles, limiting readability.
Utilize the full surface area of each key for symbol and number sizing.	Small or thin symbols lacked clarity, larger symbols improved readability.
Angle the keypad or product at a minimum of 10° toward the user when mounted at standard height (1 m).	Improves readability from a typical standing position by aligning the viewing angle with the user's line of sight.

6.1.2 Clarity

A clear and intuitive keypad layout helps users interact without confusions. By minimizing cognitive load and supporting quick recognition, the design can reduce errors and create a smoother experience. Recommendations to improve clarity are listed in Table **9**.

Table 9: Design recommendations for enhancing clarity of use.

Recommendation	**Reasoning**
Use a standard 3x3+1 keypad layout and avoid alternative configurations.	Familiar layouts support spatial memory and reduce cognitive load, enabling faster and more intuitive interaction.
Clearly differentiate special function keys from numeric keys using size, shape, color, and/or placement.	Visual and tactile distinction minimizes input errors and improves recognition of key functions.
Combine color (e.g., red and green) with intuitive, universally recognizable symbols for lock/unlock functions.	Supports users with visual impairments and color blindness, enhancing clarity and trust.
Place symbols and numbers directly on the key, avoiding placements beside or above.	Off-key placement can cause confusion and delays, especially under time pressure or in high-stress situations.
Avoid unnecessary symbols or background patterns.	Reducing visual clutter helps lower the cognitive load and minimizes distractions during interaction.
Provide tactile feedback, such as a mechanical click or vibration, upon key press.	Immediate physical feedback increases user confidence and satisfaction.
Implement auditory feedback with distinct tones for successful and unsuccessful inputs.	Reinforces correct interaction and supports users with limited vision or in low-light environments.
Use visual feedback cues (e.g., indicator lights) to confirm input status.	Visual confirmation improves clarity, reduces uncertainty, and builds user trust.

6.1.3 Tactility

Tactile features guide the user through the physical interactions, supporting precise input even without visual confirmation. Well considered key shape, spacing and feedback enhance usability and comfort. Key recommendations are presented in Table **10**.

Table 10: Design recommendations to enhance the physical interaction.

Recommendation	**Reasoning**
Place tactile markings clearly in the center of the “5” key.	This widely recognized feature helps users orient themselves on the keypad, especially for visually impaired users.
Use tactile markings to differentiate special function keys from each other.	Tactile distinctions help users identify and distinguish between functions by touch alone, reducing input errors and improving accessibility, especially for users with visual impairments or in low-visibility conditions.
Ensure keys are large enough to accommodate a fingertip without slipping, minimum width of 10 mm is recommended.	Small keys were harder to distinguish, required fine motor control, and were difficult to use with gloves, reducing accuracy and accessibility.
Maintain a minimum spacing of 2.5 mm between keys to balance keypad width and product size.	Adequate spacing improved tactile separation, reduced input errors, and enhanced user confidence.
Use slightly concave key surfaces with approximately 1 mm depth.	Concave shapes guided finger placement, improved grip, and enhanced control during input.
Provide a distinct tactile contrast between keys and the surrounding surface.	Differentiating textures helped users locate keys by touch, improving usability without visual reliance.
Design keys with well-defined edges and avoid overly rounded corners.	Clear key boundaries improved tactile distinction and input precision.
Ensure keys protrude sufficiently to support both tactile discovery and input confirmation, approximately 2 mm is recommended.	Keys with minimal protrusion lacked clear tactile feedback and were harder to feel, while overly tall keys reduced precision and felt less refined.

6.2 Application of Recommendations

Based on the design recommendations established in Chapter 6.1, improvements were applied to the reference product in two distinct design concepts. Both concepts follow to the identified recommendations while enhancing overall usability, each exploring two different approaches to implementing the same set of user-centered design principles.

6.2.1 Concept 1

The first concept, incorporating the design recommendations into the reference product, is illustrated in Figure **40** and **41**. While the main body of the product remains unchanged, the new key layout and updated key design have been integrated. The keys are angled at 10° towards the user, with a key width of 10 mm and 3 mm spacing between them. This angled configuration significantly enhances visibility without compromising the integrity of the original design.



Figure 40: Rendering of concept 1



Figure 41: Rendering of concept 1, angled

The key symbols have been increased to a height of 6 mm, representing a 150% enlargement compared to the existing product. The color scheme features white symbols on black keys, enhancing legibility. To further improve contrast and visibility, each key is illuminated by LEDs: white LEDs for the numeric and doorbell keys, and green and red LEDs for the unlock and lock keys, respectively. This lighting creates a glowing outline around each key (see Figure **42**), making them more visually identifiable, especially in low-light conditions.

For improved tactile feedback and orientation, the keys have a concave surface and follow the curvature of the product's surface. This curvature not only provides ergonomic comfort but also enhances usability.



Figure 42: Special function keys

To enhance both visual and tactile differentiation, the keys feature a glossy finish while the background surface is matte with a slight texture. This contrast helps users easily distinguish keys by touch, particularly in dim environments. The lock and unlock keys, which have the same size and shape, also feature tactile markings in the form of a filled and unfilled circle to help differentiate them from each other by touch.

6.2.2 Concept 2

The second concept aims to enhance usability by redesigning the body of the device, not just the interface. In this concept the keypad face is angled 10° towards the user, resulting in a thicker body as seen in Figure **43**. From the front, the product keeps its familiar silhouette, maintaining brand recognition and user familiarity. The keys remain flat and parallel to the front surface, preserving a clean and straightforward interface.

This concept involves a more structural redesign of the body, allowing for additional internal space, which could enable further functional improvements. The keys follow the same design as in the first concept, with consistent size, spacing, and layout. They feature white backlit symbols on a black background, with special function keys offering visual confirmation through both color and clear symbols. The choice of textures and materials remains unchanged, keeping the visual and tactile consistency across both designs.



Figure 43: Rendering of concept 2

6.2.3 Physical Prototypes

To evaluate the visual and tactile qualities of the two design concepts, full-scale physical prototypes were developed. Both concepts were constructed using the same fabrication method to ensure consistency in form, scale, and surface finish

Each prototype was first modeled in CAD and consists of three main components: the backside, the body, and the front face, along with a handle. These parts were 3D-printed using fused deposition modeling (FDM), then manually filed and spray-painted to achieve a more polished appearance. Once finished, the components were assembled to form the bodies of the two concepts. Figures **44** and **45** illustrate the prototypes before and after painting and assembly.



Figure 44: 3D-printed model parts

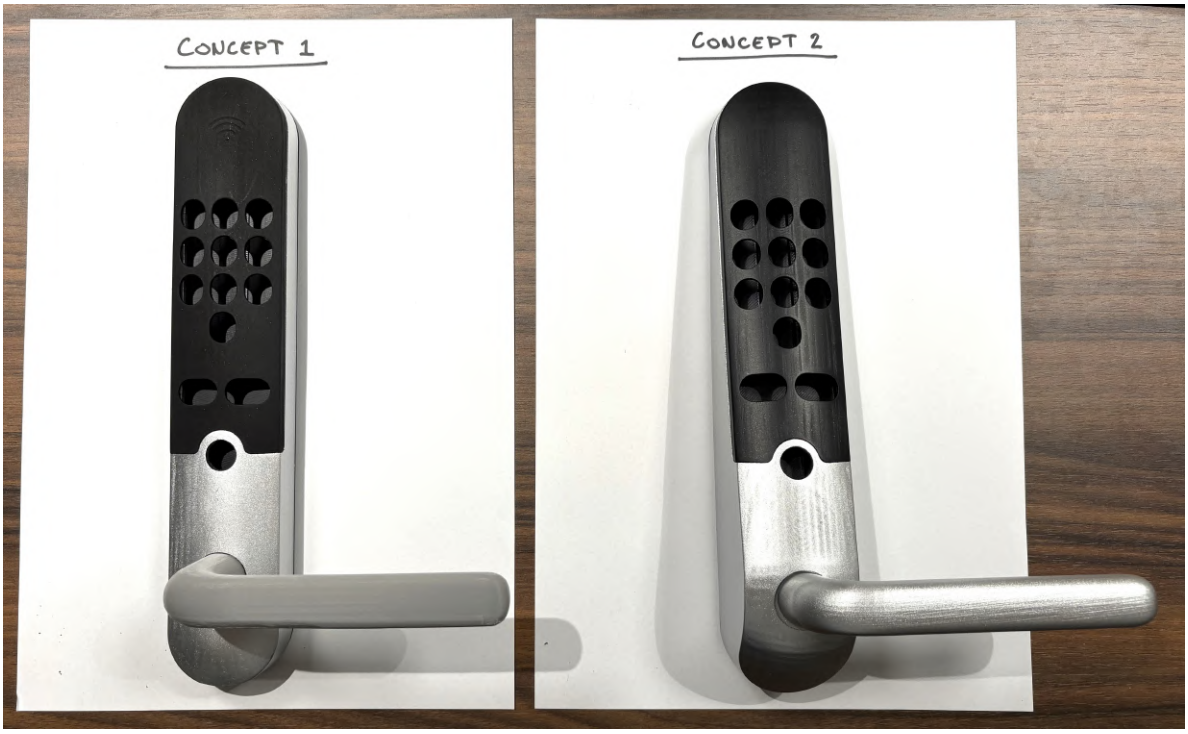


Figure 45: Assembled bodies of the concepts

The keys were designed as two-part components: an inner core featuring the number or symbol, and a hollow outer shell with a concave surface and tactile markings. These parts were printed in

clear resin using Stereolithography (SLA) to ensure high precision and surface quality. The inner cores were hand-painted with black and white acrylic paint to enhance contrast between the symbol and background. The outer shells were coated with a layer of UV-cured polish to create a glossy finish, improve transparency, and smooth out surface irregularities. The parts were then assembled by applying clear polish to the base of the outer shell, inserting the painted core, and curing the parts under UV light to bond them together. The different stages of the key fabrication process are shown in Figure **46**.

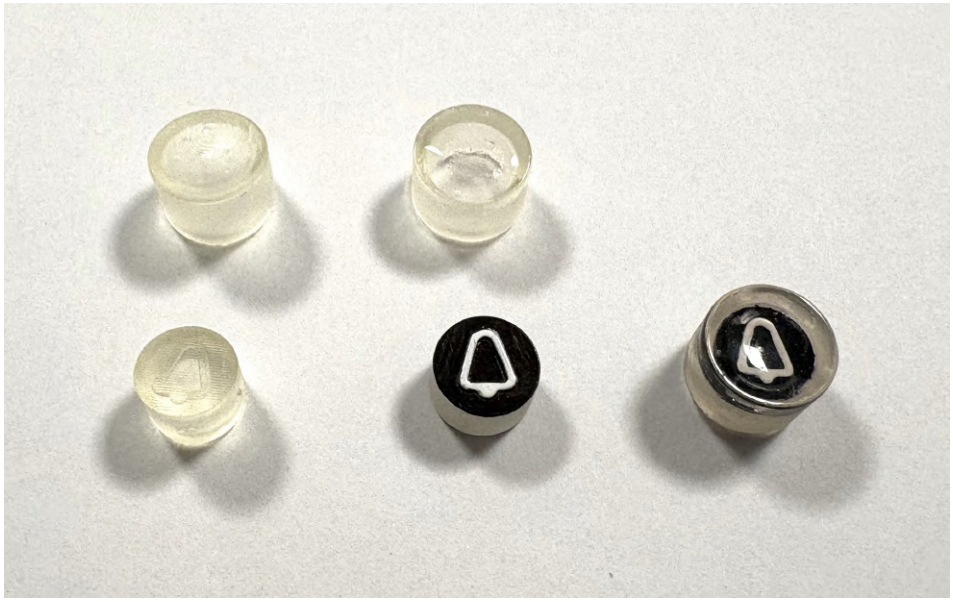


Figure 46: Key parts during process

To simulate realistic interaction, mechanical key switches were attached to the keys before mounting them onto the prototypes. This allowed the keys to be pressed, providing both tactile and auditory feedback.

The physical models were then mounted next to the reference product for comparison, see Figure **47** and Figure **48**.



Figure 47: Concept models next to reference product, front view



Figure 48: Concept models next to reference product, angled view

Overall, the prototypes successfully embody the intended design improvements, particularly in terms of visibility and clarity. The increased symbol size, thickness, new placement, and angle contribute to improved readability, especially when viewed from an angle. Additionally, the redesigned lock and unlock keys stand out more clearly than on the reference product, due to their enhanced size, separate positioning, and use of color, which may support more intuitive interaction.

7 Discussion

In this chapter, the results, methods, and proposed directions for future work are discussed. The aim is to reflect on the choices made throughout the project, evaluate their impact on the outcomes, and identify areas for improvement.

7.1 Results

This section discusses the findings in relation to the research question, previous research, and the purpose of the study.

RQ1: What common obstacles or limitations do users face when interacting with keypads?
The most frequently reported obstacles were related to visibility, feedback, and physical attributes such as key size ans spacing. These findings are consistent with the study by Muraki et al. (2010), which showed that elderly users are more likely to press the wrong keys on mobile phones of similar size to the reference product, compared to younger users. Contributing factors included reduced dexterity and larger finger sizes. In the usability tests (Chapters 3.2.1 and 3.3), participants struggled with poor contrast, lack of feedback, and small keys, especially when wearing gloves or simulating reduced dexterity. Participants also reported confusion when symbols were not clearly visible or intuitively placed.

These results suggest that many usability issues stem from a mismatch between user expectations and the keypad's visual and physical attributes.

RQ2: What key factors need to be considered when designing keypads for inclusive use?

The results from both the literature and the user studies highlight that inclusive keypad design must account for a wide range of user capabilities. Theories presented by Persad et al. (2007) and Keates & Clarkson (2004) emphasize the importance of designing for sensory, cognitive, and motor diversity. This was confirmed in the user testing and interviews, where participants with reduced vision or motor capabilities expressed clear preferences for tactile markings, high contrast, and intuitive layouts.

The preference for a 3x3+1 layout for numbered keys, found in both the layout comparison test (Chapter 3.4) and in the literature (Goodman et al., 1983), reinforces the importance of familiarity in reducing cognitive load. Similarly, the ISO/IEC Guide 71 and EN 301 549 standards presented in Chapter 3.1.4, which advocate for recognizability, visibility, and operability, were reflected in user feedback that emphasized the need for clear symbols, visible symbol placement, and tactile feedback.

These results show that inclusive design is not only about physical accessibility features, but also about reducing product complexity and enhancing intuitiveness for all users.

RQ3: How can inclusive design be applied to existing products to improve keypad usability?

The concept prototype tests (Chapter 5.2) showed that oval-shaped keys were preferred by both older and younger participants, as they aligned with the shape of the fingertip. However, preferences for key size and spacing varied between the two test groups: older users generally preferred larger keys (12-15 mm) and spacing between 2.5-4 mm, while younger users preferred smaller keys (9-11 mm) with a spacing of 2.5 mm. This variation highlights the challenge of designing for a broad user base.It reflects a dilemma in inclusive design; how to balance the goal of designing for everyone with the reality that no single solution can fully meet the needs of all users. As a result, compromises are often required between general solutions and those tailored to specific user groups. For the presented concepts, a compromise with a 10 mm key width and 3 mm spacing was selected, balancing usability and visual expression.

The angle exploration (Chapter 5.3) showed that the tilt of the keypad significantly improved the visibility and readability of numbers, especially for taller users. Although this test was conducted with only one participant, it suggests that keypad angle could be an important factor in inclusive design and warrants further investigation with a more diverse test group.

Finally, the use of color-coded padlock symbols (Chapters 3.4 and 5.4) was found to reduce cognitive load and increase user confidence. The majority of participants strongly associated these symbols with locking and unlocking, supporting the use of familiar visual cues in inclusive design.

7.2 Method Discussion

This section discusses the methods used during the project, along with other relevant factors that may have influenced the results.

7.2.1 Participants

One of the most significant challenges encountered during this project was recruiting participants with a diverse range of disabilities and capabilities for product testing and interviews. Despite outreach efforts to various organizations and advocacy groups, responses were limited. Consequently, the representation of disabilities in this study was minimal, comprising one participant with visual impairment and another with severe motor limitations. While their contributions provided invaluable insights and highlighted critical aspects of inclusive design, a broader and more varied participant base would have enabled

comparisons and more reliable findings.

The majority of participants across all tests, studies, and interviews were able-bodied. This lack of diversity likely influenced the outcomes, as much of the data reflects perspectives that do not encompass the full spectrum of user needs. Although we made conscious efforts to consider how designs might be experienced by individuals with different abilities, such assumptions cannot replace direct input. Truly inclusive design requires a deep understanding of the lived experiences of people with disabilities.

Later in the project, we established contact with a group of seniors who engaged with us in testing and discussing 3D-printed prototypes. Their feedback was highly constructive, particularly in relation to age-related cognitive and physical changes. Had this group been involved earlier, especially during the exploration of trigger materials, it might have opened new areas for investigation and design iteration.

Notably absent from the study were participants with cognitive disabilities. This gap resulted from either a lack of response or concerns from caregivers regarding potential exploitation. As a result, the design recommendations could not be evaluated for their suitability for individuals with cognitive impairments, raising questions about their inclusivity.

Future research should prioritize the inclusion of individuals with cognitive disabilities, ensuring that ethical considerations are central to the recruitment and participation process. Special care must be taken when working with individuals who may not be fully autonomous, to avoid any perception or risk of exploitation. Even if some individuals in this group may not directly use the tested products, it is essential that everyone can understand and operate basic functions, such as opening doors, especially in emergency situations.

Although this study did not include participants with diagnosed cognitive disabilities, it did address some cognitive challenges indirectly. For instance, older adults often experience reduced working memory and slower processing speeds, which can lead to hesitation and increased input errors.

7.2.2 Trigger Material Tests

During the evaluation of the products used on the trigger board (Chapter 3.3), the devices were evaluated and compared without being connected to electricity. All of the products in question are designed to operate with power, with some using LEDs and auditory feedback to guide user interaction. By excluding these elements, the focus of the test shifted entirely to the physical and tactile aspects of the keypads, such as key size and shape, spacing, surface texture, and mechanical feedback.

This approach allowed for a more isolated analysis of form and ergonomics as well as giving the user

room to express what kind of feedback they wished to receive, which proved valuable. However, it also introduced limitations. Several of the devices rely heavily on visual or auditory cues to communicate status or confirm input. Without these signals, participants were unable to experience the full interaction cycle, which may have influenced their perception of usability.

Additionally, since the keypads were not connected to any functional system, mispresses could not be registered or measured. As a result, the evaluation was limited to participants' own impressions and their confidence in having pressed the correct keys. This means that the results reflect perceived usability rather than actual input accuracy.

Another constraint was the inability to wall-mount the trigger board during testing. Since these devices are typically installed vertically at a fixed height, testing them on a flat surface may have altered how participants interacted with them, particularly in terms of reach, angle, and hand positioning. Despite this, efforts were made to simulate realistic use as closely as possible within the constraints of the test environment, by either holding it vertically or leaning it against something.

In future iterations, incorporating both visual and auditory feedback would provide a more comprehensive and realistic testing experience. This would be especially important for evaluating how users with different sensory capabilities interpret and respond to multiple feedback cues. Additionally, enabling wall-mounted testing setups would help better replicate real-world conditions and improve the validity of the results.

7.2.3 Literature

The literature collected during the desktop research largely aligned with the findings from the user tests, reinforcing the validity of the proposed design recommendations. This consistency strengthens the credibility of the results and suggests that the principles found in existing research are also reflected in users' real-world experiences of inclusive design. For instance, key insights such as the importance of high visual contrast and the inclusion of tactile feedback, discussed in Chapters 3.1.4, 3.1.5, and 3.1.6, were repeated by participants during the trigger material interviews (Chapter 3.3).

While this alignment is encouraging, it also raises a potential concern, the risk of confirmation bias. The strong agreement between literature and empirical findings could be interpreted as a sign that only sources supporting the intended design direction were selected. However, it is important to emphasize that the literature review was conducted with the aim of being as comprehensive and objective as possible. A wide range of sources was considered, and the selection process was guided by relevance, credibility, and applicability to the project's goals, not by whether the findings aligned with preconceived ideas.

That said, it would have been preferable to identify literature that contradicted or challenged the project's assumptions. Such contrasting perspectives could have led to more exploratory testing and a broader understanding of inclusive design, rather than primarily confirming existing beliefs. There is, of course, a possibility that relevant contradictory literature was overlooked, either due to limitations in search scope or availability. Being mindful of this possibility is important, as it highlights the need for continuous critical engagement with both literature and empirical data.

7.2.4 Validation Tests

A final validation test with a prototype with the recommendations applied would have provided valuable insights into the effectiveness of the proposed design recommendations. Unfortunately, due to time constraints, it was not possible to develop and carry out a test that incorporated the full set of recommendations in a realistic context. Ideally, such a test would involve a high-fidelity prototype equipped with both visual and auditory feedback, allowing for a more accurate simulation of real-world interaction and a more comprehensive assessment of the overall user experience.

The absence of this step is a limitation, as it leaves some uncertainty about how well the recommendations translate into practice. A validation phase would have enabled the identification of unexpected usability issues, as well as the opportunity to refine the design based on direct user feedback. It would also have strengthened the conclusions of the project by demonstrating how the proposed solutions perform under more realistic conditions.

However, the groundwork laid in this project provides a foundation for future validation efforts. The design recommendations are based on both literature and user input, and the structure of the project lends itself well to continued development. Future work could build on this by developing a fully functional prototype and conducting user testing with a more diverse participant group, ideally including individuals with a range of physical and cognitive abilities.

In retrospect, allocating more time for a validation phase would have added significant value to the project. However, the insights gathered throughout the process still contribute to the field of inclusive design.

7.2.5 Use of Generative AI

Generative AI was used to refine and enhance the readability and flow of this report. The programs used are Microsoft Copilot, ChatGPT, and Writefull, the AI writing assistant in Overleaf. For Writefull no prompts have been used, it flags words and sentences automatically and suggests changes to improve

the text. When using Copilot or ChatGPT, sections of the written text were pasted into the programs with variations of the following prompts:

- Is this language and tone suitable for a master thesis report?
- How can this text be refined to fit a master thesis report?
- How can the readability of this text be improved?
- How can this be phrased in a more clear and concise way?

The suggestions made by the AI programs were then taken into consideration and applied or disregarded. These tools have had an impact on the readability and overall flow of the text. The AI tools helped make the written text clearer and easier to understand by removing redundant words, rephrasing, spell checking, as well as keeping a consistent tone.

7.3 Delivery

This section reflects on the final outcomes of the project and evaluates how well they align with the goals and insights established during the earlier phases, particularly the Define phase.

7.3.1 Handbook

The handbook was developed to refine the most relevant findings into actionable and accessible design guidelines. Its primary value lies in its clarity and usability. Internal goals and feedback from the company emphasized the need for a tool that could effectively communicate inclusive design principles across development teams. To support this, the handbook uses visual examples and concise summaries rather than theoretical discussions, focusing on recommendations grounded in the results from this project.

Overall, the handbook aligns well with the project's objectives and reflects the empirical findings from both the Discover and Develop phases. Structuring the content around the themes of Visibility, Clarity, and Tactility. These themes emerged directly from user feedback—particularly through usability testing, observations, and prototype evaluations. For example, visibility challenges were consistently raised in both the reference product test (Chapter 3.2.1) and the trigger material evaluations (Chapter 3.3), where users struggled to locate and distinguish keys. Similarly, preferences for tactile features, such as concave surfaces and clearly separated function keys, were identified early in the process and carried through to the final recommendations.

That said, the final form of the handbook inevitably simplifies and generalize inclusive design. While it captures core insights, it cannot account for every nuanced scenario. Preferences around key shape,

size, and spacing varied not only between user groups but also within them, for instance, between different age groups in the concept prototype test (Chapter 5.2). Additionally, the applicability of the recommendations depends on the specific product context, including existing design guidelines, branding constraints, and physical limitations.

For this reason, the handbook is not intended to serve as a rigid set of rules, but rather as a flexible foundation for context-specific adaptation. This is also emphasized within the handbook itself. It provides direction while encouraging further iteration and refinement based on the target audience and use case.

7.3.2 Concepts

The concepts developed during the project aimed to illustrate the design recommendations in a tangible form. The purpose of presenting two design concepts was to demonstrate the flexibility of the proposed design recommendations and to explore how they could be implemented at different levels of intervention. Although both concepts share most design changes, the key difference lies in the degree of modification. Concept 1 keeps the original product housing and integrates the new keypad design with minimal structural changes, while Concept 2 involves a more extensive redesign of the product body to improve the viewing angle and internal space. Presenting both concepts demonstrates how inclusive design recommendations can be implemented within existing product constraints as well as through more extensive modifications to the overall form.

Throughout this process, intentional trade-offs were made to balance functionality, usability, and the product's visual identity. The recommended keypad layout followed a 3x3+1 configuration, with special function keys clearly distinguished by size, shape, and color, along with tactile markings. These recommendations were implemented without compromise, as they aligned well with both functional requirements and user preferences identified during testing.

Key spacing was another area where user feedback directly influenced the design. Testing from the concept prototypes (Chapter 5.2) indicated a preferred spacing range of 2.5 mm to 4 mm. A spacing of 3 mm was ultimately chosen to balance usability with visual harmony. Similarly, a key width of 10 mm was selected to ensure ease of use without compromising the overall design proportions. While larger keys might have further improved accessibility, they would have disrupted the product's visual balance, highlighting the need to weigh inclusive design goals against aesthetic considerations.

A similar trade-off was also made regarding symbol size. While the design recommendations (Chapter 4.1.11) advocate for maximizing symbol size to improve readability, especially for users with visual impairments, symbols that filled the entire key surface did not align with the desired visual expression.

Instead, the symbol size was significantly increased from previous versions, striking a balance between improved usability and visual consistency. This compromise was seen as a successful integration of inclusive design principles within the constraints of product identity.

In addition to these design considerations, the physical prototypes provided an opportunity to assess how the concepts perform in practice. The physical prototypes clearly demonstrate improved visibility and readability compared to the reference product, without making too drastic changes to the overall appearance. Clarity also appears to be enhanced, through more distinguishable special function keys and the use of color. This suggests that inclusive design enhancements can be implemented in a subtle yet effective manner. However, while the prototypes shows indications of improvement, they have not yet been evaluated by a user group. A comparative user study involving the reference product would be necessary to validate whether the concepts truly enhance usability in practice, and to identify areas where the concepts themselves could be further refined.

8 Conclusion

This chapter summarizes the key outcomes of the project and reflects on how the research questions have been addressed through the design process. It also discusses the findings for inclusive product development, and outlines potential directions for future research and implementation.

8.1 Summary of Key Findings

The purpose of this thesis was to explore how keypads can be designed to be more inclusive and accessible to a diverse user group. The study addressed three research questions related to design considerations, user challenges, and inclusive solutions.

The findings show that inclusive keypad design must account for a wide range of sensory, cognitive, and motor capabilities. Key aspects include intuitive layouts, tactile and visual clarity, and clearly distinguishable special function keys. The study identified common obstacles such as low visibility, small key sizes, and insufficient feedback, which often led to uncertainty and errors. Furthermore, the results show that keypads can be adapted to accommodate a wide range of users through ergonomic key shapes, balanced sizing, familiar symbols, and thoughtful placement of keys and icons. These design choices enhance accessibility while maintaining usability for the general population, demonstrating that inclusive design benefits all users without significantly deviating from the current mainstream design.

8.2 Practical Contribution

The project resulted in two main deliverables, a handbook with design recommendations, and two conceptual designs where the recommendations have been applied. The handbook compiles the most relevant findings into actionable recommendations, structured around the themes of *Visibility*, *Clarity*, and *Tactility*. It is intended to support designers and developers working on physical interfaces with keypads.

The two concepts demonstrate how the recommendations can be applied in practice. Concept 1 integrates the new keypad design into the existing product housing, while Concept 2 explores a more extensive redesign of the product body. While the two concepts illustrate how inclusive design recommendations can be applied at different levels of intervention, their effectiveness has not yet been validated through user testing. Initial observations suggest improved visibility and clarity. However, further evaluation is needed to confirm whether these improvements translate into better usability in practice.

8.3 Theoretical Contribution

This study reinforces existing theories in inclusive design by validating them through empirical tests. The results highlight the importance of making functions easy to understand through design, ensuring that users can intuitively recognize how to interact with the product. They also emphasize the value of providing feedback through multiple senses, such as touch, sight, and sound, to confirm user actions. By combining literature with user-centered design methods, the project contributes new insights into how inclusive design principles can be applied to physical keypad interfaces. The findings also highlight the need to balance standardized design practices with individual user preferences, especially when designing for diverse capabilities.

8.4 Future Work

Future work should focus on validating the proposed design recommendations through the development of high-fidelity prototypes and testing them in real-world contexts under varied conditions and with a diverse test group. Although the current prototypes suggest improved visibility and clarity, impressions remain speculative without user evaluation. Comparative testing with the reference product would be essential to determine whether the concepts offer measurable improvements in usability and to identify areas for further refinement.

In addition to refining the current keypad-based solutions, future research could explore alternative access methods that reduce or eliminate the need for physical interaction. Technologies such as biometric authentication, RFID tags, and other low- or no-touch systems are optimal for enhancing accessibility. These approaches can significantly improve usability, particularly for individuals with limited motor capabilities, visual impairments, or cognitive challenges.

Throughout the project, both test participants and interviewees expressed a clear preference for smoother, more intuitive interactions with fewer required steps. This feedback highlights the importance of designing systems that minimize cognitive and physical effort. Offering multiple access options, rather than relying on a keypad, aligns with inclusive design principles and better accommodates the diverse needs of users.

Additionally, future iterations of this work would benefit from a more iterative design and testing process, allowing for continuous refinement based on user feedback. Incorporating emerging technologies and maintaining a user-centered approach will be key to developing solutions that are not only functional but also suitable for a wide range of users.

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A Interview Guide, Customer Support

Intervjuguide: Customer Support (Original version in Swedish)

Introduktion
Vi introducerar oss, och syftet med intervjun
Kolla om det är okej att vi spelar in intervjun? Den kommer inte att delas, utan endast användas för att transkribera samtalet.
Fråga honom om hans roll som customer support manager, hur deras customer support ser ut.

Användarupplevelse

1. Vad skulle du säga är de vanligaste frågorna/meddelandena till supporten? Finns det frågor som förekommer särskilt ofta? Vilka produkter förekommer oftast?
2. Får ni någonsin feedback om hur kunder upplever användandet av produkten? Och i så fall, vad brukar den vara? Vilken typ av feedback får ni?
3. Är det något som ofta upplevs oklart / inte intuitivt som ni får feedback på?
4. Har ni fått någon feedback från kunder som berört designen/förslag på hur de önskat att den var utformad?
5. Har det hänt att ni fått feedback som berör knappsatsen eller interaktionen med den? Vet du vad det rört sig om?
6. Finns det någon data/rapportering vi hade kunnat ta del av som rör kundernas upplevelse vid användning/interaktion?

Tillgänglighet

1. Vet du hur användargruppen ser ut när det gäller mångfald (ålder, funktionsvariationer, språk)? Finns det en typisk användare för produkten? Vet du vilka områden produkten förekommer som mest i?
2. Erbjuder produkten tillgänglighetsanpassningar så som inställningar för ljus och ljudvolym? Om ja/nej, vet du om det kommit in feedback eller frågor kring detta?
3. Baserat på de kundupplevelser du fått ta del av, vad ser du för potentiella förbättrings/förändringsområden för produkten?

Tack så mycket för att du tog dig tid att prata med oss!

Interview Guide: Customer Support (Translated)

Introduction
We introduce ourselves and explain the purpose of the interview.
Check if it's okay that we record the interview. It will not be shared, only used to transcribe the conversation.
Ask him about his role as a Customer Support Manager and what their customer support looks like.

User Experience

1. What would you say are the most common questions/messages to support? Are there any questions that occur particularly often? Which products come up most often?
2. Do you ever receive feedback about how customers experience using the product? If so, what is that feedback typically like? What type of feedback do you receive?
3. Is there anything that is often perceived as unclear/not intuitive that you receive feedback about?
4. Have you received any feedback from customers regarding the design/suggestions on how they would have liked it to be designed?
5. Has it happened that you've received feedback related to the keypad or the interaction with it? Do you know what that was about?
6. Is there any data/reporting we could access regarding customers' experience with use/interaction?

Accessibility

1. Do you know what the user group looks like in terms of diversity (age, disabilities, language)? Is there a typical user for the product? Do you know which areas the product is most common in?
2. Does the product offer accessibility adaptations such as light and sound volume settings? If yes/no, do you know if there has been feedback or questions about this?
3. Based on the customer experiences you've been exposed to, what potential areas for improvement or changes do you see for the product?

Thank you very much for taking the time to talk with us!

B Observation and Interview Guide, Trigger Material

Trigger Material: Observations- och intervjuguide (Original version in Swedish)

Denna guide används som ett hjälpmedel vid intervjuerna, målet är att få ett flytande samtal med den intevjuade. De listade underfrågorna ställs endast om de ej blivit besvarade naturligt i samtalet.

Syfte:
Syftet med denna intervju är att samla insikter om hur deltagarna interagerar med knappsatserna, samt deras upplevelser och preferenser.

Introduktion:
Vi heter Johanna och Linnéa och går vårt sista år på KTH där vi studerar design och produktframtagning. Vi skriver för närvarande vår masteruppsats på XX, vårt uppdrag är att utvärdera och utveckla designrekommendationer för hur knappsatser kan utformas på ett mer inkluderande sätt, med fokus på användarupplevelsen.

De data och svar vi samlar in från denna intervju kommer att användas i vår rapport, som kommer att publiceras. Vi kommer inte att dela ditt namn och du kommer att vara helt anonym. Samtycker du till detta? När vi skrivit ett stycke där vi användt åsikter och/eller citat du bidragit med kommer du att få läsa texten och ha möjlighet att göra ändringar om du vill.

Innan vi börjar, skulle det vara okej om vi spelade in denna intervju? Inspelningen kommer inte att delas med någon och kommer endast att användas av oss för att korrekt komma ihåg vad som sades.

Instruktion för deltagaren:
Vänligen lås upp dörren genom att använda de angivna koderna. Ta din tid och ta en liten paus mellan varje lås, vi kommer eventuellt ställa någon fråga.

"Om dessa knappsatser var kopplade till ett dörrlås, hur skulle du låsa upp dörren?"

Observationer:
- Hur anger deltagaren koden? (fingerval, teknik)
- Trycker deltagaren på rätt knappar?
- Finns det några specifika knappar som verkar svårare att trycka på än andra?
- Hur säker verkar deltagaren under interaktionen?
- Ändrar deltagaren sin position för att bättre komma åt eller se knappsatsen?
- Använder deltagaren några hjälpmedel för att interagera med knappsatsen?

Uppföljningsfrågor:
1) Vilken knappsats tyckte du var lättast att hantera?
    a) Vad gjorde att den kändes lätt att använda?
    b) På vilket sätt kändes den lättare än de andra?
    c) Vad tyckte du om knappsatsens storlek och layout? Vänligen utveckla.
    d) Var det någon knapp som var svår att hitta? Vilken?
    e) Var det något du inte gillade med knappsatsen? Vad?
    f) Om du kunde ändra något för att göra den ännu lättare att använda, vad skulle det vara och varför?

2) Vilken knappsats tyckte du var svårast att hantera?
    a) Vad gjorde att den kändes svår att använda?
    b) På vilket sätt kändes den svårare än de andra?
    c) Vad tycker du om knappsatsens storlek och layout? Vänligen utveckla.
    d) Var det någon knapp som var extra svår att hitta? Vilken?
    e) Var det något du gillade med knappsatsen? Vad?
    f) Om du kunde ändra något för att göra den ännu lättare att använda, vad skulle det vara och varför?

3) Vilka knappar var mest behagliga att trycka på?
    a) Hur skulle du beskriva känslan av att trycka på knapparna? (mjuka, hårda, klickiga)
    b) På vilka sätt var de mer behagliga än de andra? (storlek, form, konkav/konvex)
    c) Fick du någon feedback från knappen? Vilken typ? Bidrog det till dess känsla och ditt val?
    d) Om du kunde ändra något med knapparna för att göra dem ännu mer behagliga, vad skulle det vara och varför?

4) Vilka knappar var minst behagliga att trycka på?
    a) Hur skulle du beskriva känslan av att trycka på knapparna? (mjuka, hårda, klickiga)
    b) På vilka sätt var de mindre behagliga än de andra? (storlek, form, konkav/konvex)
    c) Stötte du på några specifika problem med dessa knappar?
    d) Fick du någon feedback från knappen? Vilken typ? Bidrog det till dess känsla och ditt val?
    e) Om du kunde ändra något med knapparna för att göra dem ännu mer behagliga, vad skulle det vara och varför?

5) Finns det något du vill tillägga, som du tror kan vara relevant för oss? Det kan vara vad som helst!

Tack så mycket för att du deltog och hjälpte oss, varje input är värdefull.

Trigger Material: Observation & interview guide (Translated)

Purpose:
The purpose of this interview is to evaluate how participants interact with different keypads when entering a code on different keypads. The observations and follow-up questions aim to gather insights into the usability and ergonomics of the keypads.

Introduction:
Our names are Johanna and Linnéa and we are in our last year of KTH where we are studying design and product realization. We are currently writing our master thesis at XX, our mission is to evaluate and develop design recommendations on how keypads can be designed in a more inclusive way, focusing on user interaction and experience.

The data and answers we collect from this interview will be used in our master thesis, which will be published. We will not share your name or any personal information. Do you consent to this?

Before we start, would it be okay if we recorded this interview? The recording will not be shared with anyone and will only be used by us to accurately remember what was said.

Instruction for the participant:
Please unlock the door by using the codes provided. Take your time and take a small break in between the keypads, we might ask some questions.

“If these keypads were connected to a door lock, how would you unlock the door?”

Observations:
- How does the participant enter the code? (finger choice, technique)
- Does the participant press the right buttons?
- Are there any specific buttons that seem harder to press than others?
- How confident does the participant appear during the interaction?
- Does the participant change their position to better access or see the keypad?
- Does the participant use any aids to interact with the keypad?

Follow-up questions:
1. Which keypad did you find the easiest to handle?
    a. What made it feel easy to use?
    b. In what way did it feel easier that the others?
    c. What did you think of the keypad’s size and layout? Please elaborate.
    d. Was there any button that was hard to find? Which one?
    e. Was there anything you didn’t like about the keypad? What?
    f. If you could change anything to make it even easier to use, what would it be and why?

2. Which keypad did you find hardest to handle?
    a. What made it feel difficult to use?
    b. In what way did it feel harder than the others?
    c. What do you think of the keypads size and layout? Please elaborate.
    d. Was there any button that was extra hard to find? Which one?
    e. Was there anything you liked about the keypad? What?
    f. If you could change anything to make it even easier to use, what would it be and why?

3. Which buttons were the most pleasant to press?
    a. How would you describe the feeling of pressing the buttons? (soft, hard, clicky)
    b. In what ways were they more pleasant than the others? (size, shape, concave/convex)
    c. Did you receive any feedback from the button? What type? Did it contribute to its feeling and your choice?
    d. If you could change anything about the buttons to make it even more pleasant, what would it be and why?

4. Which buttons were the least pleasant to press?
    a. How would you describe the feeling of pressing the buttons? (soft, hard, clicky)
    b. In what ways were they less pleasant than the others? (size, shape, concave/convex)
    c. Did you encounter any specific issues with these buttons?
    d. Did you receive any feedback from the button? What type? Did it contribute to its feeling and your choice?
    e. If you could change anything about the buttons to make them even more pleasant, what would it be and why?

5. Is there anything you would like to add, that you believe could be relevant to us? It could be anything!

Thank you so much for participating and helping us, every input is valuable.

C Interview Guide, Keypads

Intervjuguide: Knappsatser (Original version in Swedish)

Denna guide används som ett hjälpmedel vid intervjuerna, målet är att få ett flytande samtal med den intevjuade. De listade underfrågorna ställs endast om de ej blivit besvarade naturligt i samtalet.

Syfte:
Syftet med denna intervju är att samla insikter om när och hur knappsatser används, människors känslor, erfarenheter och preferenser, samt eventuella problem de stöter på med knappsatser vid ingångar och i vardagen.

Introduktion:
Vi heter Johanna och Linnéa och går vårt sista år på KTH där vi studerar design och produktframtagning. Vi skriver för närvarande vår masteruppsats på XX, vårt uppdrag är att utvärdera och utveckla designrekommendationer för hur knappsatser kan utformas på ett mer inkluderande sätt, med fokus på användarupplevelsen.

De data och svar vi samlar in från denna intervju kommer att användas i vår rapport, som kommer att publiceras. Vi kommer inte att dela ditt namn och du kommer att vara helt anonym. Samtycker du till detta? När vi skrivit ett stycke där vi användt åsikter och/eller citat du bidragit med kommer du att få läsa texten och ha möjlighet att göra ändringar om du vill.

Innan vi börjar, skulle det vara okej om vi spelade in denna intervju? Inspelningen kommer inte att delas med någon och kommer endast att användas av oss för att korrekt komma ihåg vad som sades.

Allmänt:

1. Stöter du på några hinder i din vardag när det gäller att ta dig in och ut ur rum och byggnader?
    a. I vilka situationer uppstår dessa?
    b. Hur löser du dom?

Knappsaster vid ingångar:

2. Hur ofta behöver du använda en kod för att komma in i ett rummeller byggnad?
    a. I vilka situationer är det?
    b. Vilken typ av knappsatser används? ("klassisk", tangenter på handtag, app, etc.)

3. Vad tycker du om att använda koder för att komma in i byggnader och rum? Varför?
    a. Finns det andra lösningar som du skulle föredra? Varför?

4. Vad känner du när du måste ange en kod för att komma in/ut? Varför?
    a. Känner du dig trygg när du använder koder vid entreer? Varför?
    b. Finns det någon situation där användandet av kod har kännts otryggt? Varför?

Dagnig användning av knappsatser:

5. I vilka situationer använder du knappsatser i vardagen?
    a. Hur ofta inträffar dessa situationer?
    b. Vilken typ av knappsatser används i dessa situationer?
    c. Finns det situationer där du uppskattar att det finns en kod för att få åtkomst/göra något? Vilka situationer? Varför?
    d. Finns det situationer där du skulle föredra att det inte finns någon kod för att få åtkomst/göra något? Vilka situationer? Varför?

   e. Har du stött på knappsatser som du tyckte var särskilt lätta att använda?
      i. I vilken situation var detta?
      ii. Vad gjorde den så bra?
   f. Har du stött på knappsatser som du tyckte var svåra att använda?
      i. I vilken situation var detta?
      ii. Vad gjorde den svår?

6. Vad känner du när du måste ange en kod för att göra något? Varför?
   a. Har du någonsin haft problem med att se, nå eller trycka på en knappsats?
      i. I vilken situation var detta?
      ii. Vad bidrog till det?
      iii. Hur kände du dig i situationen?

7. Har du någonsin behövt ange en kod på en knappsats i en stressig situation? (händerna är fulla, kö bakom, följd, etc.)
   a. I vilken situation var detta?
   b. Hur påverkade det din förmåga att ange koden?
   c. Hur kände du dig i situationen?

Framtid:

8. Om du kunde designa en knappsats baserat på dina erfarenheter, hur skulle den se ut? Varför?
9. I ett drömscenario, hur skulle du vilja att ingångs- och åtkomstlösningar ser ut i framtiden?

Övrigt:

10. Finns det något du vill lägga till, som du tror kan vara relevant för oss? Det kan vara vad som helst!

Tack så mycket för ditt deltagande och din hjälp, varje bidrag är värdefullt!

Interview guide: Keypads (Translated)

This guide is used as an aid during interviews, the goal is to have a flowing conversation with the interviewee. The listed sub-questions are only asked if they have not been answered naturally in the conversation.

Purpose:
The purpose of this interview is to gather insights on when and how keypads are used, people's feelings, experiences and preferences, and any pain points they encounter with keypads at entrances and in daily life.

Introduction:
Our names are Johanna and Linnéa and we are in our last year of KTH where we are studying design and product realization. We are currently writing our master thesis at XX, our mission is to evaluate and develop design recommendations on how keypads can be designed in a more inclusive way, focusing on user experience.

The data and answers we collect from this interview will be used in our master thesis, which will be published. We will not share your name, and you will remain completely anonymous. Do you consent to this? When we have written a section using opinions and/or quotes you have contributed, you will get to read the text and have the opportunity to make changes if you wish.

Before we start, would it be okay if we recorded this interview? The recording will not be shared with anyone and will only be used by us to accurately remember what was said.

General:

1. Do you encounter any obstacles when entering and exiting buildings or rooms in your daily life??
    a. In what situations do these arise?
    b. How do you solve them?

Keypads at entrances:

2. How often do you have to enter a code to enter a room or building?
    a. In what situations?
    b. What kind of keypads are used? (“classic”, keys on handle, app, etc.)

3. What are your thoughts on using codes as an access system at entrances? Why?
    a. Are there other solutions that you would prefer? Why?

4. What do you feel when you have to enter a code to get in/out? Why?
    a. Do you feel safe when using keypads at entrances? Why?
    b. Are there any situations where it has felt unsafe? Why?

Daily use of keypads:

5. In what situations do you use keypads in your daily life?
    a. How often do these situations occur?
    b. What kind of keypads are used in these situations?
    c. Have you encountered keypads that you found particularly easy to use?
        1. In what situation was this?
        2. What made it so good?

   d. Have you encountered keypads that you found difficult to use?
      1. In what situation was this?
      2. What made it difficult?
   e. Are there situations where you appreciate that there is a code to access/do something? What situations? Why?
   f. Are there situations where you would prefer that there is no code to access/do something? What situations? Why?

6. What do you feel when you have to enter a code to do something? Why?
   a. Have you ever had trouble seeing, reaching or pressing a keypad?
      1. In what situation was this?
      2. What contributed to it?
      3. How did you feel in the situation?

7. Have you ever had to enter a code into a keypad in a stressful situation? (hands are full, line behind, followed, etc.)
   a. In what situation was this?
   b. How did it affect your ability to enter the code?
   c. How did you feel in the situation?

Future:

8. If you could design a keypad based on your experiences, what would it look like? Why?
9. In a dream scenario, how would you like entrance and access solutions to look in the future?

Other:

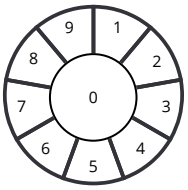
10. Is there anything you would like to add, that you believe could be relevant to us? It could be anything!

Thank you so much for your participation and help, every input is valuable.

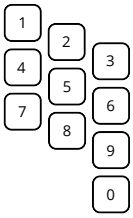
D Layouts for comparison

The different layouts created for layout comparisons.

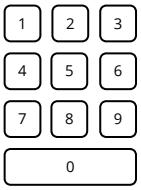
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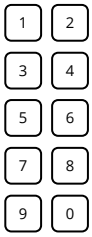
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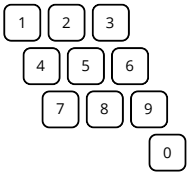
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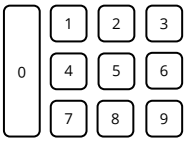
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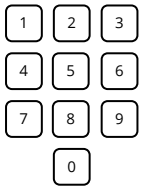
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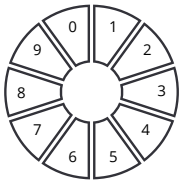
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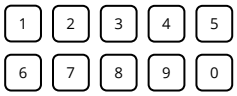
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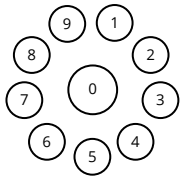
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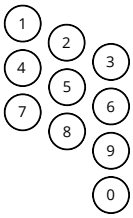
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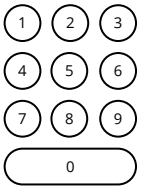
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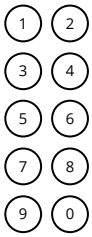
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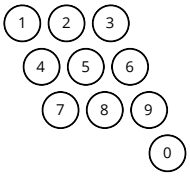
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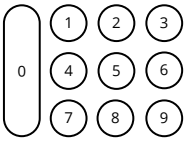
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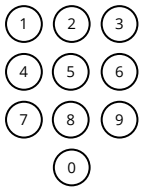
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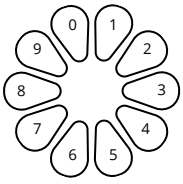
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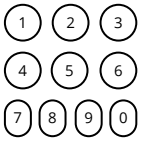
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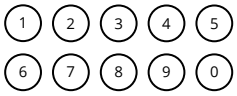
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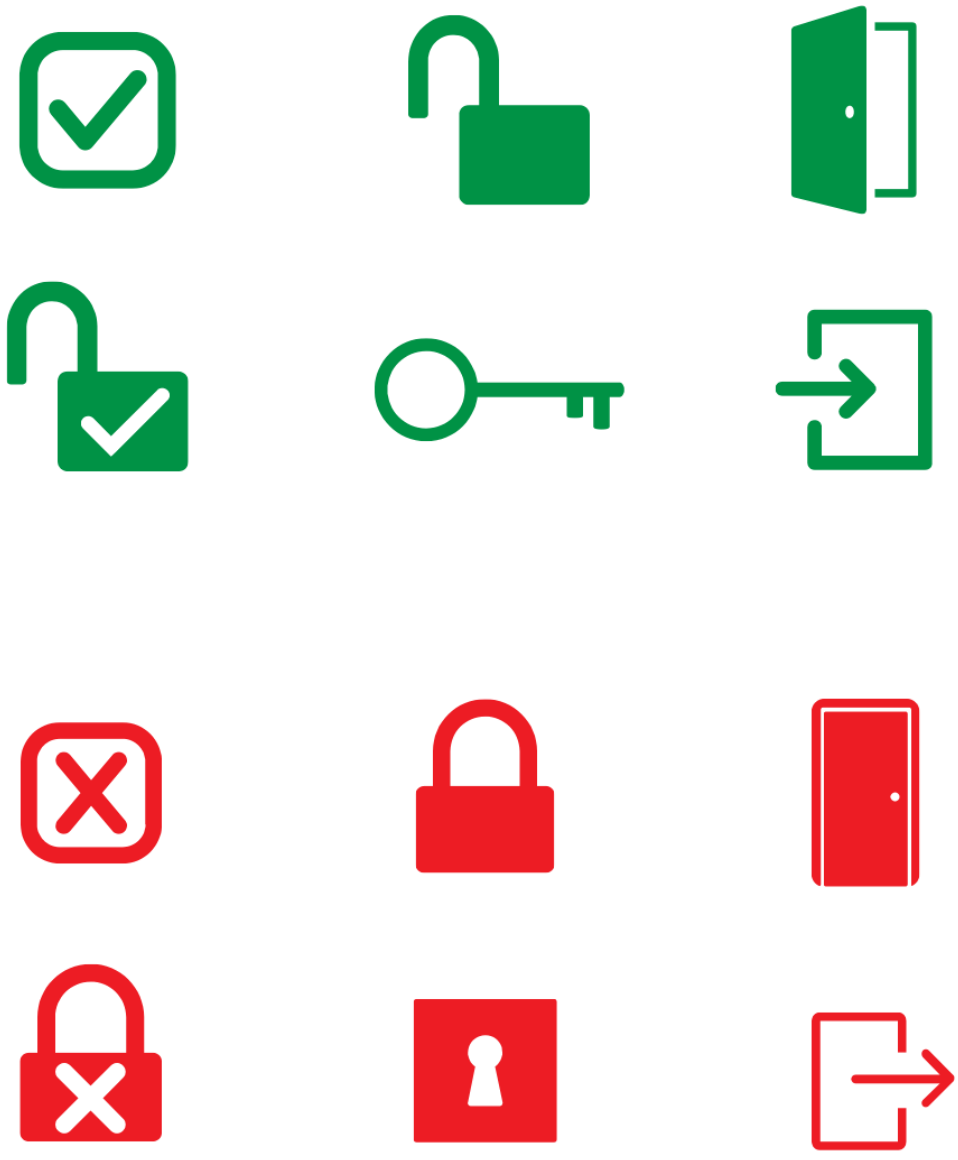


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E Symbol Preference Study

The different symbols presented in the symbol preference study.



F Translated and Original Quotes

This Appendix includes all translated quotes from the report and their original form in the Table **11** below.

Table 11: Translated and Original Quotes

Page	**Translated Quote**	**Original Quote**
32	”I can’t see anything on keypad number 1. Absolutely nothing. I can feel the keys, so I can count where they are. ... Sometimes it’s stubbornness that helps you get through, and now, here key 5 is marked, even though it’s poorly marked, so you can figure out where the other keys are” - P2	”Jag ser ju ingenting på 1an. Absolut ingenting. Jag kan känna knapparna så jag kan räkna it vart de sitter. ... Ibland är det tjurskallighet som gör att man tar sig fram, och nu, här är knapp 5 markerad, även om den är dåligt markerad, så då kan man lista ut vart de andra knapparna är” - P2
32	”There was no text on it [keypad no. 2], so you didn’t really know what you were clicking on. Sometimes the numbers go from top to bottom, and sometimes the other way around, so I thought it was a bit unclear. ... Now I just assumed it starts at the top, but you don’t know for sure.” - P5	”Det var dock ingen text på [2an], så man visste inte riktigt vad vam klickade på. Ibland går siffrorna uppifrån och ner, och ibland tvärtom, så jag tyckte att det var lite otydligt. ... Nu utgick jag bara från att det börjar högst upp, men det vet man ju inte.” - P5
33	”If I saw keypad 2 when I was stressed, I would get so angry, because I think clarity is very important when you’re stressed. I would just look at it and think, ’What is this?!’” - P5	”Om jag skulle se 2an när jag var stressad skulle jag bli så arg, för att jag tror att tydlighet är jätteviktigt om man är stressad. Jag hade bara kolla på den och tänkt ’vad är det här?!’” - P5
33	”I don’t like keypad 1 and I don’t like keypad 2, and I don’t understand how I can have so much trouble with the fact that there are no numbers on them?!” - P3	”Gillar inte 1an och gillar inte 2an, och jag fattar inte hur jag kan ha så mycket problem med att det inte är siffror på dom?!” - P3

Page	Translated Quote	Original Quote
33	”Sometimes when the numbers are above the keys like this, I find it a bit unclear. If the numbers are placed under the keys or above the keys, I don’t know if the five is connected to [key] five or if it is connected to [key] two” - P11	”Ibland när det är siffror över knapparna så här så kan jag tycka att det är lite otydligt. Ifall det är siffror under knappen eller över knappen, så jag vet inte om femman sitter på femman eller om en sitter på tvåan” - P11
34	”I would probably say that I would have wanted the numbers on the keys and not above. Now there’s a millisecond where I wonder which number belongs to which button.” - P4	”Jag skulle nog säga att jag hade velat ha siffrorna på knappen och inte över. Nu finns det en millisekund där jag undrar vilken siffra som hör till vilken knapp.” - P4
34	”Maybe an emergency button. I can imagine this [keypad] being in an elevator, if you need to go to a specific floor, so if you get stuck or can’t get out, you might press button 8 to send a signal somewhere.” - P2	”Kanske någon nödklnapp. Sånna här kan jag tänka mig finns i hiss, om du ska in på någon speciell våning, så om du fastnar eller inte kommer loss kanske man kan trycka på 8an så att det kommer en signal någonstans.” - P2
34	”Around key 8, I see a lot of parentheses, which I don’t know why. ... the large parentheses should be in the same place as key 5, since it is always marked.” - P1	”Under runt 8an ser jag att det är massa paranterer, vilket jag inte vet varför. ... dom stora parantererna borde vara på samma plats som knapp 5, eftersom den alltid är markerad.” - P1
34	”Oh, I just assumed the star was there [on the hashtag], it feels intuitive, then I’ve been entering the wrong code every time.” - P4	”Jaha jag bara antog att stjärnan var där [på fyrkantsknappen], det kändes intuitivt då har ju jag klickat fel kod varje gång.” - P4
35	”I would click on it last, and then it felt most natural to click on the key that is placed last, to finish, you know. ... I didn’t check what the symbol was ... But it feels more instinctive to click on the last one, I think.” - P5	”Jag skulle ju klicka på den sist, och då kändes det mest naturligt att klicka på knappen som ligger sist, för att avsluta liksom. ... Jag kollade ju inte vad det var för symbol ... Men det känns mer instinktivt att klicka på den sista, tycker jag.” - P5

Page	Translated Quote	Original Quote
34	”But you can’t see the star or the square! Have they switched places? [sees the lock symbols] One is filled in and the other isn’t. ... The meaning of the symbols is very unclear.” - P7	”Men man ser inte stjärna eller fyrkant! Har de bytt plats? [ser lås-symbolerna] Den ena är ifylld och den andra är inte. ... Det [symbolernas betydelse] är jätteotydligt.” - P7
35	”Where the hell is the star? The star isn’t there? [sees the lock symbol] Yeah, I could have figured that out, but it would have made me irritated.” - P11	”Vart fan är stjärnan? Stjärnan finns ju inte? [ser lås-symbolen] Ja, det hade man ju kunnat klura ut, men det hade gjort mig irriterad.” - P11
35	”...one should consider that in a keypad like this, there should be a contrasting color around the keypad, otherwise it blends in.” – P2	“...så bör man tänka på att i en sån här knappsats ska man ha en avvikande färg runt om knappsatsen, för annars smälter det ihop.” – P2
35	”...on the third keypad, I have to search more because the keys are close together and the same color throughout. ... this one [keypad 4], is quick in a crisis, because the keys are a different color than the background.” - P8	“...på trean [knappsats tre] måste jag leta mer för att knapparna är nära och samma färg genomgående. ... den här går snabbt i en kris [knappsats 4], det är också för att knapparna är en annan färg än bakgrunden..”
35	”Contrast is nice. I can not understand how someone can enter things [pin codes] without contrast [distinctive symbols], like the 5 [keypad no.5], why is there not a contrasting color in the number [embossed on the key]? But on the 4 [keypad no.4], there is a dark gray background and white text, that is great.” - P3	”Men det är ju najs med konstrast. Jag kan inte förstå hur man kan skriva saker utan konstrast, som 5an, varför är det ingen kontrasterande färg i siffran? Men på 4an är det mörkgrå bakgrund och vit text, det är toppen.” – P3
35	”...in a doorway where the outdoor lighting has broken, I would never be able to get in. [Speaking of symbol contrast and visibility on keypad no. 6]” – P10	“...i en port där utebelysningen har pajat skulle jag aldrig komma in.” -P10

Page	Translated Quote	Original Quote
36	"...on one of those modern keypads, they are very hard to read due to the contrast. For example, searching for names is very difficult if you can't see. ... It's very difficult to read because there is no contrast against the background. It's always [dark] gray letters on a [light] gray background. And it's impossible, for practically anyone."- P1	"...vid en sån där modern knappsats, dom äe väldigt svårlästa pga kontrastern. Tex söka på namn, det är jättesvårt om man inte ser. ... Det är väldigt svårt att läsa eftersom att de inte är konstrast mot bakgrunden, det är alltid (mörk)grå bokstäver mot (ljus)grå bakgrund. Och det är omöjligt, för vem som helst i princip. – P1
36	"Yes, it also depends on the general lighting. If it's too bright, it causes glare, but if it's like now [overcast], it might work. However, I still can't see the screen properly." – P2	"Ja, det beror också på hur allmänljuset är. Om det lyser för mycket så blir det bländande, om det är som nu så kan det funka, men jag kan likt förbannat inte se på skärmen." – P2
36	"Yes, overall disgusting, it [the key] sticks out like a joystick, you don't want that. It shouldn't be able to move sideways at all." - P4	"Ja overall disgusting, det sticker ju ut som en joystick, det vill man ju inte ha. Den ska inte kunna röra sig alls i sidled." - P4
36	"The keypad [keypad no. 4] doesn't feel the same, maybe not the same quality as number 3 [keypad no.3]. Number 3 was very soft and the keys felt sturdier. And then it also feels like the keys are connected on number 4, it feels a bit off, I think. ... It feels like the keys are attached to some plate behind the cover, it feels a bit cheap." - P5	"Att den [knappsats nr. 4] inte känns lika, kanske inte samma kvalitet som 3an. 3an var väldigt mjuk och det kändes som att knapparna var mer sturdy. Och sen känns det också som att knapparna hänger ihop på 4an, det känns lite off tycker jag. ... Det känns som att knapparna sitter ihop med någon platta bakom höljet, det känns lite billigt kanske." - P5
37	"The keys here [keypad no. 3] are nice. I like the keys, they have a nice feedback. It's a bit less aggressive than the other two [keypads no. 1 and 2], I mean." - P6	"Det är sköna knappar här [knappsats 3]. Jag tycker om knapparna, det är en skön feedback tycker jag. Den är lite mindre aggressiv än de två [knappsats 1 och 2] menar jag alltså." - P6

Page	Translated Quote	Original Quote
37	”I think I like the feedback on keypad 4, but I want a bit more sturdiness. When you press the keys on keypads 3 and 6, it feels like the button goes straight in, but on keypad 4, it feels a bit wobbly. Keypad 3 was probably more comfortable than keypad 6. So, the softness of keypad 3 but the feedback from keypad 4.” - P5	”Jag tror jag gillar feedbacken i 4an, men jag vill ha lite mer sturdyness. När man klickar på 3an och 6an känns det som att knappen går rätt in, men på 4an känns den lite wobbly. 3an var nog skönare är 6an. Så mjukheten i 3an men feedbacken från 4an.” - P5
37	Keypad 6 was sluggish in the keys. I don’t know if I pressed the button at all.” - P11	”Den här [knappsats 6] den var seg i knapparna. Jag vet inte om jag har tryckt på knappen överhuvudtaget.” - P11
37	”I must say that if we’re talking about shape as well, I think rectangles are better than round ones. I think there’s a clearer contrast between the keys then.” - P11	”Jag måste ändå säga att om man ska snacka om form också så tycker jag att rektanglar är bättre än runda. Jag tycker att det är tydligare kontrast mellan knapparna då.” - P11
37	...and the spacing should also be good, I think number 4 [keypad no. 4] has good spacing, so you don’t feel the keys next to it when you press.” - P3	”...och avståndet ska också vara bra, 4an har bra avstånd tycker jag, att man inte känner knapparna bredvid när man trycker.” - P3
37	”I think the keys [keypad no. 6] are a bit tight, they are a bit close to each other, I might accidentally click on more than one.” - P4	”Jag tycker nog att knapparna [knappsats 6] är lite tajta, dom sitter lite nära varandra, jag kanske råkar klicka på mer än en.” - P4
37	”Yes, keypad 1 was a terrible experience. The keys are too high, they tip over when you press them.” - P7	”Ja 1an var en hemsk upplevelse. De är för höga knapparna, de tippar när man trycker på dem.” - P7
37	”It goes faster if it doesn’t protrude that much. Keypad 3 is faster to press than keypad 1 and keypad 2.” - P6	”Det går ju snabbare om det inte sticker ut så mycket alltså. Trean går ju snabbare att knappa på än ettan och tvåan.” - P6
38	”However, the keys [keypad no. 2] were a bit slippery, your finger slips a bit.” - P1	”Däremot var dom [knappsats 2] lite hala, man halkar lite med fingret.” - P1

Page	Translated Quote	Original Quote
38	”I would have liked some texture on the keys too, because I liked feeling of something, I don’t know if it was the numbers or the texture.” - P3	”Jag hade nog önskat lite textur i knapparna också, för jag gillade när man kände någonting, vet inte om det var siffrorna eller om det var texturen.” - P3
38	”I like keypad 3 and 6, that they have this slightly glossy surface. You can slide over them quickly.” - P6	”Jag tycker om [knappsats] 3 och 6, att de har den här lite glossiga ytan. Att man kan glida snabbt över.” - P6
38	”The feeling of the material [keypad 3] is nicer than keypad 2” - P9	”Känslan av materialet [knappsats 3] är trevligare än [knappsats] 2” - P9
39	”Something I personally find very difficult is touch keys. It’s one of the reasons why I don’t use a smartphone. I have a lot of trouble with these touch keys, you accidentally press two buttons at the same time. ... Just because I’m a bit jittery, it goes wrong, so it’s good to keep that in mind.” - P1	”Någonting jag själv har väldigt svårt för är touchknappar. Det är en av anledningarna till att jag inte använder någon smartphone. Jag har jättesvårt för dessa touchknappar, man råkar trycka på två knappar samtidigt. ... Just eftersom jag är lite spattig blir det fel, så det kan vara bra att ha med sig i det här.” - P1
39	”Yes, I remember having difficulty when touch was new, so when there’s a slight delay, you get confused about whether you’ve pressed or not. Those were bad experiences. And with these induction cooktops, where you place your finger, I find it annoying, I want the feedback in the key.” - P3	” Ja jag minns att jag hade svårt för när touch var nytt, så när det blir någon liten fördröjning blir man förvirrad om man tryckt eller inte. Det var dåliga upplevelser. Och sånna här induktionshällar, att man lägger fingret på, jag tycker det är jobbigt, jag vill ha knappfeedbacken.” - P3
39	”...then the feedback is very clear, both in sound and feel, and it really is a step you press. It goes to a certain depth, not deeper or shallower. [Keypad no. 4]” - P6	”...sen är feedbacken väldigt tydlig, att det både låter och det känns, och det är verkligen ett steg man trycker. Den går en viss djup inte djupare eller grundare. -P6”

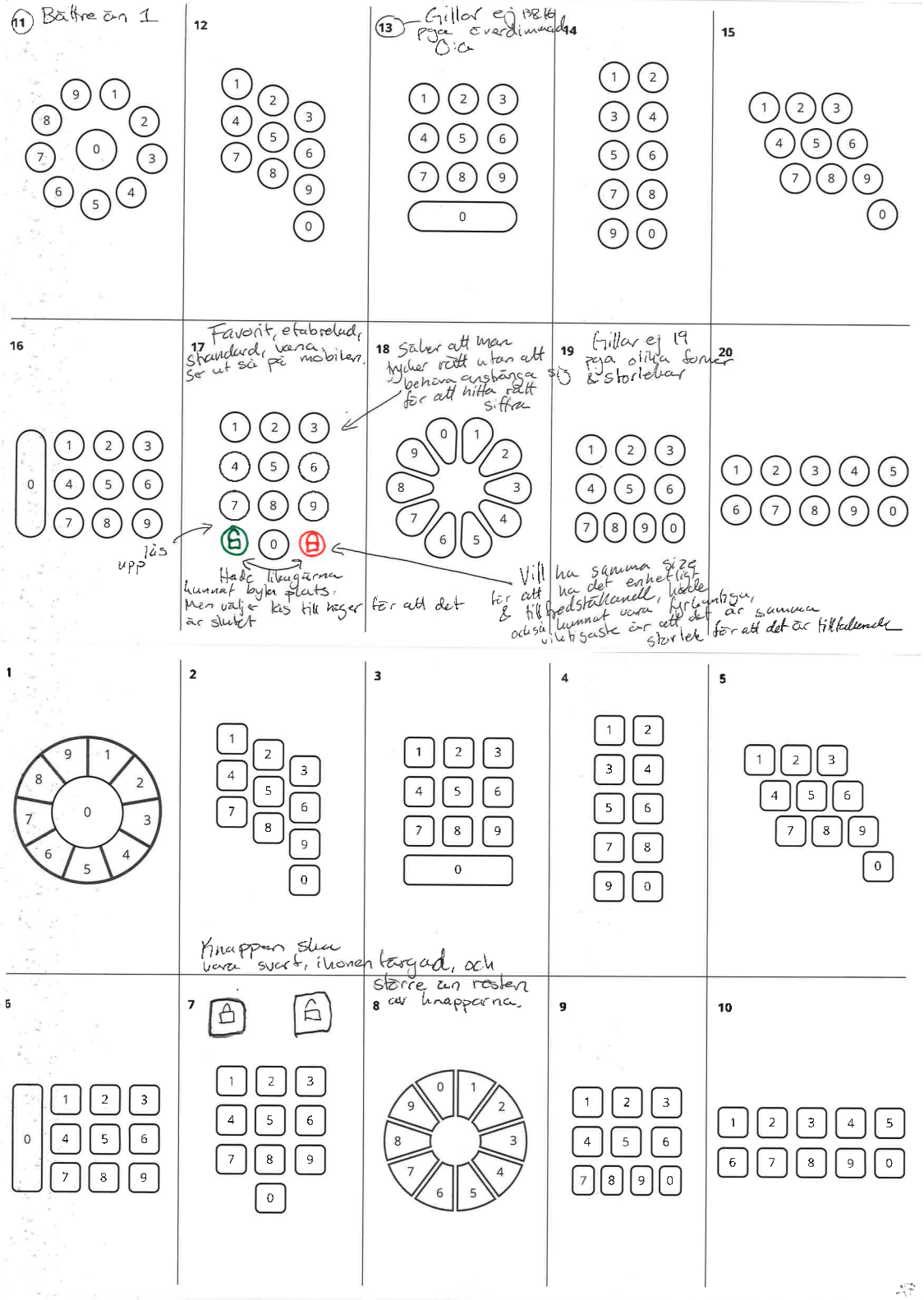
Page	Translated Quote	Original Quote
39	”But keypad 6, ugh, the keys, pressing there was like a swamp. ... It gives no response, you can press and press and press and nothing happens. ... This feels like an old remote control that has given up.” - P2	”Men 6an, usch, knapparna, att trycka där var som en sunkmark. ... Den ger ingen respons, du kan trycka och trycka och trycka och det händer inget. ... Det här känns som en gammal fjärrkontroll, som har sagt upp sig.” - P2
39	”The click confirms to me, yes, you have pressed [key] three” – P8	”Knäppet bekräftar för mig, ja du har tryckt på trean” – P8
40	”Yes, especially if it’s dark, you want to feel that you’ve pressed [a key], with tactile feedback. And if it also made a beep, it would be incredibly clear.” – P4	”Ja speciellt om det är mörkt, då vill man känna att man tryckt, med taktil feedback. Och sen om den skulle göra något pip också blir det otroligt tydligt.” – P4
40	”And then I want feedback, it should light up around the button. ... they light up green when I click so that I can see. That’s good feedback. I like feedback, maybe even both sound and light.” – P4	”Och sen vill jag ha feedback, det ska då lysa upp runt knappen. ... de lyser grönt när jag klickar så att jag ser. Så det är bra respons. Jag gillar feedback, kanske till och med både ljud och ljus.” – P4
40	”And if it’s a bit dark, there’s no chance of seeing [keypad no.6]. ... Unless they [the keys] light up, of course, but you don’t know that.” – P10	”Och skulle det vara lite mörkt. Det finns ju inte en chans att se. ... Om de inte lyser förstås, och det vet man ju inte.” – P10
40	“Keypad 4 is the clearest because you can see the keys and the spacing is clear. I would prefer to use this one if there was a fire.” – P8	”Den är tydligast för att man ser knapparna, och tydligt avstånd. Jag hade helst använt denna om det brinner.” - P8
40	“Yes, I’ve had this type [keypad no. 5] at work, and since more people use it, there are dirty fingers and other things that get into the numbers. You can almost see what the code is.” – P10	”Ja, jag har haft den här på jobbet och då blir det liksom fler som trycker. Det är lortiga fingrar och annat som kommer in de här siffrorna. Du kan nästan se vad koden är.” - P10

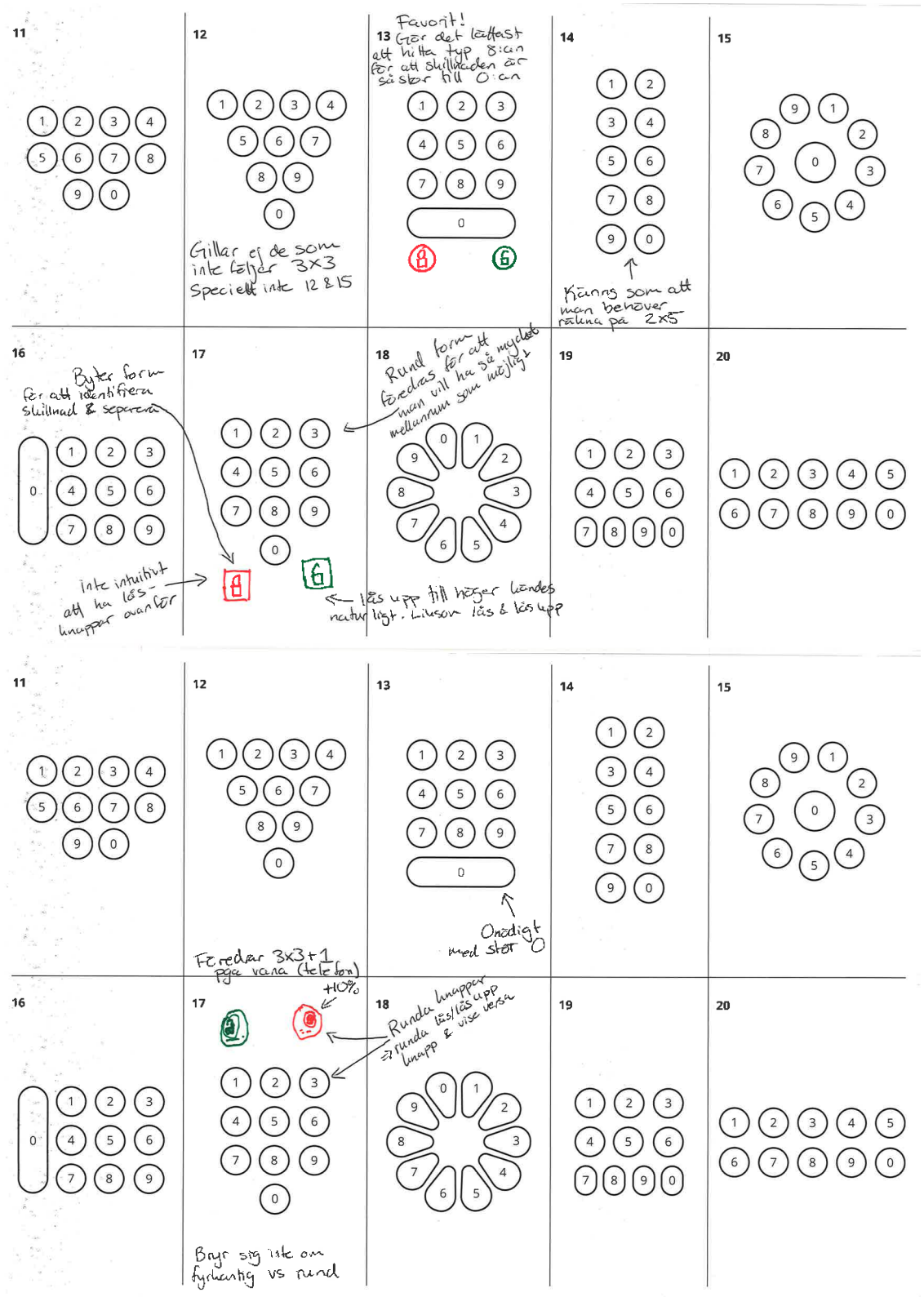
Page	Translated Quote	Original Quote
41	“I would probably like to have a code at the entrance, because it’s nice to be able to give one to visitors. But maybe it should only work until 10 p.m., and after that, you have to use a key. . . . And for my apartment door, I want a key because then I know exactly how many exist and who has one, then I know who can get in.” – P3	”Jag hade nog gärna haft det [portkod], för det är najs att kunna ge en kod om någon kommer på besök. Men kanske då att den bara ska funka fram tills klockan 10 på kvällen, och efter det måste man ha nyckel. ... Och sen till min dörr till hemmet vill jag ha en nyckel, för då vet jag hur många som finns och vilka som har en, då vet jag vilka som kan komma in.” - P3
41	”Keypad 6 is nice to look at. I probably wouldn’t want keypad 4 on my own door, I want a nice and stylish design on it” - P5	”Den [knappsast 6] är snygg att kolla på. 4an hade jag nog inte velat ha på min egna dörr, utan jag vill ha en sjysst och stilren design på det.” - P5
41	”Keypad 6 is the best looking, it has the nicest keys, they are round and feel a bit designed.” - P3	”Snyggast är ju 6an, den har snyggast knappar, dom är runda och känns lite genomtänkta.” - P3
41	”Keypad 4 was the easiest, but it’s boring. It’s not nice looking in any way.” - P7	”Den [knappsats 4] var ju lättast, men den är ju tråkig. Den är ju inte snygg på något sätt.” - P7
42	”Keypad 4 is ugly and feels outdated.” - P9	”Den [knappsats 4] är ful och känns omodern” - P9
42	”Then I want, number one [keypad no. 1], but with contrast between the key and the background, and the number and the background. Then I want the number on the key, not somewhere next to it, and then it should be in a different color, there should be contrast. Then there should be an angle, so that it is angled upwards. And then I want feedback, it should light up around the button, it should be tactile. And absolutely no wobble.” - P4	Då vill jag ha, nummer ett, men med kontrast mellan knapp och bakgrund och siffra och bakgrund. Sen vill jag ha en siffra på knappen inte någonstans bredvid, och sen ska den vara i en anna färg, det ska vara kontrast. Sen ska det vara en vinkel, så att den är vinklad uppåt. Och sen vill jag ha feedback, det ska då lysa upp runt knappen, det ska vara taktilt. Och absolut inget wobbel. - P4

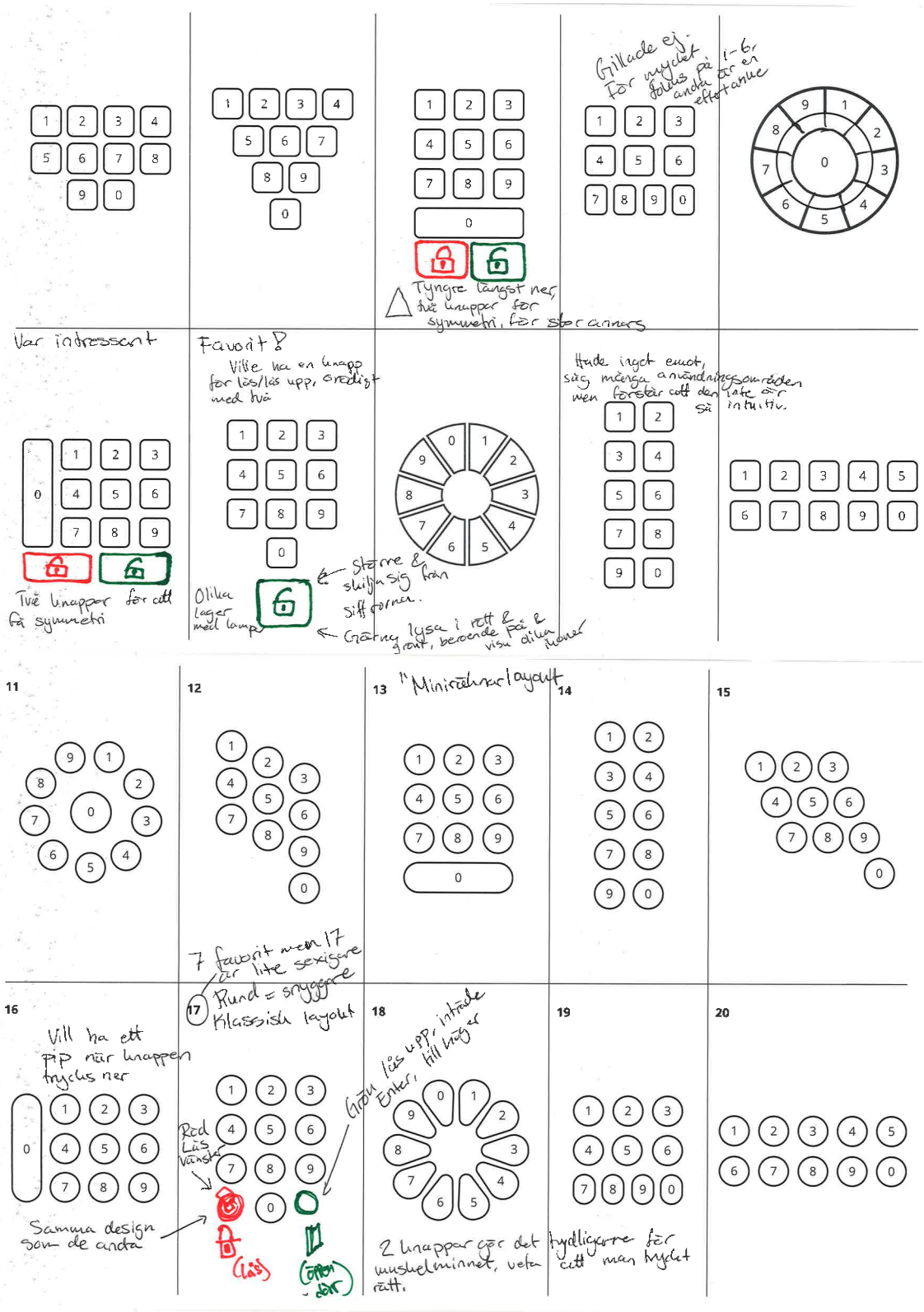
Page	Translated Quote	Original Quote
42	”I would want it to be clear, each key should be clear about what it is and what it does. I think each key should be easy to see. ... And then it’s important with feedback, that you know you’re clicking the key. And then it should be nice to look at. ... And that get a good feeling, that it doesn’t feel wobbly.” - P5	Jag skulle nog ha den tydlig, varje knapp ska vara tydlig med vad den är och vad den gör. ... Och sen är det viktigt med feedbacken, att man vet at tman klickar på knappen. Och sen att dsen är snygg att kolla på. ... Och sen att man får en bra känsla, att den inte känns wobbly. - P5
42	”No, I think that if only the numbers on keypad 4 had been tactile, it would have been fine. And that it is at the right height and that there is lighting around it. And that you could use your tag, especially if you have something to carry.” - P2	”Nä, jag tror att om siffrorna på knappsats 4 hade varit taktila hade det varit okej. Och att det är på rätt höjd, och att det är ljus runt. Och att du man kan använda sin tag, speciellt om man bär på saker.” - P2
43	“Well, I don’t really like these new ones with smooth buttons, they don’t have enough contrast. The new payment terminals. ... I don’t like them, you become so damn dependent on others.” - P2	”Asså jag gillar ju inte de här nya med släta knappar, det sätter för dålig kontrast. De nya betalningsdosorna. ... Jag gillar inte dom, man bli så jäkla beroende av andra.” -P2
43	”Yes, now I use an automatic door opener, and I have an automatic keypad instead of a key. Otherwise, with those modern keypads, they are very hard to read because of the contrast. For example, searching for a name is very difficult if you can’t see well.” - P1	”Jo nu använder jag en automatisk dörröppnare, jag har en automatisk knappsats istället för nyckel. Annars vid en sån där modern knappsats, dom äe väldigt svårlästa pga kontrastern. Tex söka på namn, det är jättesvårt om man inte ser.” - P1
43	”I just remembered that we have a keypad like this at the office, but it also has a tag, so it’s a nice interaction with the keypad since I don’t need to enter the code and remember anything.” – P3	”Nu kom jag på att vi har en sån här knappsats på kontoret, men den har ju också sån här blipp, så det är ju en najs interaktion med knappsaten så ,eftersom jag inte behöver slå in koden och minnas något.” – P3
43	”The best thing would be if it could recognize who you are in some other way.” - P5	”Det bästa hade varit om den kan förstå vem man är på något annat sätt.” – P5

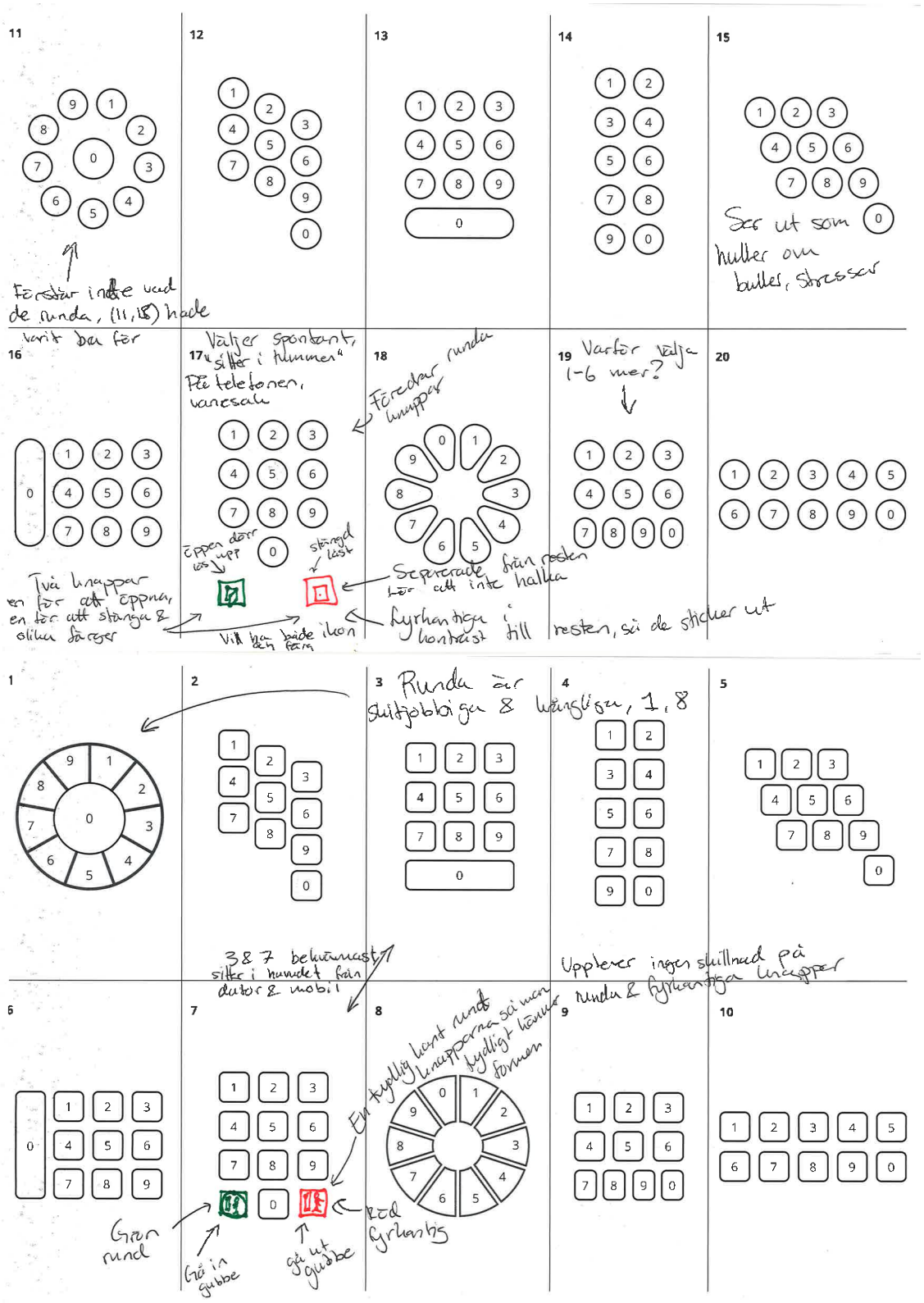
G Chosen Layouts

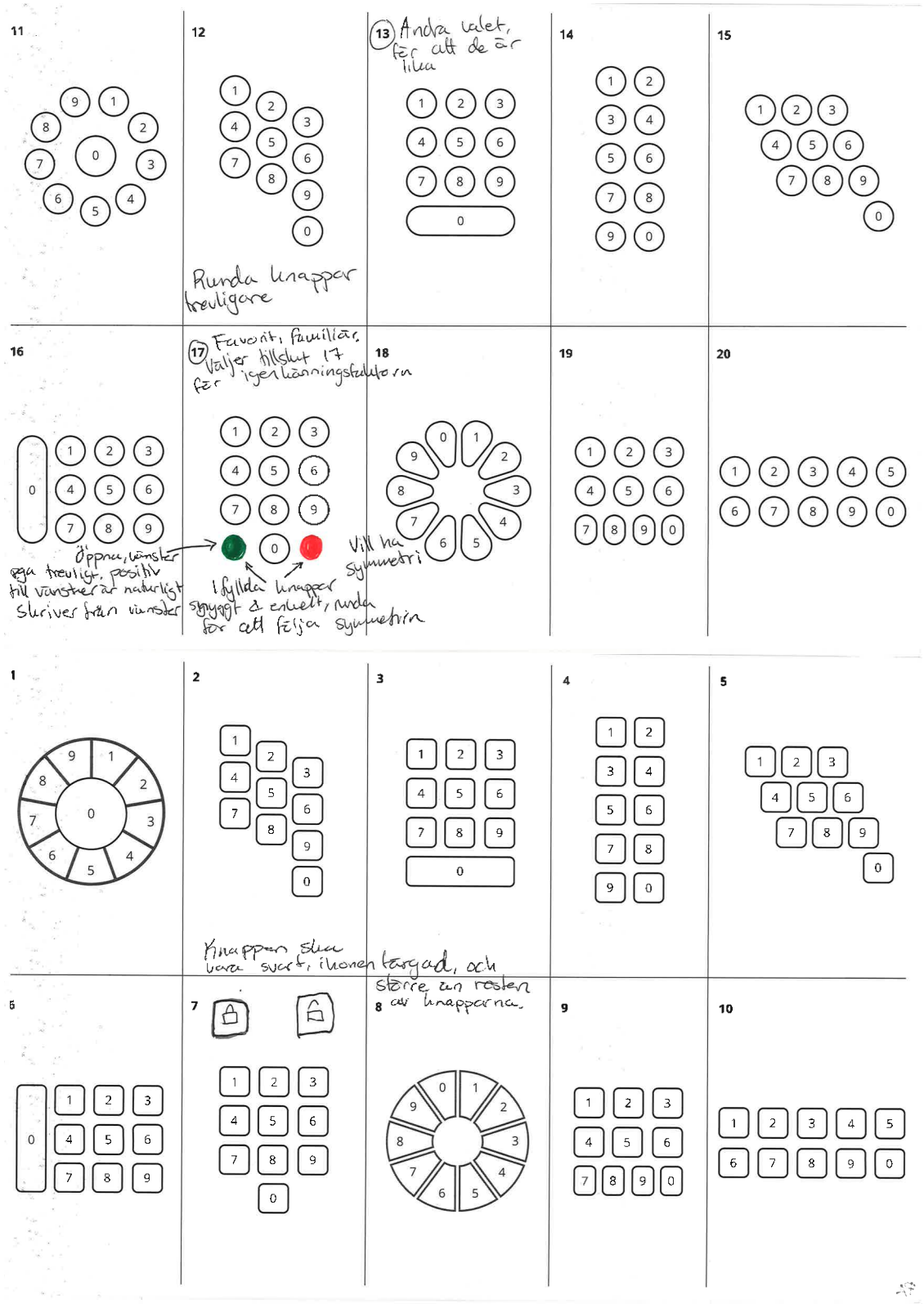
The different layouts chosen by the participants with notes in Swedish.

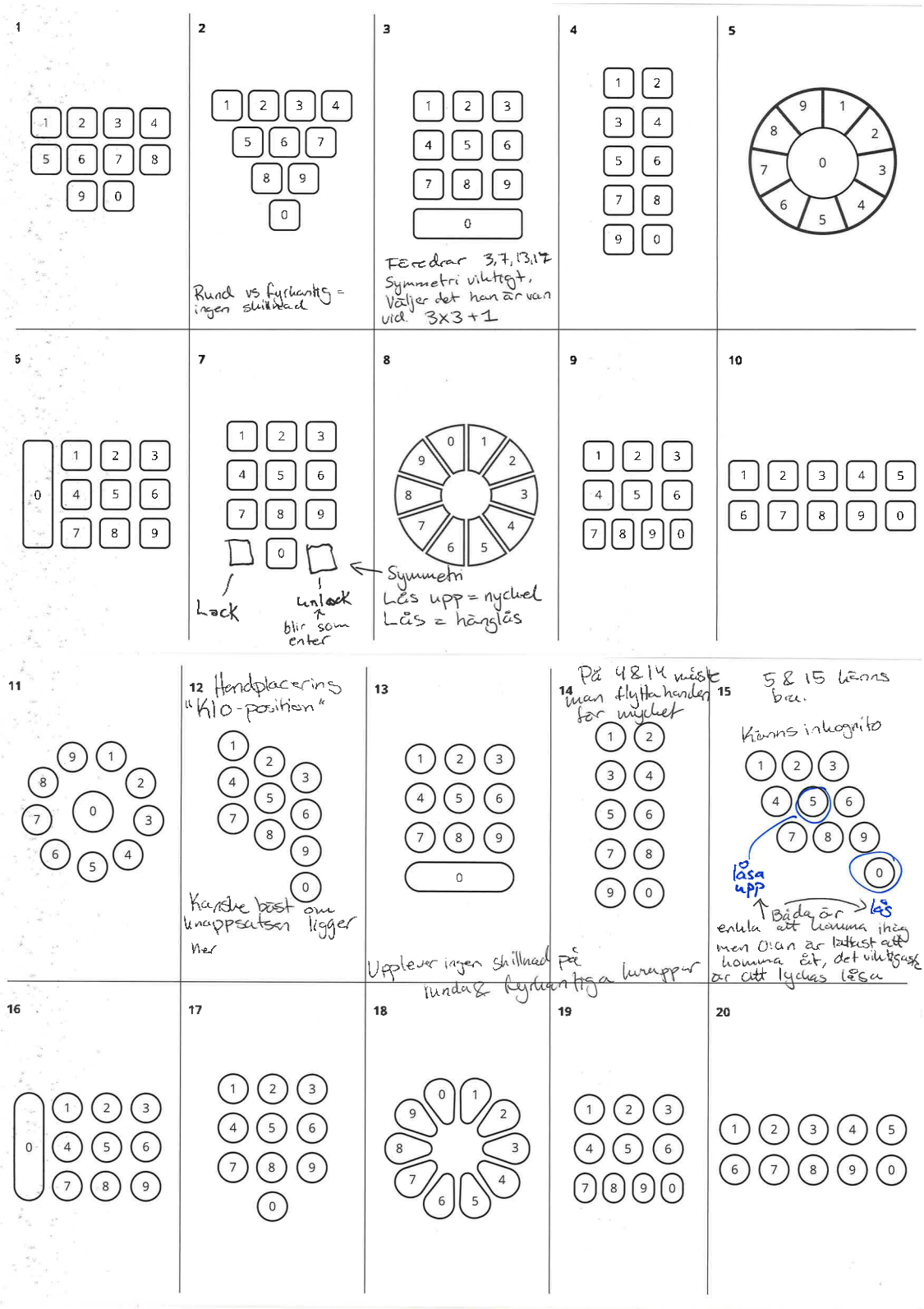


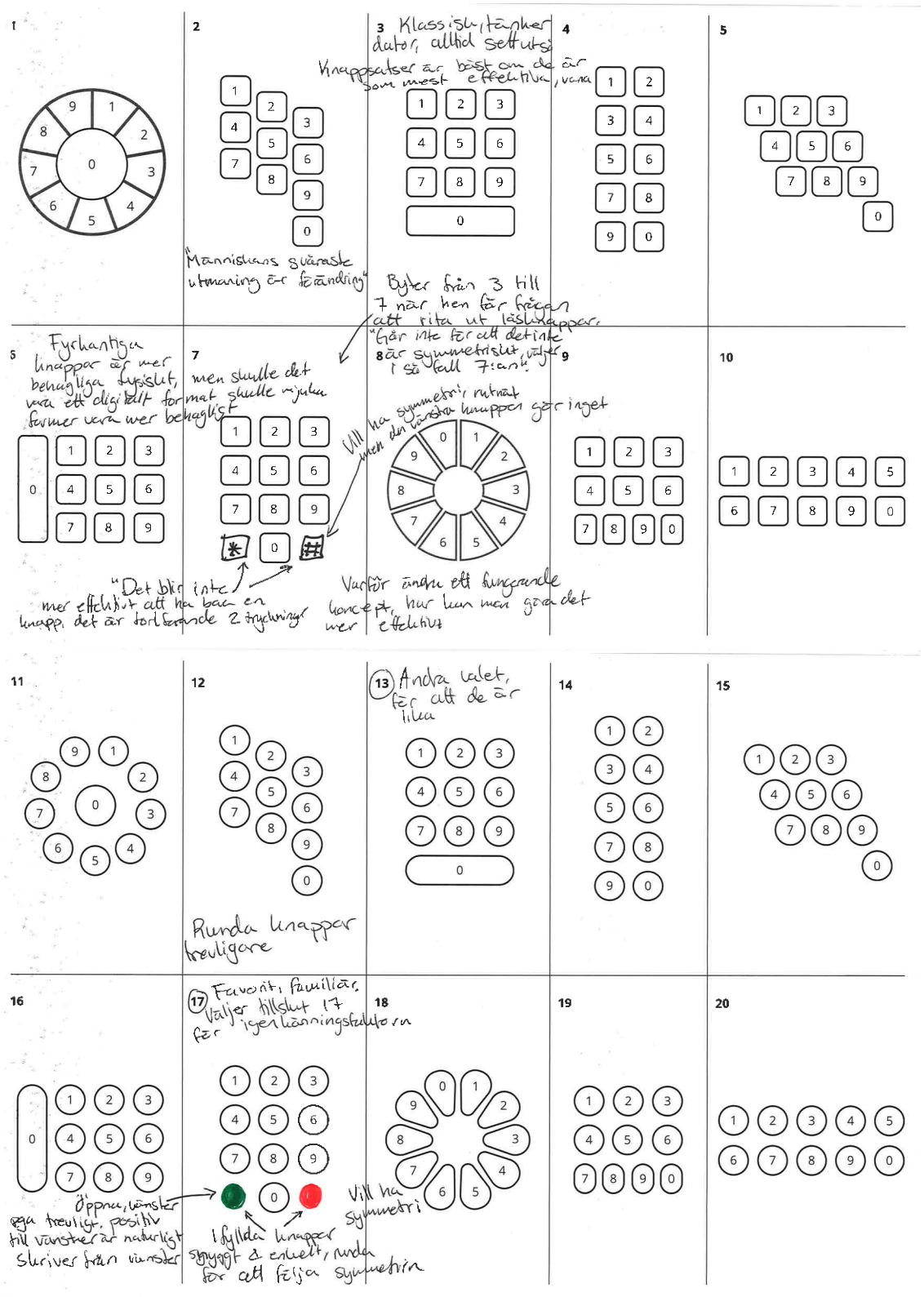

11
12
Gillar ej de som inte följer 3x3
Speciellt inte 12 & 15
13
Favorit!
Gör det lättast att hitta typ siffran för att skillnaden är så stor till 0:an
14
Känns som att man behöver räkna på 2x5
15
16
Byter form för att identifiera skillnad & seperera
inte intuitivt att ha läs-knapper ovanför
17
18
Rund form föredras för att man vill ha så mycket mellanrum som möjligt
läs upp till höger känns naturligt. Liksom läs & läs upp
19
20
11
12
Föredrar 3x3+1 pga vana (telefon)
13
Onödigt med stor 0
14
15
16
17
+10%
Bryr sig inte om fyrkantig vs rund
18
Runda knappar
=> runda läs/läs upp knapp & vise versa
19
20

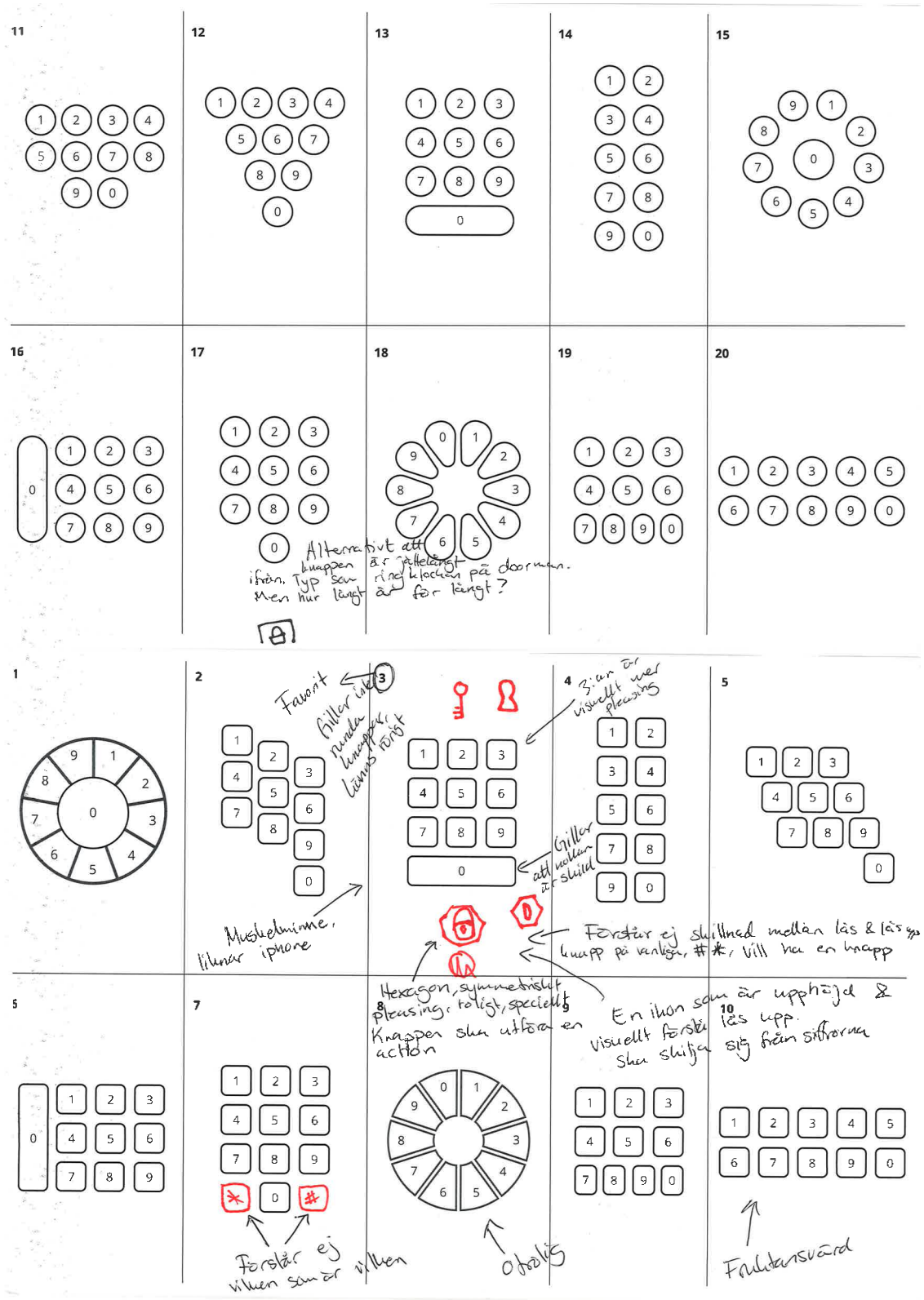

Gillade ej för mycket fokus på 1-6r andra är en effektanse
Tyngre längst ner två knappar för symmetri, för storamars
Var intressant
Två knappar för att få symmetri
Favorit?
Ville ha en knapp för lås/lås upp, enedigt med två
Olika lager med lampa
Större & skilja sig från sifferna.
Grönt lysa i rätt & grönt, beroende på & visa olika nivåer
Hade inget emot, säg många användningsområden men förstår att den kan vara så intuitiv.
11
12
13 "Miniräknar layout
14
15
7 favorit men 17 är lite sexigare
17 Rund = snygg Klassisk layout
16
Vill ha ett pip när knappen trycks ner
Samma design som de andra
Röd Lås vänster
Grön lås upp, inträde Enter, till höger
18
2 knappar gör det minnesvärt, veta rätt.
19
tydligare för att man trycket
20

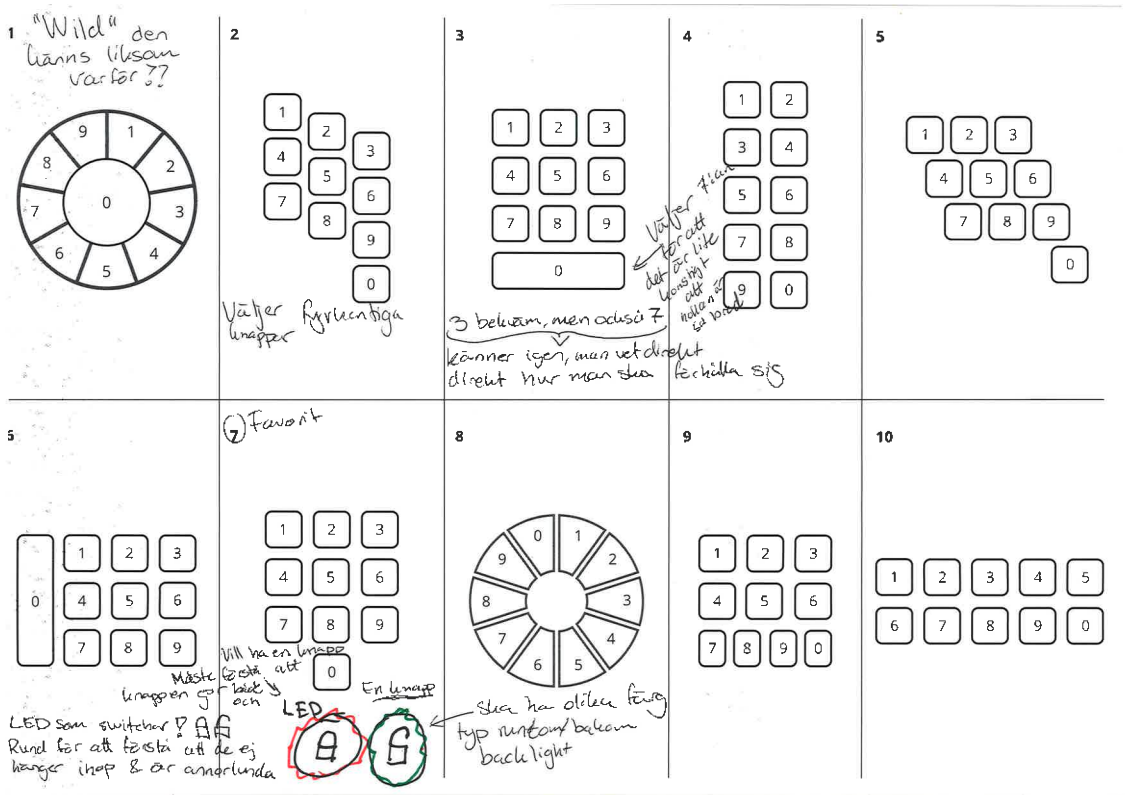

11
12
13
14
15
Förstår inte vad de runda, (11, 15) hade varit bra för
Ser ut som 0
knäppar om bollar, stressar
16
17 "sitter i klumpen"
Väljer spontant, Får telefonen, vanesak
18
Föredrar runda knappar
19 Varför välja 1-6 mer?
20
Två knappar en för att öppna, en för att stänga & olika färger
öppen dörr upp
stängd låst
Separerade från resten för att inte hallka
Vill ha den färg
Fyrhantigt i kombinat till
resten, så de sticker ut
1
2
3 Runda är skiftjobbigast 8 knäppligast, 1, 8
4
5
6
7
8
9
10
387 behövs sitter i huvudet från datorn & mobil
Upplever ingen skillnad på runda & fyrhantiga knäpper
En skjutlig kant runt knapparna så man tydligt känner formen
Grön rund
Gå in gubbe
gå ut gubbe
Röd fyrhantig


11
12
Runda knappar trevligare
13 Andra valet, för att de är lika
14
15
16
Öppna, vänster
Vill ha symmetri
17 Favorit, familjär. Väljer tillslut 17 för igenkänningsfaktorn
I fyllda knappar snyggt å enkelt, runda för att följa symmetrin
18
19
20
1
2
3
4
5
6
Knappen Stan bara svart, ikonen färgad, och
7
8 är knapparna.
9
10


1
2
3
4
5
Rund vs fyrkantig = ingen skillnad
Föredrar 3,7,13,17
Symmetri viktigt,
Väljer det han är van vid. 3x3+1
5
7
8
9
10
Symmetri
Lås upp = nyckel
Lås = hänglås
Lock
Unlock
blir som enter
11
12 Handplacering "Klo-position"
Kanske bäst om knappsatsen ligger ner
13
14
På 4 & 14 måste man flytta handen för nyckel
15
5 & 15 känns bra.
Känns inloggad
låsa upp
lås
Båda är enkla att komma ihåg men Olan är lättast att komma åt, det viktigaste är att lyckas låsa
Upplever ingen skillnad på
16
17
18
19
20

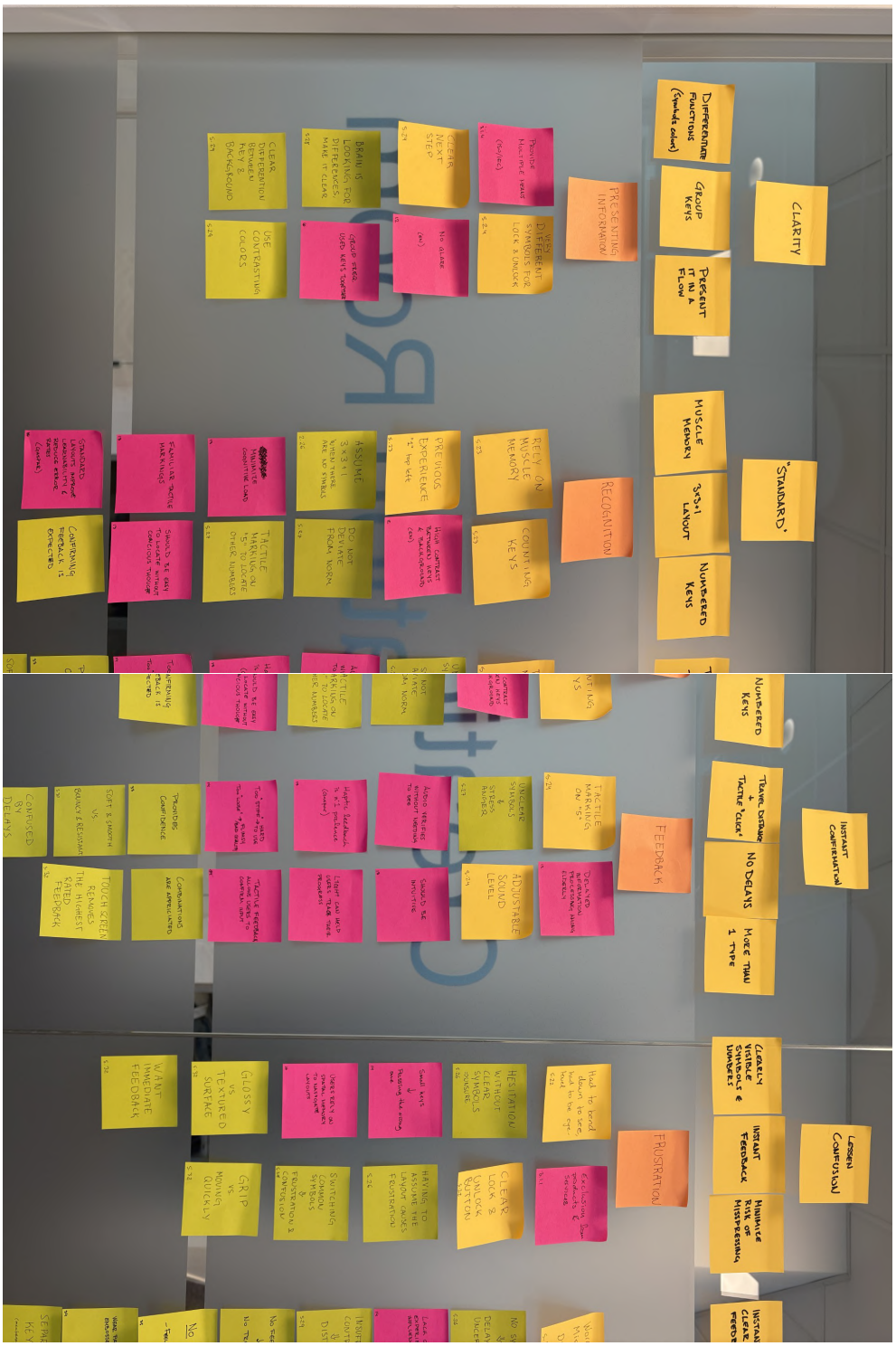

1
2
3 Klassisk, tänker dator, alltid settuts
Knappsatser är bäst om de är som mest effektiva, vana
4
5
"Människans svåraste utmaning är förändring"
Byter från 3 till 7 när hen får frågan att rita ut låsknappar. "Går inte för att det inte är symmetriskt, väljer i så fall 7:an!"
Fyrkantiga knappar är mer behagliga typiskt, vara ett digitalt format skulle mjuka former vara mer behagligt
6
7
men skulle det
8
9
10
Vill ha symmetri, rutnät men den översta knappen gör inget
"Det blir inte mer effektivt att ha bara en knapp, det är fortfarande 2 tryckningar
Varför ändra ett fungerande koncept, hur kan man göra det mer effektivt
11
12
13 Andra valet, för att de är lika
14
15
Runda knappar trevligare
16
17 Favorit, familjär. Väljer tillslut 17 för igenkänningsfaktorn
18
19
20
Öppna, vänster ega trevligt, positiv till vänster är naturligt Skriver från vänster
Ifyllda knappar snyggt å enkelt, runda för att följa symmetrin
Vill ha symmetri

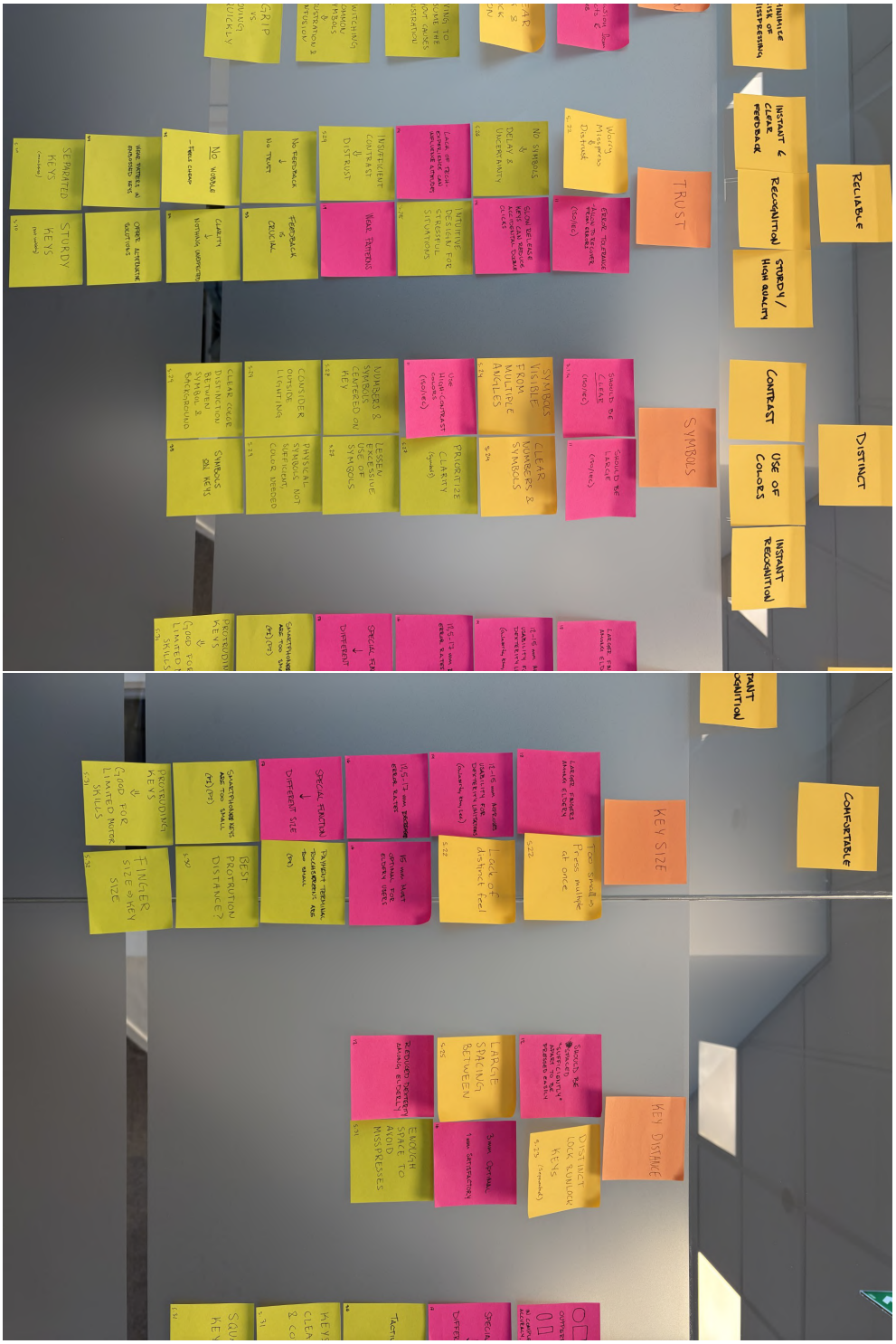

11
12
13
14
15
16
17
18
19
20
Alternativt att knappen är jättelångt ifrån. Typ som ring klockan på dörrmen. Men hur långt är för långt?
1
2
Favorit
Gillar inte runda knappar, känns fel
3
4
Gillar att mellan är skilld
Visuellt mer pleasing
5
Muskelminne, liknar iphone
Förstår ej skillnad mellan lås & lås upp knapp på vänster, #*, Vill ha en knapp
Hexagon, symmetriskt pleasing, roligt, speciellt
Knappen som utför en action
En ikon som är upphöjd & visuellt förstår den skall skjutas sig från siffrorna
6
7
8
9
10
lås upp
Förstår ej vilken som är vilken
Ofokus
Fruktansvärd

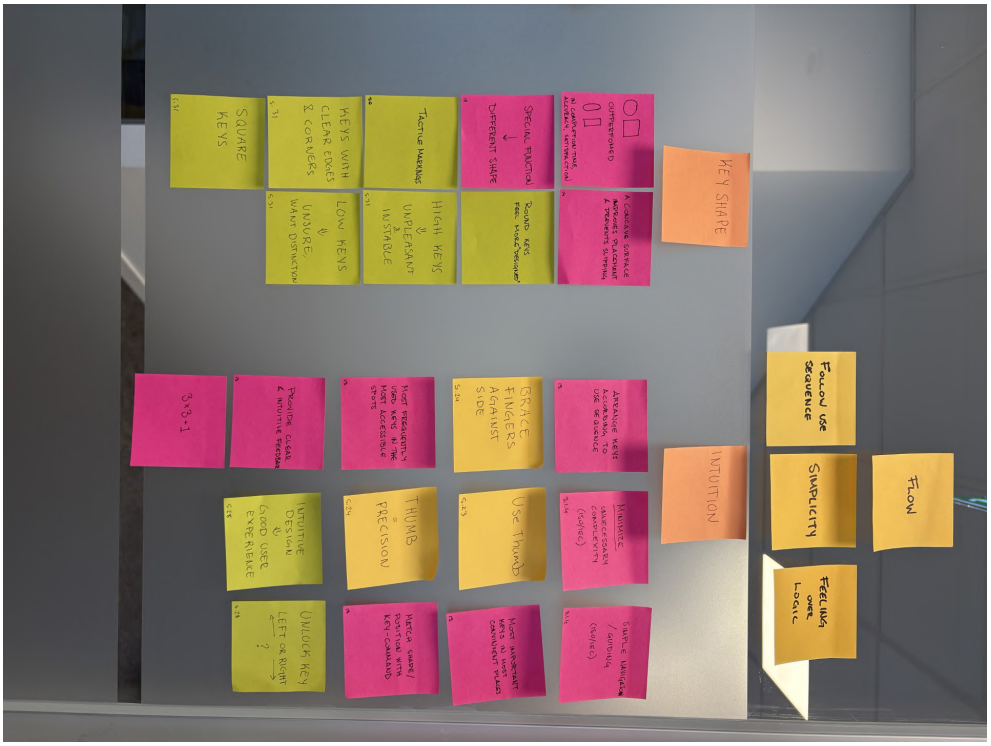

1
"Wild" den känns likson vaför??
2
Väljer fyrlaenliga knappar
3
3 bekvam, men också 7
känner igen, man vet direkt
direkt hur man ska
Väljer för att det är lite lättare att hålla sig bredd
4
5
5
LED som switchar? Rund för att förstå att de ej hänger ihop & är annorlunda
Måste knapparen gå bakåt och
7 favorit
Vill ha en knapp för att förstå att
En knapp
LED
8
Ska ha olika färg typ runtomkring bakom backlight
9
10

H Clustering

Clustering of insights from the Discover phase.

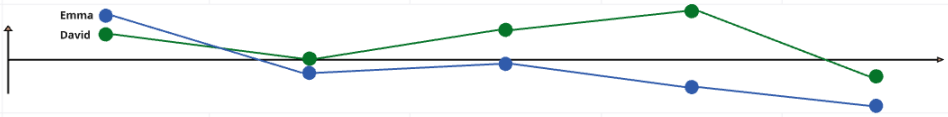


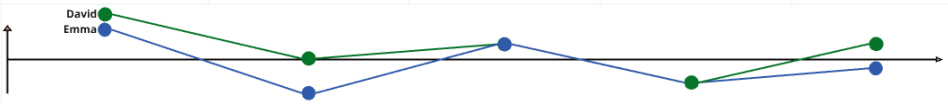

TRUST
SYMBOLS
KEY SIZE
KEY DISTANCE
DISTINCT
CONFORTABLE
RELIABLE
STURDY / HIGH QUALITY
RECOGNITION
INSTANT & CLEAR FEEDBACK
CONTRAST
USE OF COLORS
INSTANT RECOGNITION
SEPARATED KEYS
STURDY KEYS
FINGER SIZE ⇒ KEY SIZE
BEST PROTRUTION DISTANCE?
Too small & Pressed at once
Lack of distinct feel
LARGE SPACING BETWEEN
DISTINCT LOCK &UNLOCK KEYS
ENOUGH SPACE TO AVOID MISSPRESSES
SPECIAL FUNCTION ⇒ DIFFERENT SIZE
SYMBOLS VISIBLE FROM MULTIPLE ANGLES
CLEAR NUMBERS & SYMBOLS
PRIORITIZE CLARITY
NUMBERS & SYMBOLS CENTERED ON KEY
LESSEN EXCESSIVE USE OF SYMBOLS
CONSIDER OUTSIDE LIGHTING
PHYSICAL SYMBOLS NOT SUFFICIENT
CLEAR CONTRAST BETWEEN SYMBOL & BACKGROUND
SYMBOLS ON KEYS
NO TRUST
FEEDBACK IS CRUCIAL
CLARITY
NO MISTAKE
INSUFFICIENT CONTRAST ⇒ DISTRUST
NEAR PATTERNS
INTUITIVE DESIGN FOR SPECIFIC SITUATIONS
NO SYMBOLS ⇒ DELAY & UNCERTAINTY
Working Misspress & Distrust
CLEAR TACTILE FEEDBACK (DESIRED)
SHOULD BE CLEAR (DESIRED)
SHOULD BE LARGE (DESIRED)
LARGE FINGERS ARRIVE EASILY
12-16 mm KEYCAPS (SUBJECTIVE PREFERENCE)
15 mm MOST OPTIMAL FOR EVERY USERS
Sweet Spot
Too Satisfaction
REDUCED PROPERTY ARRIVE EASILY
SPECIAL FUNCTION ⇒ DIFFERENT SIZE
SMARTPHONE KEYS ARE TOO SMALL (PC) (TP)
PROTRUDING KEYS ⇒ GOOD FOR LIMITED MOTOR SKILLS
SQUARE KEYS


KEY SHAPE
SQUARE KEYS
KEYS WITH CLEAR EDGES & CORNERS
TACTILE MARKING
SPECIAL FUNCTION → DIFFERENT SHAPE
LOW KEYS ↓ UNSURE, WANT DISTINCTION
HIGH KEYS ↓ UNPLEASANT & UNSTABLE
RAISED KEYS
A CONCAVE SURFACE IMPROVES PLACEMENT & PREVENTS SLIPPING
FOLLOW USE SEQUENCE
SIMPLICITY
FLOW
FEELINGS OVER LOGIC
INTUITION
ARRANGE KEYS ACCORDING TO USE FREQUENCY
BRACE FINGERS AGAINST SIDE
MINIMIZE UNNECESSARY COMPLEXITY (SHAPE)
USE THUMB
THUMB = PRECISION
MOST FREQUENTLY USED KEYS IN THE MOST ACCESSIBLE SPACE
MATCH SHAPE/ KEY TO KEY COMMAND
MOST IMPORTANT KEYS IN MOST CONVENIENT PLACE
SIMPLE VARIATION / MOTIVEN (SHAPE)
3x3-1
INTUITIVE DESIGN = GOOD USER EXPERIENCE
UNLOCK KEY
LEFT OR RIGHT ?

I Customer Journey Mapping

Customer journey maps for two different users, at work and at home.

Customer Journey	Leaving home	On the bus	Arriving at work	Buying lunch	Lock up office
Context & Touchpoints	Leaving for work, locking door with digital keypad or phone.	Pressing stop button to get off, no feedback in button or any signs lighting up.	Entering office, automatic door with code-based keypad beside it.	Using an up-tilted payment terminal on high counter.	Activating security alarm via wall-mounted keypad.
Emma (wheelchair user)	Keypad positioned around eyelevel, sees the numbers and symbols clearly. Uses phone or keypad, smooth interaction.	May struggle to reach button. Unsure if press registered, no light/sound confirmation.	Keypad may be out of reach or angled awkwardly, glare can make it hard to read.	Counter height may block access, arm strain to reach. Card terminal may be out of reach or angled awkwardly.	May not be able to reach or clearly see the keypad, complex process adds frustration.
David (able-bodied)	Keypad positioned around waist height, struggles to see numbers & symbols, uses muscle memory. Uses phone or keypad easily, relatively smooth interaction.	Unsure if press registered, no light/sound confirmation.	Code entry is fine, but sometimes mistypes happen.	Terminal is well-positioned for standing height, clearly visible and easy to tap with card or phone.	Can access keypad but forgets the multi-step process, steps are not intuitive.
Opportunities	Angle the keys or surface for standing visibility.	Add visual & auditory feedback. Duplicate buttons at accessible locations.	Angle keypad for seated visibility, lower height, or offer app/bluetooth entry.	Use flexible mounts for terminals, or lower height.	Simplify interface, offer other alternatives such as mobile apps, ensure keypad is accessible.
Emotion					
		Lack of feedback causes a feeling of uncertainty and worry.	Being placed out of reach demands extra work of the user, feeling excluded.	Visibility is important to feel confident when using the product.	Excessive amount of inputs can cause confusion and stress when performing important tasks

Customer Journey	Arriving at home	Cooking	Using Phone	Watching TV	Adjusting the lights
Context & Touchpoints	Smart lock with keypad, app, and NFC/blip tag entry.	Using induction cooktop with flat touch-only controls (no tactile feedback). Possible wet hands	Unlocking with Face ID (primary) or passcode.	Using a remote with soft buttons and no click, sensitive to long presses.	Touch dimmer with press & hold gesture to brighten or dim.
Emma (wheelchair user)	Prefers using the tag as it's faster and less work, but the keypad also works.	Bad viewing angle, and glare. Difficult to reach and verify activation, no tactile and poor visual confirmation. Confusing auditory signals.	Face ID fails in low light or odd angles (lying down), passcode is fallback.	Frustrated when pressing too long, presses multiple times accidentally.	Dimmer may be placed high or require too much precision.
David (able-bodied)	Uses app or blip tag most of the time, keypad used as backup.	Touch zones unclear, occasionally triggers wrong zone or nothing at all. Confusing auditory signals.	Face ID fails in low light or odd angles (lying down), passcode is fallback.	Frustrated when pressing too long, presses multiple times accidentally.	No clear signal on brightness level hard to time release.
Opportunities	Ensure keypad is reachable and visible by all.	Position controls in more accessible positions. Add haptic/audio feedback, clear contrast, voice control.	Use fingerprint, trusted devices, or voice unlock options.	Switch to haptic/voice remote; simplify layout; ensure button spacing and contrast.	Use app or voice to adjust lights, use dimmers with physical knob, tactile stops or light indicators.
Emotion					
		Not being able to operate the product as intended, and overly sensitive products causes frustration.		Lack of feedback and delays can cause errors and makes the user redo the operation, causing frustration.	Slow response messes with precision causing frustration.

J Needs Analysis

This Appendix included the full Need Analysis, categorized after physical, cognitive and emotional needs.

Table 12: Needs Analysis

Category	**Insight**	**Need**	**Requirement**
Physical	Users with reduced dexterity struggle with small, cramped keys.	Users need the keypad to be physically comfortable.	Ensure keys are sized appropriately with enough spacing to prevent mispresses.
Physical	Users with visual or physical limitations face usability challenges.	Users need the keypad to be accessible to a wide range of abilities.	Incorporate tactile features (e.g. braille) and low-effort keypress mechanisms.
Physical	Users may struggle if all keys are not within reach.	Users need to reach all keys comfortably regardless of hand size.	Ensure the layout so all keys are easily accessible without strain.
Physical	Users worry about keypad quality and key wobble.	Users need the keypad to feel durable and stable.	Incorporate materials and mechanisms that endure frequent use without degrading.
Physical	Users are discouraged by either overly stiff or too soft keys.	Users need the keypress to require active effort but not excessive force.	Incorporate keys with balanced resistance suitable for varied strength levels.
Physical	Users may face visibility or usability issues in different conditions.	Users need the keypad to function reliably in varied environments.	Ensure the keypad resists water, dust, and glare, and remains readable in all lighting.
Cognitive	Users feel confused by low-contrast or unclear symbols.	Users need to easily understand symbols and instructions.	Design high-contrast, large symbols that are clearly visible from various angles.

Category	**Insight**	**Need**	**Requirement**
Cognitive	Users feel more at ease with simple, intuitive layouts.	Users need to navigate the keypad without confusion.	Ensure a clean layout with distinct functional areas and minimal distractions.
Cognitive	Users feel frustrated by cluttered layouts and poor key placement.	Users need to quickly grasp how to operate the keypad.	Arrange keys in a logical sequence with guiding visual cues.
Cognitive	Users feel disoriented when unfamiliar layouts are used.	Users need to easily recognize and understand common functions.	Incorporate familiar symbols and standardized key arrangements to reduce learning time.
Cognitive	Users feel limited when access is restricted to the keypad only.	Users need to choose from multiple secure and convenient access methods.	Ensure the product supports access via mobile app, RFID tag, or biometric input.
Emotional	Users feel frustrated when they don't trust the keypad to register inputs.	Users need to feel confident that the keypad is reliable and responsive.	Ensure the product performs consistently and gives a sense of robustness.
Emotional	Users feel more confident when the keypad gives clear feedback.	Users need to feel reassured that their input has been recognized.	Ensure each key press triggers a visual (e.g. LED) and audible response.
Emotional	Users feel discomfort using keypads that are hard to press or lack feedback.	Users need to feel at ease while interacting with the keypad.	Incorporate keys with distinct tactile feedback to confirm input.
Emotional	Users feel more satisfied when feedback is clear and immediate.	Users need to feel that the keypad performs reliably and intuitively.	Incorporate both tactile and auditory feedback to enhance satisfaction.
Emotional	Users feel more positively toward keypads with thoughtful, attractive design.	Users need the keypad to feel visually considered and appealing.	Ensure the visual layout with balanced composition and appealing aesthetics.

K Handbook

The handbook containing the design recommendations.

DESIGN RECOMMENDATIONS

FOR INCLUSIVE KEYPAD DESIGN

JOHANNA ABRAHAMSSON & LINNÉA BERGGREN
SPRING 2025

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CHAPTER 01.

INTRODUCTION

ABOUT THE PROJECT

This handbook is the result of a master's thesis project carried out by students in the Integrated Product Design program at the Royal Institute of Technology (KTH) in Stockholm. Conducted in collaboration with a company specializing in access systems, the project explores how smart lock keypads for domestic use can be made more inclusive and accessible to users with diverse physical and cognitive abilities.

As smart locks become increasingly common in homes around the world, the need for intuitive and universally usable interfaces grows. Through user research, design analysis, and the application of inclusive design principles, the project aimed to identify key usability challenges and propose concrete improvements.

01. INTRODUCTION

ABOUT THE DESIGN RECOMMENDATIONS

The design recommendations presented in this handbook are the outcome of the project's research and development process. They are organized into three key focus areas, **Visibility**, **Clarity**, and **Tactility**, which reflect the main aspects influencing user interaction with smart lock keypads.

Each recommendation addresses common barriers to usability and aims to support confident, accurate, and accessible keypad use for a broad range of users. While not all suggestions will apply to every design context, they offer a flexible framework that can be adapted to improve both the usability and inclusivity of keypad interfaces. By promoting inclusive and user-centered design, these recommendations contribute to UN Sustainable Development Goals 9 (Industry, Innovation and Infrastructure), 10 (Reduced Inequalities), and 11 (Sustainable Cities and Communities).



Image Source: UN, 2015

WHAT IS INCLUSIVE DESIGN?

Inclusive design is a design approach that aims to make products and services usable by as many people as possible, regardless of age, ability, or experience. Rather than focusing on an "average" user, it considers the full spectrum of human diversity, including permanent, temporary, and situational limitations.

By addressing a wide range of user needs from the beginning, inclusive design improves accessibility and enhances overall usability for everyone.

UNDERSTANDING USER CAPABILITIES

User capabilities refer to the sensory, cognitive, and motor functions that affect how people interact with products. These vary widely between individuals and may be influenced by disabilities, aging, injury, or fatigue.

- **Sensory:** Vision, hearing, and touch. For example, reduced vision may make it difficult to read small symbols or see low-contrast elements
- **Cognitive:** Memory, attention, and comprehension. Complex or inconsistent interfaces can be challenging for users with cognitive impairments.
- **Motor:** Movement, strength, and coordination. Small buttons or tightly spaced layouts may hinder users with limited dexterity.

Designing with these differences in mind helps create keypads and interfaces that are more intuitive, accessible, and usable for all.

CHAPTER 02.

VISIBILITY

Visibility is essential for ensuring keypads are easy to see and use, especially for users with varying levels of vision. Challenges such as low contrast, unclear symbol placement, and poor lighting can reduce readability and slow interaction. Enhancing visibility improves recognition speed, lowers cognitive effort, and builds user confidence. The following recommendations focus on addressing these issues to support clearer, more inclusive keypad design.

02. VISIBILITY



BEST PRACTICE



TOO SIMILAR COLORS



TOO LOW COLOR CONTRASTS

CONTRAST BETWEEN SYMBOL & KEY

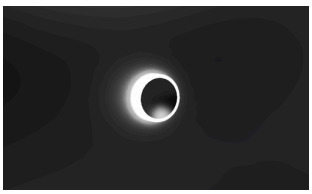
Ensure high color contrast between symbols and the key surface to maintain readability, especially in bright or varying lighting. Low contrast can make symbols difficult to distinguish, leading to confusion, slower input, and user frustration.

CONTRAST BETWEEN KEY & BACKGROUND

Visually separate keys from the background using contrasting colors, surface finishes, or perimeter lighting to improve key recognition and reduce cognitive load. When possible, use backlighting to enhance visibility in low-light conditions.







02. VISIBILITY



BEST PRACTICE



GOOD, SYMBOL
CLOSE TO SURFACE



TOO FAR BACK, NOT
READABLE

SYMBOL PLACEMENT

Position symbols on the front-facing surface of each key to ensure clear visibility during use. Avoid placing symbols on the rear of acrylic keys, as they can be obscured or visible only from limited angles.

Ensure symbols are not hidden by the keypad housing, which can cause confusion or hesitation, especially for users with visual impairments or limited experience. Clear, visible symbols improve recognition and overall usability.

SYMBOL SIZE

Use the full surface area of each key to maximize symbol and number size. Small or thin symbols can be difficult to read, especially in low light or for users with visual impairments. Larger, bolder symbols improve clarity and reduce misinterpretation.

Avoid fine details or thin lines, which may blur at small sizes or from angled views. Prioritize high contrast and ample sizing to ensure readability across varying lighting, distances, and user abilities.



BEST PRACTICE



TOO THIN & SMALL,
NOT UTILIZING THE
SPACE



BEST PRACTICE



TOO THIN & TOO MANY
DETAILS



ANGLED KEYS

Keys should be angled at least 10° toward the user when mounted at a standard height of about one meter. This tilt enhances symbol visibility and readability from a natural standing position, while remaining clear at eye level for wheelchair users or shorter individuals, reducing the need to adjust posture or viewing angle.

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ANGLED BODY

Instead of angling individual keys, tilting the entire keypad surface about 10° toward the user can achieve similar benefits. This orientation improves readability and symbol visibility from a typical standing position by aligning with the user's natural line of sight, reducing the need to adjust posture or viewing angle during use.

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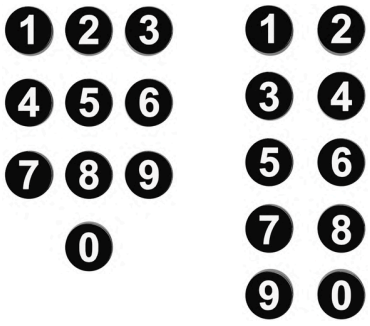
CHAPTER 03.

CLARITY

Clarity in keypad design ensures users can easily understand and operate the interface without confusion or hesitation. An intuitive layout, distinct functions, and clear feedback reduce cognitive load and simplify interaction.

Clear design helps users of all ages, abilities, and experience levels navigate the keypad confidently and efficiently. Easily recognizable symbols and labels minimize errors and create a smoother, more accessible user experience.

03. CLARITY



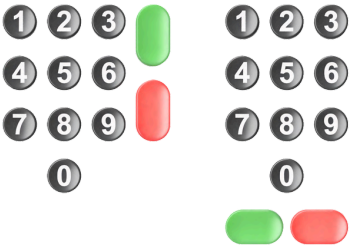
BEST PRACTICE NOT AS RECOGNIZABLE

LAYOUT

Use a **3x3+1** layout for numeric keys to ensure familiarity, reduce input errors, and enable quick interaction. Avoid alternative layouts, as they increase cognitive load and may cause confusion.

03. CLARITY

BEST PRACTICE



SPECIAL FUNCTION KEYS

Design special keys (e.g., lock/unlock) to stand out from numeric keys through distinct shape, size, color, and symbols. Position them near, but visually and tactilely separate from, the 3x3+1 layout to prevent confusion. Clear differentiation supports quick recognition and reduces errors.

TOO SIMILAR TO NUMERIC KEYS



03. CLARITY



BEST PRACTICE



TOO SIMILAR WITHOUT DISTINGUISHING COLORS



DOOR ICONS CAN BE INTERPRETED AS EMERGENCY EXITS

LOCK & UNLOCK KEYS

Design lock and unlock keys to be instantly recognizable and clearly distinct from numeric keys and each other. Combine intuitive symbols (closed/open padlock) with color cues (red for lock, green for unlock) to support quick decisions and reduce confusion. Color also assists users who may struggle to interpret symbol details.

BEST PRACTICE



SYMBOL PLACEMENT

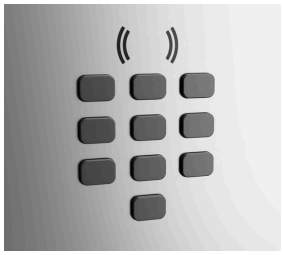
Place symbols directly on keys to clearly indicate their function. Avoid positioning symbols beside keys, as this can cause confusion, increase errors, and slow down key recognition.

AVOID PLACING SYMBOLS NEXT TO KEYS

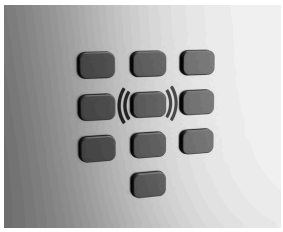




03. CLARITY



BEST PRACTICE



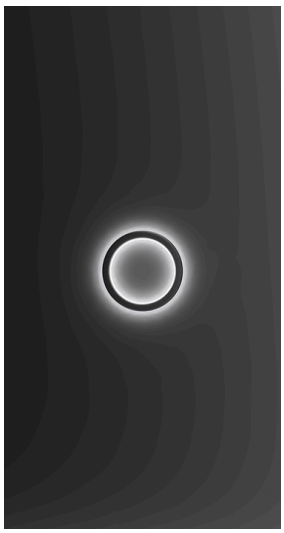
AVOID VISUAL CLUTTER

SYMBOL USE

Use only essential symbols to avoid visual clutter. If extra icons are necessary, position them away from keys and functions to prevent confusion.

FEEDBACK

Clear and immediate feedback is crucial for confirming user input. Without visual, auditory, or tactile feedback, users may hesitate, repeat actions, or assume a malfunction. Effective feedback reinforces clarity by confirming actions, guiding next steps, and reducing uncertainty, especially for users with limited experience or sensory impairments. Keypad feedback should be intuitive, consistent, and perceivable in various conditions.



CHAPTER 04.

TACTILITY

04. TACTILITY

Tactility is vital for accurate, confident keypad use, especially for users with reduced vision or motor skills. Flat surfaces, unclear key boundaries, and poorly placed tactile markers hinder key identification. Enhancing tactile features, such as key shape, texture, and tactile landmarks, improves physical feedback and navigation. The following recommendations aim to make keypads more touch-perceptible, supporting inclusive and accessible use.

04. TACTILITY



BEST PRACTICE



NOT WIDELY
RECOGNIZED

TACTILE MARKINGS

Place tactile markings prominently in the center of the “5” key. This widely recognized reference helps users navigate the keypad without visual input. Centered placement aligns with user expectations and established patterns, improving key location, especially for visually impaired users.





TACTILE MARKINGS ON SPECIAL FUNCTION KEYS

Use tactile markings to differentiate special function keys. These variations help users identify functions by touch, crucial for those with visual impairments or in low-visibility settings. Clear tactile cues reduce input errors and enhance overall accessibility.

04. TACTILITY



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DIFFERENCE IN TEXTURE

Incorporating texture in keypad design helps users distinguish keys from the surrounding surface by touch alone. Subtle changes in material, surface finish, or raised edges provide valuable tactile cues, especially for users with visual impairments or in low-visibility settings.

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04. TACTILITY

KEY SIZE

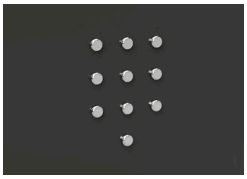
Keys should be wide enough to fit a fingertip comfortably, with a minimum width of **10 mm** recommended. Smaller keys are harder to distinguish by touch and demand greater fine motor control, reducing accuracy and accessibility, especially for users with limited dexterity.



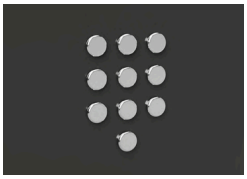
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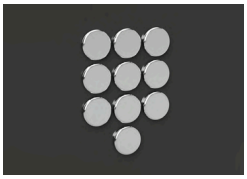
04. TACTILITY



TOO SPARSE



BEST PRACTICE



TOO CLOSE

KEY DISTANCE

Provide sufficient space between keys to prevent accidental presses and support accurate input, particularly for users with reduced dexterity or larger fingers. Avoid excessive spacing, which can slow interaction and increase effort. Balanced spacing enhances comfort, input speed, and overall accessibility. **2 mm** to **4 mm** spacing is recommended depending on key size.

04. TACTILITY

KEY EDGE & SURFACE

Design keys with clearly defined edges to support tactile distinction and improve input precision. Avoid overly rounded corners, as they can make it harder to distinguish keys by touch, especially for users with reduced vision or dexterity. A concave surface further enhances accuracy by guiding the fingertip toward the center of the key, providing a stable resting point that reduces slips and supports confident interaction. Recommended depth, approximately **1 mm**.

BEST PRACTICE

TOO ROUNDED



04. TACTILITY



KEY PROTRUTION

Keys should protrude enough to support tactile discovery and provide clear physical feedback upon press. A protrusion of around **2 mm** is recommended to balance detectability and input precision. Too little protrusion makes keys hard to locate by touch, while too much can reduce input accuracy and affect perceived quality.

DESIGN RECOMMENDATIONS

FOR INCLUSIVE KEYPAD DESIGN

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